

Below are **important basic MCQs** created **directly from your uploaded file**:

File: *Introduction to Molecular Cell Biology*

(Cited as:

1-Introduction to Cell Biology

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Based strictly on the content of the PPT)

Module 1 – Introduction to Cell Biology

1. Cell Biology primarily focuses on the study of:

- A. Ecosystems
- B. Atoms and subatomic particles
- C. Cells and their molecular processes
- D. Evolution of species

Answer: C. Cells and their molecular processes

1-Introduction to Cell Biology

2. The course progresses in which sequence according to the outline?

- A. Molecules → Atoms → Cells → Cell Structures
- B. Cell Structures → Cells → Atoms → Molecules
- C. Atoms → Molecules → Cell Structures → Cells
- D. Cells → Organisms → Molecules → Atoms

Answer: C. Atoms → Molecules → Cell Structures → Cells

1-Introduction to Cell Biology

3. Which of the following is NOT listed as a major Cell Function in the lecture?

- A. Propagation / Heredity
- B. Expression Regulation
- C. Cell Communication
- D. Photosynthesis

Answer: D. Photosynthesis

1-Introduction to Cell Biology

4. Applied Cell Biology focuses on:

- A. Only plant cells
- B. Only human diseases
- C. Use of cellular knowledge for practical applications
- D. Genetic engineering only

Answer: C. Use of cellular knowledge for practical applications

1-Introduction to Cell Biology

Module 2 – Course Objectives

5. One major objective of the course is to understand:

- A. How living organisms evolve
- B. The molecular logic of the living state
- C. How planets support life
- D. How humans adapt to climate

Answer: B. The molecular logic of the living state

1-Introduction to Cell Biology

6. According to the lecture, a molecule's function arises from its:

- A. Origin
- B. Size
- C. Structure
- D. Color

Answer: C. Structure

1-Introduction to Cell Biology

7. The course aims to explain how molecular information is conveyed:

- A. To the environment
- B. To the rest of the cellular machinery
- C. To neighboring organisms
- D. To the immune system only

Answer: B. To the rest of the cellular machinery

1-Introduction to Cell Biology

8. System manipulation in cell biology refers to:

- A. Designing new planets
- B. Altering cellular components for study or application
- C. Changing species
- D. Producing artificial intelligence

Answer: B. Altering cellular components for study or application

1-Introduction to Cell Biology

Module 3 – General Concepts

9. Cell Biology connects molecular structure to:

- A. Planetary cycles
- B. Cellular function
- C. Global ecosystems
- D. Physical chemistry only

Answer: B. Cellular function

1-Introduction to Cell Biology

10. Propagation or heredity in cells concerns:

- A. Production of body heat
- B. Passing genetic information to daughter cells
- C. Transporting oxygen
- D. Eliminating waste products

Answer: B. Passing genetic information to daughter cells

1-Introduction to Cell Biology

11. Expression regulation involves controlling:

- A. Cell death timing
- B. Which genes are turned on or off
- C. Nutrient absorption
- D. Atomic stability

Answer: B. Which genes are turned on or off

1-Introduction to Cell Biology

12. Cell communication refers to:

- A. Talking between humans
- B. Interactions and signaling between cells
- C. Signals from the nervous system only
- D. Weather patterns

Answer: B. Interactions and signaling between cells

1-Introduction to Cell Biology

Below are **important basic MCQs** created **directly from your uploaded file:**

File: What is a Cell?

(Cited as:

2. What is a cell

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on the PPT content)

Module 1 – Introduction to Cells

1. A cell is defined as the:

- A. Largest unit of life
- B. Smallest independent unit of life
- C. Basic unit of atoms
- D. Component of only plants

Answer: B. Smallest independent unit of life

2. What is a cell

2. Which of the following is *not* a characteristic of a cell?

- A. Can grow
- B. Can divide
- C. Can synthesize biomolecules
- D. Cannot function independently

Answer: D. Cannot function independently

2. What is a cell

3. A cell is considered both a:

- A. Structural and chemical unit
- B. Structural and functional unit
- C. Chemical and mechanical unit
- D. Mechanical and thermal unit

Answer: B. Structural and functional unit

2. What is a cell

Module 2 – Types of Cells

4. How many basic types of cells are mentioned?

- A. One
- B. Two
- C. Three
- D. Four

Answer: B. Two (Eukaryotic & Prokaryotic)

2. What is a cell

5. Eukaryotic cells differ from prokaryotic cells mainly because they:

- A. Have no organelles
- B. Lack DNA
- C. Have a true nucleus
- D. Are always unicellular

Answer: C. Have a true nucleus

2. What is a cell

Module 3 – Structure of the Cell

6. How many basic structural parts of a cell are listed?

- A. Two
- B. Three
- C. Four
- D. Five

Answer: B. Three (Plasma membrane, Cytoplasm, Nucleus)

2. What is a cell

7. The outer boundary of a cell is called:

- A. Nucleus
- B. Cytoplasm
- C. Plasma membrane
- D. Cell fluid

Answer: C. Plasma membrane

2. What is a cell

8. The jelly-like material inside the cell is the:

- A. Membrane
- B. Cytoplasm
- C. Nucleus
- D. Chromosome

Answer: B. Cytoplasm

2. What is a cell

9. The control center of the cell is the:

- A. Plasma membrane
- B. Cytoplasm
- C. Nucleus
- D. Mitochondria

Answer: C. Nucleus

2. What is a cell

Module 4 – Functions of Cells

10. Cells are responsible for:

- A. Only digestion
- B. Making all body parts
- C. Only reproduction
- D. Only movement

Answer: B. Making all body parts

2. What is a cell

11. Which of the following is NOT listed as a function of cells?

- A. Secretion
- B. Absorption
- C. Transportation
- D. Photosynthesis

Answer: D. Photosynthesis

2. What is a cell

12. All cellular activities are performed by:

- A. Organs
- B. Organisms
- C. Cells
- D. Nuclei only

Answer: C. Cells

2. What is a cell

Module 5 – Number of Cells in Organisms

13. A unicellular organism contains:

- A. No cells
- B. Two cells
- C. One single cell
- D. Hundreds of cells

Answer: C. One single cell

2. What is a cell

14. Multicellular organisms contain:

- A. Only one cell
- B. A few to trillions of cells
- C. Only plant cells
- D. Only animal cells

Answer: B. A few to trillions of cells

2. What is a cell

15. Humans are considered:

- A. Unicellular organisms
- B. Non-cellular beings
- C. Multicellular organisms
- D. Prokaryotic organisms

Answer: C. Multicellular organisms

2. What is a cell

Below are **important basic MCQs** created **directly from your uploaded file:**

File: *Cell Theory – Development and Modern Concepts*

(Cited as:

3.Cell Theory

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

Module 1 – Modern Cell Theory

1. According to the Modern Cell Theory, all living organisms are composed of:

- A. Tissues only
- B. One or more cells
- C. Only plant cells
- D. Only animal cells

Answer: B. One or more cells

3.Cell Theory

2. The basic structural and functional unit of life is the:

- A. Tissue
- B. Organ
- C. Cell
- D. Organ system

Answer: C. Cell

3.Cell Theory

3. According to the Modern Cell Theory, all cells arise from:

- A. Chemical reactions
- B. Dead tissue
- C. Airborne particles
- D. Pre-existing cells

Answer: D. Pre-existing cells

3.Cell Theory

Module 2 – Early Discoveries in Cell Theory

4. Robert Hooke first described cells in which year?

- A. 1600
- B. 1665
- C. 1750
- D. 1800

Answer: B. 1665

3.Cell Theory

5. Hooke observed cells in:

- A. Human tissues
- B. Blood
- C. Cork
- D. Leaves

Answer: C. Cork

3.Cell Theory

6. The scientist who observed living microorganisms (“animalcules”) in pond water was:

- A. Robert Hooke
- B. Anton van Leeuwenhoek
- C. Robert Brown
- D. Lamarck

Answer: B. Anton van Leeuwenhoek

7. Jean-Baptiste de Lamarck proposed that no organism can live without:

- A. Water
- B. Light
- C. Cellular tissues
- D. Food

Answer: C. Cellular tissues

8. Robert Brown discovered the:

- A. Cell membrane
- B. Ribosome
- C. Nucleus
- D. Mitochondria

Answer: C. Nucleus

Module 3 – Major Contributions to Cell Theory

9. Matthias Schleiden concluded that all _____ are made of cells.

- A. Animals
- B. Fungi
- C. Plants
- D. Bacteria

Answer: C. Plants

10. Theodor Schwann declared that all _____ are made of cells.

- A. Plants
- B. Animals
- C. Protists
- D. Viruses

Answer: B. Animals

11. Rudolf Virchow proposed which famous concept?

- A. Spontaneous generation
- B. All organisms evolve from one ancestor
- C. Cells arise from pre-existing cells
- D. Cells appear from nonliving matter

Answer: C. Cells arise from pre-existing cells (Omnis cellula e cellula)

12. Louis Pasteur provided experimental proof for:

- A. Abiogenesis
- B. Biogenesis
- C. Natural selection
- D. Genetics

Answer: B. Biogenesis

Below are **important basic MCQs** created directly from your uploaded file:**

File: Unit of Life

(Cited as:

4-Unit of life

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on the PPT content)

Module 1 – Basic Concept of Unit of Life

1. The basic unit of life is the:

- A. Tissue
- B. Cell
- C. Organ
- D. Molecule

Answer: B. Cell

4-Unit of life

2. According to the slide, all cells come from:

- A. Chemical reactions
- B. Preexisting cells
- C. Artificial synthesis
- D. Non-living material

Answer: B. Preexisting cells

4-Unit of life

3. The study of the unit of life falls under:

- A. Anatomy
- B. Botany
- C. Cell Biology
- D. Ecology

Answer: C. Cell Biology

4-Unit of life

Module 2 – Cellular Organization

4. The "Unit of Life" includes studying which of the following?

- A. Atmosphere and climate
- B. Cell structure, cellular compartments, organelles
- C. Evolution of species
- D. Chemical bonding

Answer: B. Cell structure, cellular compartments, organelles

4-Unit of life

5. Organelles within a cell are related to:

- A. Only structural functions
- B. Only energy production
- C. Specific structures with specific functions
- D. Random cellular positions

Answer: C. Specific structures with specific functions
4-Unit of life

Module 3 – Nature of Life

6. Life is described as:

- A. A random collection of molecules
- B. A collection of macromolecules performing unique functions
- C. Only based on chemical reactions
- D. Independent of structural compartments

Answer: B. A collection of macromolecules performing unique functions
4-Unit of life

7. Structural compartments inside cells help maintain:

- A. Instability
- B. Randomness
- C. Homeostasis (consistency)
- D. Flexibility only

Answer: C. Homeostasis (consistency)
4-Unit of life

8. Cellular compartments are important because they:

- A. Store DNA only
- B. Prevent all reactions
- C. Allow macromolecules to function efficiently
- D. Reduce cellular complexity

Answer: C. Allow macromolecules to function efficiently
4-Unit of life

Module 4 – Cell Theory Connection

9. The concept “all organisms are composed of cells” is part of:

- A. Mendel’s laws
- B. Darwin’s theory
- C. Cell Theory
- D. Biomolecule theory

Answer: C. Cell Theory
4-Unit of life

10. Cells are not just grouped molecules but:

- A. Energetic waves
- B. Random structures
- C. Organized functional systems
- D. Non-living assemblies

Answer: C. Organized functional systems
4-Unit of life

Your uploaded file “**Journey into the Cell**” contains **very limited content**, but I have created **important basic MCQs** strictly from the available information.

(All content cited from the file:

5-Journey into the Cell

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Based strictly on the PPT content)

Module 1 – Introduction

1. The topic “Journey into the Cell” falls under which broader subject?

- A. Genetics
- B. Ecology
- C. Introduction to Molecular Cell Biology
- D. Evolution

Answer: C. Introduction to Molecular Cell Biology

5-Journey into the Cell

2. The presentation is part of which course section?

- A. Cell Theory
- B. Cell Chemistry
- C. Unit of Life
- D. Plant Physiology

Answer: C. Unit of Life

5-Journey into the Cell

Module 2 – Cell Biology Context

3. Journey into the Cell is categorized under:

- A. Microbiology
- B. Astronomy
- C. Cell Biology
- D. Environmental Biology

Answer: C. Cell Biology

5-Journey into the Cell

4. The phrase “Unit of Life” in the file refers to:

- A. Atoms
- B. Tissues
- C. Organ systems
- D. Cells

Answer: D. Cells

5-Journey into the Cell

Module 3 – Visualization / Animation

5. The slide includes a reference to:

- A. Diagrams
- B. Scientific models
- C. An animation to explore the cell
- D. Video lectures

Answer: C. An animation to explore the cell (“View animation”)

5-Journey into the Cell

6. The purpose of the “Journey into the Cell” is likely to:

- A. Study tissues
- B. Visualize internal cell structures
- C. Learn about ecosystems
- D. Analyze DNA mutations

Answer: B. Visualize internal cell structures

5-Journey into the Cell

Module 4 – General Concepts

7. The phrase “Cell Biology” indicates the study of:

- A. Whole organisms
- B. Chemical reactions only
- C. Cellular structure and function
- D. Evolutionary patterns

Answer: C. Cellular structure and function

5-Journey into the Cell

8. The term “Unit of Life” implies that cells are the:

- A. Largest biological units
- B. Smallest living units
- C. Basis of genetics only
- D. Components of ecosystems

Answer: B. Smallest living units

5-Journey into the Cell

Below are **important basic MCQs** created **directly from your uploaded file:**

File: Size Matters

(Cited as:

6-Size Matters

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on the PPT content)

Module 1 – Cell Size and Surface Area

1. Cells are small mainly to:

- A. Reduce DNA content
- B. Maintain a large surface area to volume ratio
- C. Store more organelles
- D. Increase cell division rate

Answer: B. Maintain a large surface area to volume ratio

6-Size Matters

2. A large surface area to volume ratio helps the cell by:

- A. Slowing transport of materials
- B. Increasing exchange of materials
- C. Reducing membrane function
- D. Preventing photosynthesis

Answer: B. Increasing exchange of materials (implied by importance of SA/V ratio)

Module 2 – Prokaryotic Cell Structure

3. Prokaryotic cells lack:

- A. Cytoplasm
- B. DNA
- C. Membrane-enclosed internal compartments
- D. Ribosomes

Answer: C. Membrane-enclosed internal compartments

6-Size Matters

4. Which structure in prokaryotes regulates the traffic of materials in and out of the cell?

- A. Cell wall
- B. Plasma membrane
- C. Nucleoid region
- D. Flagella

Answer: B. Plasma membrane

6-Size Matters

5. The nucleoid region of a prokaryotic cell contains:

- A. RNA
- B. Ribosomes
- C. DNA
- D. Lipids

Answer: C. DNA

6-Size Matters

6. Most prokaryotes have a cell wall which helps in:

- A. ATP production
- B. Protection from phagocytes
- C. Protein synthesis
- D. Communication

Answer: B. Protection from phagocytes

6-Size Matters

7. The cell wall of prokaryotes is involved in:

- A. Photosynthesis
- B. Support and cell shape
- C. DNA replication
- D. Hormone secretion

Answer: B. Support and cell shape

6-Size Matters

Module 3 – Special Features of Prokaryotes

8. Cyanobacteria contain chlorophyll and have:

- A. Nuclei
- B. Folds of plasma membrane
- C. Lysosomes
- D. Mitochondria

Answer: B. Folds of plasma membrane
6-Size Matters

9. Mesosomes in some prokaryotes function in:

- A. Cell communication
- B. Energy processes
- C. Protein packaging
- D. DNA storage

Answer: B. Energy processes (implied from slide)

6-Size Matters

10. Some prokaryotes contain actin-like filaments which are important for:

- A. Enzyme synthesis
- B. Cell structure and movement
- C. Food storage
- D. Reproduction

Answer: B. Cell structure and movement (function of actin-like filaments)

6-Size Matters

11. Prokaryotic flagella are made of:

- A. Tubulin
- B. Myosin
- C. Flagellin
- D. Actin

Answer: C. Flagellin

6-Size Matters

Here are **important basic MCQs** created directly from your uploaded file:

File: *Why Most Cells Are Small?*

(Cited as:

7. why most cell are small

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on the PPT content)

Module 1 – Average Size of Cells

1. Most cells range in size between:

- A. 1–10 mm
- B. 5–150 μm
- C. 1–2 cm
- D. 0.1–0.5 m

Answer: B. 5–150 μm

7. why most cell are small

2. The largest known single cell mentioned in the slides is:

- A. Human neuron
- B. Red blood cell
- C. Ostrich egg
- D. Liver cell

Answer: C. Ostrich egg (15 × 13 cm)

7. why most cell are small

3. The approximate length of a human neuron is:

- A. 1 cm
- B. 1 m
- C. 10 m
- D. 150 μm

Answer: B. 1 m

7. why most cell are small

4. The approximate length of an elephant neuron is:

- A. 1 m
- B. 2 m
- C. 4 m
- D. 10 cm

Answer: C. 4 m (approx.)

7. why most cell are small

Module 2 – Reasons for Smallness of Cells

5. Increasing cell size increases:

- A. Surface area only
- B. Volume enormously
- C. Both surface area and volume equally
- D. Neither surface area nor volume

Answer: B. Volume enormously

7. why most cell are small

6. Two basic reasons why cells remain small are:

- A. Temperature and pressure
- B. Nucleus control and membrane transport
- C. Osmosis and diffusion
- D. Movement and reproduction

Answer: B. Nucleus control and membrane transport

7. why most cell are small

Module 3 – Control of the Nucleus

7. As cell size increases, the ratio of nucleus volume to cytoplasm volume:

- A. Increases
- B. Remains constant
- C. Decreases
- D. Doubles

Answer: C. Decreases

7. why most cell are small

8. If the cytoplasm becomes too large, the nucleus will:

- A. Gain more control
- B. Lose control over cellular activities
- C. Divide more rapidly
- D. Shrink in size

Answer: B. Lose control over cellular activities

7. why most cell are small

Module 4 – Reduced Surface Area Problem

9. When a cell grows, its volume increases:

- A. Less than surface area
- B. Greater than surface area
- C. At the same rate as surface area
- D. Only when cell divides

Answer: B. Greater than surface area

7. why most cell are small

10. A reduced surface area relative to volume makes it difficult for a cell to:

- A. Produce ATP
- B. Exchange gases efficiently
- C. Replicate DNA
- D. Perform cell division

Answer: B. Exchange gases efficiently

7. why most cell are small

11. Reduced surface area also hinders:

- A. Hormone release
- B. Uptake of nutrients
- C. Chromosome formation
- D. Protein synthesis

Answer: B. Uptake of nutrients

7. why most cell are small

12. Increased volume with limited surface area affects:

- A. Waste removal
- B. Mitochondrial function only
- C. Photosynthesis
- D. Cell differentiation only

Answer: A. Waste removal

7. why most cell are small

Here are **important basic MCQs** created **directly from your uploaded file:**

File: *Composition of Matter*

(Cited as:

8-Composition of Matter

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on the PPT content)

Module 1 – Atomic Structure

1. Matter is composed of:

- A. Cells
- B. Atoms

C. Molecules only

D. Electrons only

Answer: B. Atoms

8-Composition of Matter

2. The smallest unit of matter that retains the properties of an element is the:

A. Molecule

B. Cell

C. Atom

D. Proton

Answer: C. Atom

8-Composition of Matter

3. Atomic structure refers to:

A. Arrangement of cells

B. Arrangement of protons, neutrons, and electrons

C. Folding of proteins

D. Bonding patterns only

Answer: B. Arrangement of protons, neutrons, and electrons

8-Composition of Matter

Module 2 – Carbon Atom

4. The element essential for all organic molecules is:

A. Hydrogen

B. Oxygen

C. Carbon

D. Sodium

Answer: C. Carbon

8-Composition of Matter

5. Carbon is important in biology mainly because it can form:

A. Only ionic bonds

B. Four covalent bonds

C. Only hydrogen bonds

D. Metal ions

Answer: B. Four covalent bonds (implied by Carbon atom slide)

8-Composition of Matter

Module 3 – Electronegativity

6. Electronegativity is defined as:

A. Ability to lose protons

B. Tendency of an atom to attract electrons

C. Ability to split molecules

D. Ability to form ions

Answer: B. Tendency of an atom to attract electrons

8-Composition of Matter

7. A highly electronegative atom will:

A. Repel electrons

B. Attract electrons strongly

- C. Lose neutrons
 - D. Become positively charged always
- Answer:** B. Attract electrons strongly

8-Composition of Matter

Module 4 – Polarity of Water & Hydrogen Bonding

8. The polarity of H₂O results in:

- A. Covalent bonding
- B. Hydrogen bonding
- C. Ionic bonding
- D. Metallic bonding

Answer: B. Hydrogen bonding

8-Composition of Matter

9. Polarity in water molecules is caused by:

- A. Equal sharing of electrons
- B. Unequal sharing of electrons
- C. Absence of hydrogen
- D. Absence of oxygen

Answer: B. Unequal sharing of electrons (implied by polarity concept)

8-Composition of Matter

Module 5 – Ionic Bond

10. An ionic bond is formed when:

- A. Atoms share electrons equally
- B. Atoms share electrons unequally
- C. One atom donates electrons to another
- D. Protons move between atoms

Answer: C. One atom donates electrons to another

8-Composition of Matter

11. Ionic bonds typically form between:

- A. Two nonmetals
- B. Two metals
- C. A metal and a nonmetal
- D. Carbon atoms only

Answer: C. A metal and a nonmetal (general chemical concept)

8-Composition of Matter

Below are **important basic MCQs** created **directly from your uploaded file:**

File: *Molecules of Life*

(Cited as:

9- Molecules of life

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

Module 1 – Composition of Life

1. Molecules of life are responsible for:

- A. Cell division only
- B. Composition of living organisms
- C. Movement only
- D. Photosynthesis only

Answer: B. Composition of living organisms

9- Molecules of life

2. Biological molecules are also called:

- A. Minerals
- B. Nutrients
- C. Molecules of life
- D. Enzymes only

Answer: C. Molecules of life

9- Molecules of life

Module 2 – Building Blocks of Life

3. Cells contain different molecules such as:

- A. Water, gases, minerals
- B. Polymers, monomers, ions
- C. Proteins, lipids, carbohydrates, nucleic acids
- D. Vitamins, hormones, enzymes

Answer: C. Proteins, lipids, carbohydrates, nucleic acids (implied by “Building Blocks of Life”)

9- Molecules of life

4. Macromolecules are made from smaller repeating units called:

- A. Elements
- B. Polymers
- C. Monomers
- D. Acids

Answer: C. Monomers

9- Molecules of life

Module 3 – Polymers & Monomers

5. Macromolecules are:

- A. Simple sugars only
- B. Polymers of smaller monomers
- C. Always made of lipids
- D. Found only in plant cells

Answer: B. Polymers of smaller monomers

9- Molecules of life

6. The reaction in which monomers join together by removing water is called:

- A. Hydrolysis
- B. Condensation

C. Ionization
D. Reduction
Answer: B. Condensation
9- Molecules of life

7. Hydrolysis reactions involve:

- A. Removing water from polymers
- B. Adding water to break polymers into monomers
- C. Breaking hydrogen bonds only
- D. Forming peptide bonds

Answer: B. Adding water to break polymers into monomers
9- Molecules of life

8. During condensation reactions, the product formed alongside the polymer is:

- A. Peptide
- B. ATP
- C. Water
- D. Oxygen

Answer: C. Water
9- Molecules of life

Module 4 – Importance of Chemical Reactions

9. Condensation reactions are essential for building:

- A. DNA and proteins
- B. Only carbohydrates
- C. Only lipids
- D. Mineral salts

Answer: A. DNA and proteins (general biological understanding consistent with slides)
9- Molecules of life

10. Hydrolysis reactions are important in:

- A. Breaking down food polymers
- B. Making fats only
- C. Forming amino acids
- D. Synthesizing ATP directly

Answer: A. Breaking down food polymers (consistent with hydrolysis concept)
9- Molecules of life

Here are **important basic MCQs** created directly from your uploaded file:

File: *Proteins – Polymers of Amino Acids*

(Cited as:

10-Proteins Polymers of Amino A...

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on PPT content)

Module 1 – Protein Basics

1. Proteins are polymers of:

- A. Sugars
- B. Fatty acids
- C. Amino acids
- D. Nucleotides

Answer: C. Amino acids

10-Proteins Polymers of Amino A...

2. The most abundant polymer in the cell is:

- A. DNA
- B. Lipid
- C. Protein
- D. Carbohydrate

Answer: C. Protein

10-Proteins Polymers of Amino A...

3. Which of the following is NOT a function of proteins?

- A. Catalysis
- B. Protection
- C. Genetic information storage
- D. Transport

Answer: C. Genetic information storage (proteins do NOT store genetic info)

10-Proteins Polymers of Amino A...

Module 2 – Protein Functions

4. Proteins play a role in all of the following EXCEPT:

- A. Movement
- B. Regulation
- C. Defense
- D. Storing hereditary material

Answer: D. Storing hereditary material

10-Proteins Polymers of Amino A...

5. Structural support in cells is mainly due to:

- A. Lipids
- B. Carbohydrates
- C. Proteins
- D. ATP

Answer: C. Proteins (support is one of the listed functions)

10-Proteins Polymers of Amino A...

Module 3 – Amino Acid Structure

6. The properties of each amino acid are determined by its:

- A. Amino group
- B. Carboxyl group
- C. R group (side chain)
- D. Peptide bond

Answer: C. R group (side chain)

7. Amino acids differ from one another due to variations in their:

- A. Central carbon
- B. Hydrogen atom
- C. Side chain (R group)
- D. Carboxylic acid

Answer: C. Side chain (R group)

10-Proteins Polymers of Amino A...

Module 4 – Types of R Groups

8. R groups can be:

- A. Only nonpolar
- B. Only polar
- C. Charged, polar, or hydrophobic
- D. Only charged

Answer: C. Charged, polar, or hydrophobic

10-Proteins Polymers of Amino A...

9. A special feature of cysteine R groups is that they can:

- A. Form peptide bonds
- B. Store energy
- C. Form disulfide bridges
- D. Produce ATP

Answer: C. Form disulfide bridges

10-Proteins Polymers of Amino A...

10. Nonpolar R groups are typically:

- A. Hydrophilic
- B. Hydrophobic
- C. Charged
- D. Acidic

Answer: B. Hydrophobic (implied from nonpolar nature)

10-Proteins Polymers of Amino A...

11. Polar R groups tend to be:

- A. Water-repelling
- B. Hydrophobic
- C. Water-attracting
- D. Metallic

Answer: C. Water-attracting (polar = hydrophilic)

10-Proteins Polymers of Amino A...

Here are **important basic MCQs** created directly from your uploaded file:

File: *Structural Levels of Proteins – Primary Structure*

(Cited as:

11. structural level of protein...

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

Module 1 – Primary Structure of Proteins

1. The primary structure of a protein refers to:

- A. 3D folding of the polypeptide
- B. Sequence of amino acids in a linear chain
- C. Interaction of multiple polypeptide chains
- D. Alpha helix and beta sheet arrangement

Answer: B. Sequence of amino acids in a linear chain

11. structural level of protein...

2. The primary structure is made up of:

- A. Lipids
- B. Monosaccharides
- C. Polypeptide chains
- D. Fatty acids

Answer: C. Polypeptide chains

11. structural level of protein...

3. Amino acids in the primary structure are linked by:

- A. Ionic bonds
- B. Hydrogen bonds
- C. Peptide bonds
- D. Disulfide bonds

Answer: C. Peptide bonds

11. structural level of protein...

Module 2 – Peptide Bond

4. A peptide bond is formed between:

- A. Two R groups
- B. COOH of one amino acid and NH₂ of another
- C. Two NH₂ groups
- D. Two COOH groups

Answer: B. COOH of one amino acid and NH₂ of another

11. structural level of protein...

5. The formation of a peptide bond results in:

- A. Breakdown of protein
- B. Formation of ATP
- C. Joining of two amino acids
- D. Formation of disulfide linkage

Answer: C. Joining of two amino acids

11. structural level of protein...

Module 3 – Importance of Primary Structure

6. Primary structure determines the protein's:

- A. Size of polypeptide
- B. Tertiary structure only

C. Hydrogen bonding pattern

D. Lipid content

Answer: A. Size of polypeptide (molecule size depends on amino acid chain length)

11. structural level of protein...

7. The most important characteristic of primary structure is:

A. Polarity

B. Solubility

C. Amino acid sequence

D. Color

Answer: C. Amino acid sequence

11. structural level of protein...

8. A change in the primary structure of a protein may alter its:

A. DNA only

B. Tertiary and quaternary structure

C. Inorganic content

D. Ribosome binding

Answer: B. Tertiary and quaternary structure (biological implication of sequence change)

Here are **important basic MCQs** created directly from your uploaded file:

File: *Structural Levels of Proteins – Secondary Structure*

(Cited as:

12. structural level of protein...

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on PPT content)

Module 1 – Secondary Structure Basics

1. Secondary structure of proteins refers to:

A. Overall 3D folding of the entire polypeptide

B. Linear sequence of amino acids

C. Local folding of the polypeptide chain

D. Interaction between multiple polypeptides

Answer: C. Local folding of the polypeptide chain

12. structural level of protein...

2. Secondary structure is stabilized mainly by:

A. Ionic bonds

B. Covalent bonds

C. Hydrogen bonds

D. Disulfide bonds

Answer: C. Hydrogen bonds

12. structural level of protein...

3. Secondary structure shows a repeating pattern of hydrogen bonds between:

A. R groups

B. DNA bases

C. Amino acids

D. Peptide bonds only

Answer: C. Amino acids
12. structural level of protein...

Module 2 – Alpha Helix

4. The α -helix is described as:

- A. Left-handed helix
- B. Right-handed helix
- C. Flat sheet-like structure
- D. Irregular folding

Answer: B. Right-handed helix
12. structural level of protein...

5. In an α -helix, the amino hydrogen of one amino acid bonds with the carboxyl oxygen of the:

- A. Second amino acid
- B. Third amino acid
- C. Fourth amino acid
- D. Fifth amino acid

Answer: C. Fourth amino acid
12. structural level of protein...

Module 3 – Beta-Pleated Sheets

6. In β -pleated sheets, the polypeptide chains form:

- A. A helical structure
- B. Sheets
- C. Random coils
- D. Double helices

Answer: B. Sheets
12. structural level of protein...

7. Sheets in β -pleated structures are held together by:

- A. Disulfide bonds
- B. Peptide bonds
- C. Hydrogen bonds
- D. Hydrophobic interactions

Answer: C. Hydrogen bonds
12. structural level of protein...

Module 4 – Other Secondary Structures

8. Besides alpha helices and beta sheets, other forms of secondary structure include:

- A. Supercoils
- B. β -turns and random coils
- C. Ionic loops
- D. Peptide bridges

Answer: B. β -turns and random coils
12. structural level of protein...

Here are **important basic MCQs** created directly from your uploaded file:

File: Structural Levels of Proteins – Tertiary Structure

(Cited as:

13. structural level of protein...

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on PPT content)

Module 1 – Tertiary Structure Basics

1. Tertiary structure of a protein refers to:

- A. Linear sequence of amino acids
- B. Local folding such as alpha helices
- C. Overall 3D folding of the polypeptide chain
- D. Interaction between multiple polypeptides

Answer: C. Overall 3D folding of the polypeptide chain

13. structural level of protein...

2. The tertiary structure gives the protein its:

- A. Basic amino acid sequence
 - B. Local helical structure
 - C. Final functional shape
 - D. Ability to form peptides
- Answer:** C. Final functional shape (implied from 3D folding concept)

13. structural level of protein...

Module 2 – Bonds in Tertiary Structure

3. Which of the following bonds are involved in maintaining tertiary structure?

- A. Peptide bonds only
- B. Hydrogen, ionic, hydrophobic, disulfide
- C. Only hydrophobic interactions
- D. Only disulfide bonds

Answer: B. Hydrogen, ionic, hydrophobic, disulfide

13. structural level of protein...

4. Disulfide bonds form between which amino acid residues?

- A. Lysine
- B. Glycine
- C. Cysteine
- D. Alanine

Answer: C. Cysteine (general protein knowledge consistent with tertiary structure bonding)

5. Hydrophobic interactions in proteins occur when:

- A. Nonpolar side chains avoid water
- B. Hydrophilic groups bind water strongly
- C. Disulfide bridges form
- D. Ionic groups attract each other

Answer: A. Nonpolar side chains avoid water (conceptual extension consistent with slide)

Module 3 – Examples of Tertiary Structure Proteins

6. An example of a protein with a tertiary structure is:

- A. Collagen
- B. Myoglobin
- C. Keratin
- D. Hemoglobin (quaternary)

Answer: B. Myoglobin

13. structural level of protein...

7. Another protein listed as having tertiary structure is:

- A. Insulin
- B. DNA polymerase
- C. Salivary amylase
- D. Tubulin

Answer: C. Salivary amylase

13. structural level of protein...

Here are **important basic MCQs** created directly from your uploaded file:

File: *Structural Levels of Proteins – Quaternary Structure*

(Cited as:

14. structural level of protein...

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on PPT content)

Module 1 – Quaternary Structure Basics

1. Quaternary structure of a protein is formed when:

- A. A single polypeptide folds into 3D form
- B. Two or more polypeptide chains fold together
- C. Amino acids join to form peptide bonds
- D. Alpha helices form and fold locally

Answer: B. Two or more polypeptide chains fold together

14. structural level of protein...

2. Quaternary structure is found in proteins that are:

- A. Only single polypeptides
- B. Unfolded
- C. Made of multiple subunits
- D. Always fibrous

Answer: C. Made of multiple subunits (implied by definition)

Module 2 – Bonds in Quaternary Structure

3. Which of the following bonds can help stabilize quaternary structure?

- A. Peptide bonds only
- B. Hydrogen, ionic, Van der Waals, sometimes covalent bonds
- C. Only disulfide bonds
- D. Only ionic bonds

Answer: B. Hydrogen, ionic, Van der Waals, sometimes covalent bonds

14. structural level of protein...

4. The weakest interactions that help hold subunits together are:

- A. Covalent bonds
- B. Van der Waals forces
- C. Peptide bonds
- D. Disulfide bridges

Answer: B. Van der Waals forces (listed among the bonding types)

14. structural level of protein...

Module 3 – Examples of Quaternary Structure Proteins

5. Which of the following proteins has a quaternary structure?

- A. Myoglobin
- B. Pepsin
- C. Hemoglobin
- D. Trypsin

Answer: C. Hemoglobin

14. structural level of protein...

6. Other examples of proteins with quaternary structure include:

- A. Collagen only
- B. Pepsin and keratin
- C. Keratin only
- D. Amylase

Answer: B. Pepsin and keratin

14. structural level of protein...

Module 4 – Fibrous Proteins

7. Fibrous proteins are:

- A. Water soluble and spherical
- B. Functional proteins found in blood
- C. Long fiber-like chains providing structural support
- D. Enzymes involved in catalysis

Answer: C. Long fiber-like chains providing structural support

14. structural level of protein...

8. Which of the following is NOT a fibrous protein?

- A. Keratin
- B. Collagen
- C. Silk fibroin
- D. Hemoglobin

Answer: D. Hemoglobin (Hemoglobin is globular)

14. structural level of protein...

Module 5 – Globular Proteins

9. Globular proteins are generally:

- A. Water insoluble
- B. Structural components
- C. Water soluble and functional
- D. Found only in hair and nails

Answer: C. Water soluble and functional

14. structural level of protein...

10. All enzymes belong to which category of proteins?

- A. Fibrous proteins
- B. Globular proteins
- C. Structural proteins
- D. Storage proteins

Answer: B. Globular proteins

14. structural level of protein...

11. Which protein is both globular and has quaternary structure?

- A. Keratin
- B. Collagen
- C. Hemoglobin
- D. Silk fibroin

Answer: C. Hemoglobin

14. structural level of protein...

Here are **important basic MCQs** created directly from your uploaded file:**

File: Protein Interactions

(Cited as:

15-Protein Interaction

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on the PPT content)

Module 1 – Basics of Protein Interactions

1. Protein interactions depend mainly on:

- A. Covalent bonds only
- B. Weak interactions
- C. DNA sequence
- D. ATP concentration

Answer: B. Weak interactions

15-Protein Interaction

2. Protein interactions influence the protein's:

- A. Linear amino acid sequence
- B. 3D structure
- C. RNA expression
- D. Genetic code

Module 2 – Denaturation of Proteins

3. Denaturation of proteins is caused by:

- A. Light exposure only
- B. Heat and pH change
- C. Lack of oxygen
- D. Water molecules

Answer: B. Heat and pH change
15-Protein Interaction

4. Denaturation affects which levels of protein structure?

- A. Only primary structure
 - B. Only quaternary structure
 - C. Tertiary and secondary structure
 - D. Primary and secondary structure only
- Answer: C. Tertiary and secondary structure
15-Protein Interaction
-

5. Denaturation results in the loss of:

- A. Amino acid sequence
 - B. Peptide bonds
 - C. Biological function
 - D. ATP production
- Answer: C. Biological function
15-Protein Interaction
-

Module 3 – Importance of Weak Interactions

6. Weak interactions in proteins include all of the following EXCEPT:

- A. Hydrogen bonds
 - B. Ionic interactions
 - C. Van der Waals forces
 - D. Peptide bonds
- Answer: D. Peptide bonds (peptide bonds are strong covalent bonds; not weak)
(Conceptual addition, consistent with slide's mention of "weak interactions")
-

7. Weak protein interactions are essential because they:

- A. Maintain 3D structure
 - B. Break down proteins
 - C. Form primary structure
 - D. Stop folding
- Answer: A. Maintain 3D structure
15-Protein Interaction

Here are **important basic MCQs** created directly from your uploaded file:**

File: Proteins – Importance

(Cited as:

16. Proteins, importance
)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on PPT content)

Module 1 – Structural Role of Proteins

1. Proteins form the structural components of:

- A. Bones, muscles, organs, hair, nails
 - B. DNA and RNA
 - C. Vitamins and minerals
 - D. Carbohydrate polymers
- Answer:** A. Bones, muscles, organs, hair, nails

16. Proteins, importance

2. Proteins serve as a major structural part of the animal body, including:

- A. Blood plasma only
- B. Skin only
- C. Visceral organs
- D. Only tendons

Answer: C. Visceral organs

16. Proteins, importance

Module 2 – Proteins as an Energy Source

3. Proteins produce energy after they are converted to:

- A. Lipids
- B. Nucleic acids
- C. Amino acids
- D. Glucose

Answer: C. Amino acids

16. Proteins, importance

4. Before proteins can be used for energy, the amino portion is removed by:

- A. Oxidation
- B. Hydrolysis
- C. Deamination
- D. Transamination

Answer: C. Deamination

16. Proteins, importance

5. After deamination, amino acids are:

- A. Stored in bones
- B. Oxidized to release energy
- C. Converted into antibodies
- D. Changed into DNA

Answer: B. Oxidized to release energy

16. Proteins, importance

Module 3 – Protection Against Diseases

6. Proteins that protect the body include:

- A. Vitamins
- B. Antibodies
- C. Glucose molecules
- D. Minerals

Answer: B. Antibodies

16. Proteins, importance

7. Some antibiotics are:

- A. Sugars
- B. Proteins
- C. Lipids
- D. Vitamins

Answer: B. Proteins (as shown in slide)

16. Proteins, importance

Module 4 – Transport Functions of Proteins

8. Hemoglobin transports oxygen from:

- A. Capillaries to lungs
- B. Lungs to capillaries
- C. Muscles to blood
- D. Skin to organs

Answer: B. Lungs to capillaries

16. Proteins, importance

9. Myoglobin transports oxygen in:

- A. Blood plasma
- B. Brain cells
- C. Muscles
- D. Stomach lining

Answer: C. Muscles

16. Proteins, importance

10. Protein channels in the cell membrane help in:

- A. Movement across the membrane
- B. DNA replication
- C. Enzyme denaturation
- D. Food digestion

Answer: A. Movement across the membrane

16. Proteins, importance

Module 5 – Enzymes

11. Most enzymes are:

- A. Carbohydrates
- B. Proteins
- C. Lipids
- D. Hormones

Answer: B. Proteins

16. Proteins, importance

12. Enzymes work by:

- A. Destroying cells
- B. Catalyzing biochemical reactions
- C. Absorbing sunlight
- D. Producing antibodies

Answer: B. Catalyzing biochemical reactions

16. Proteins, importance

Module 6 – Movement

13. Movement in living organisms is made possible mainly by:

- A. Lipids
- B. Proteins
- C. Minerals
- D. Nucleic acids

Answer: B. Proteins

16. Proteins, importance

14. Protein-based movements include:

- A. Bone growth only
- B. Blood clotting only
- C. Heartbeat and breathing
- D. Hormone synthesis

Answer: C. Heartbeat and breathing

16. Proteins, importance

Module 7 – Chemical Messengers

15. Some proteins act as:

- A. DNA templates
- B. Hormones and neurotransmitters
- C. Fat storage molecules
- D. Enzymes only

Answer: B. Hormones and neurotransmitters

16. Proteins, importance

16. Intracellular messaging is often carried out by:

- A. Starch molecules
- B. Protein messengers
- C. Minerals
- D. Water

Answer: B. Protein messengers

16. Proteins, importance

Here are **important basic MCQs** created directly from your uploaded file:**

File: *Carbohydrates – Sugar Polymers*

(Cited as:

17-Carbohydrates-Sugar Polymers

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on the PPT content)

Module 1 – Introduction to Carbohydrates

1. Carbohydrates are also known as:

- A. Amino acid polymers
- B. Sugar polymers
- C. Lipid chains
- D. Fatty acids

Answer: B. Sugar polymers

17-Carbohydrates-Sugar Polymers

2. One major function of carbohydrates is:

- A. Forming DNA
- B. Transporting oxygen
- C. Serving as a source of energy
- D. Forming amino acids

Answer: C. Serving as a source of energy

17-Carbohydrates-Sugar Polymers

3. In addition to energy storage, carbohydrates also play a:

- A. Genetic role
- B. Structural role
- C. Hormonal role
- D. Nervous system role

Answer: B. Structural role

17-Carbohydrates-Sugar Polymers

Module 2 – Types of Carbohydrates

4. Carbohydrates are divided into all of the following EXCEPT:

- A. Monosaccharides
- B. Disaccharides
- C. Polysaccharides
- D. Liposaccharides

Answer: D. Liposaccharides (not listed)

17-Carbohydrates-Sugar Polymers

5. Oligosaccharides contain how many sugar units?

- A. 1–2
- B. 3–20
- C. 20–50
- D. 50–100

Answer: B. 3–20

17-Carbohydrates-Sugar Polymers

Module 3 – Monosaccharides

6. Monosaccharides are:

- A. Complex sugars
- B. Simple sugars
- C. Polymers
- D. Amino acid chains

Answer: B. Simple sugars

17-Carbohydrates-Sugar Polymers

7. Glucose is produced mainly by:

- A. Animals
- B. Fungi
- C. Plants
- D. Viruses

Answer: C. Plants

17-Carbohydrates-Sugar Polymers

8. All living cells contain:

- A. Ribose only
- B. Fructose only
- C. Glucose
- D. Sucrose

Answer: C. Glucose

17-Carbohydrates-Sugar Polymers

Module 4 – Examples of Monosaccharides

9. Which of the following is a monosaccharide?

- A. Sucrose
- B. Starch
- C. Glucose
- D. Cellulose

Answer: C. Glucose

17-Carbohydrates-Sugar Polymers

10. Which monosaccharide is commonly found in fruits?

- A. Ribose
- B. Glucose
- C. Fructose
- D. Lactose

Answer: C. Fructose

17-Carbohydrates-Sugar Polymers

11. Ribose is important because it is a component of:

- A. Cell membrane
- B. RNA
- C. Enzymes
- D. Fatty acids

Answer: B. RNA (biologically known; consistent with ribose structure slide)

17-Carbohydrates-Sugar Polymers

Module 5 – Glucose Forms

12. Glucose exists in:

- A. Only one form
- B. Two forms
- C. Three forms
- D. Many random structures

Answer: B. Two forms

17-Carbohydrates-Sugar Polymers

Here are **important basic MCQs** created directly from your uploaded file:

File: *Carbohydrates – Glycosidic Linkages*

(Cited as:

18-Carbohydrates-Glycosidic lin...

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on PPT content)

Module 1 – Glycosidic Linkages Basics

1. A glycosidic linkage is a bond formed between:

- A. Two amino acids
- B. A fatty acid and glycerol
- C. Two monosaccharides
- D. A sugar and a lipid

Answer: C. Two monosaccharides

18-Carbohydrates-Glycosidic lin...

2. Glycosidic linkages result in the formation of:

- A. Proteins
- B. Lipids
- C. Disaccharides or polysaccharides
- D. Nucleic acids

Answer: C. Disaccharides or polysaccharides

18-Carbohydrates-Glycosidic lin...

Module 2 – α and β Orientation

3. The orientation of glycosidic linkages may be:

- A. Left or right
- B. Alpha (α) or Beta (β)
- C. Upstream or downstream
- D. Strong or weak

Answer: B. Alpha (α) or Beta (β)

18-Carbohydrates-Glycosidic lin...

4. The digestibility of glycosidic linkages depends on their:

- A. Temperature
- B. Length
- C. α or β orientation
- D. Color

Answer: C. α or β orientation

Module 3 – Digestibility of Carbohydrates

5. Humans can digest which type of glycosidic linkage?

- A. β (beta) linkages only
- B. Both β and α linkages equally
- C. α (alpha) linkages
- D. No glycosidic linkages

Answer: C. α (alpha) linkages

(Implied by slides showing digestible vs. non-digestible linkages.)

6. Cellulose contains which type of glycosidic linkage?

- A. α 1–4
- B. α 1–6
- C. β 1–4
- D. β 1–2

Answer: C. β 1–4 (associated with indigestibility shown in slides)

7. Which polysaccharide is easily digestible due to its α -glycosidic linkage?

- A. Cellulose
- B. Glycogen
- C. Chitin
- D. Peptidoglycan

Answer: B. Glycogen

Module 4 – Comparison of Polysaccharides

8. Cellulose is structurally similar to starch but differs in:

- A. Length
- B. Glycosidic linkage orientation
- C. Color
- D. Water content

Answer: B. Glycosidic linkage orientation (β vs. α)

9. Starch contains which type of linkage?

- A. β 1–4
- B. β 1–6
- C. α 1–4 and α 1–6
- D. α 1–2

Answer: C. α 1–4 and α 1–6 (digestible starch shown in slides)

10. Cellulose is indigestible for humans because:

- A. It is too large
- B. It dissolves in water
- C. Humans lack enzymes for β -linkages
- D. It has no glycosidic linkages

Answer: C. Humans lack enzymes for β -linkages

18-Carbohydrates-Glycosidic lin...

Here are **important basic MCQs** created directly from your uploaded file:**

File: *Different Classes of Carbohydrates – Monosaccharides*

(Cited as:

19. different classes of carboh...

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on PPT content)

Module 1 – Definition of Monosaccharides

1. Monosaccharides are defined as:

- A. Complex sugars made of many units
- B. Sugars that can be hydrolyzed into two sugars
- C. The simplest carbohydrates that cannot be further hydrolyzed
- D. Carbohydrates that always form chains

Answer: C. The simplest carbohydrates that cannot be further hydrolyzed

19. different classes of carboh...

2. The most common monosaccharide is:

- A. Fructose
- B. Sucrose
- C. Ribose
- D. Glucose

Answer: D. Glucose

19. different classes of carboh...

Module 2 – Physicochemical Properties of Monosaccharides

3. Monosaccharides are generally:

- A. Tasteless
- B. Bitter
- C. Sweet in taste
- D. Salty

Answer: C. Sweet in taste

19. different classes of carboh...

4. Which of the following is a physicochemical property of monosaccharides?

- A. Insoluble in water
- B. Readily soluble in water

C. Liquid at room temperature
D. Non-crystalline
Answer: B. Readily soluble in water
19. different classes of carboh...

5. Monosaccharides are:

- A. Non-reducing sugars
- B. Reducing sugars
- C. Always acidic
- D. Always basic

Answer: B. Reducing sugars
19. different classes of carboh...

Module 3 – Types on the Basis of Size

6. Monosaccharides typically contain:

- A. 1–2 carbons
- B. 3–7 carbons
- C. 8–12 carbons
- D. 15 carbons

Answer: B. 3–7 carbons
19. different classes of carboh...

7. A 3-carbon monosaccharide is called:

- A. Tetrose
- B. Pentose
- C. Triose
- D. Hexose

Answer: C. Triose
19. different classes of carboh...

8. A 6-carbon sugar is called a:

- A. Heptose
- B. Hexose
- C. Pentose
- D. Triose

Answer: B. Hexose
19. different classes of carboh...

Module 4 – Types on the Basis of Functional Group

9. Monosaccharides with an aldehyde group are called:

- A. Ketoses
- B. Aldoses
- C. Polysaccharides
- D. Lipids

Answer: B. Aldoses
19. different classes of carboh...

10. Monosaccharides with a ketone group are known as:

- A. Aldoses
- B. Disaccharides
- C. Ketoses
- D. Pentoses

Answer: C. Ketoses

19. different classes of carboh...

Module 5 – Chain and Ring Structures

11. Monosaccharides form linear carbon chains, but when they contain four or more carbons they typically form:

- A. Spirals
- B. Rings
- C. Sheets
- D. Double helices

Answer: B. Rings

19. different classes of carboh...

12. A six-membered ring structure in a monosaccharide is called:

- A. Furanose
- B. Ketose
- C. Pyranose
- D. Aldose

Answer: C. Pyranose

19. different classes of carboh...

13. A five-membered ring monosaccharide is called:

- A. Pyranose
- B. Furanose
- C. Hexose
- D. Tetrose

Answer: B. Furanose

19. different classes of carboh...

Module 6 – Importance of Monosaccharides

14. Monosaccharides are important mainly because they serve as:

- A. Hormones
- B. Sources of energy
- C. Lipid-forming molecules
- D. Enzymes

Answer: B. Sources of energy

19. different classes of carboh...

15. Large carbohydrates such as cellulose and starch are built from:

- A. Lipids
- B. Amino acids
- C. Monosaccharides
- D. Proteins

Answer: C. Monosaccharides

19. different classes of carboh...

Here are **important basic MCQs** created directly from your uploaded file:**

File: *Different Classes of Carbohydrates – Disaccharides*

(Cited as:

20. different classes of carboh...

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on PPT content)

Module 1 – Definition & Properties of Disaccharides

1. Disaccharides are formed when:

- A. Two lipids combine
- B. Two amino acids combine
- C. Two monosaccharides combine
- D. Two polysaccharides combine

Answer: C. Two monosaccharides combine

20. different classes of carboh...

2. Disaccharides are generally:

- A. Insoluble in water
- B. Soluble in water
- C. Proteins in nature
- D. Tasteless

Answer: B. Soluble in water

20. different classes of carboh...

3. Compared to monosaccharides, disaccharides are:

- A. Sweeter
- B. Less sweet
- C. Bitter
- D. Sour

Answer: B. Less sweet

20. different classes of carboh...

4. Disaccharides may be:

- A. All reducing
- B. All non-reducing
- C. Either reducing or non-reducing
- D. Never crystalline

Answer: C. Either reducing or non-reducing

20. different classes of carboh...

5. The general formula of disaccharides is:

- A. $C_6H_{12}O_6$
- B. $C_{12}H_{22}O_{11}$
- C. C_6H_6
- D. $C_{18}H_{32}O_{16}$

Answer: B. $C_{12}H_{22}O_{11}$

Module 2 – Common Disaccharides

6. Which of the following is a common disaccharide?

- A. Glucose
- B. Ribose
- C. Sucrose
- D. Cellulose

Answer: C. Sucrose

7. Lactose is commonly known as:

- A. Table sugar
- B. Fruit sugar
- C. Milk sugar
- D. Cane sugar

Answer: C. Milk sugar (general biological knowledge; listed as a common disaccharide)

8. Maltose is a disaccharide formed from:

- A. Glucose + Glucose
- B. Glucose + Fructose
- C. Glucose + Galactose
- D. Galactose + Fructose

Answer: A. Glucose + Glucose (standard biological concept; included under "common disaccharides")

Module 3 – Isomerism in Disaccharides

9. Disaccharides are isomers of each other because they:

- A. Have different molecular formulas
- B. Have the same molecular formula but different structures
- C. Are made of proteins
- D. Are always reducing sugars

Answer: B. Have the same molecular formula but different structures

10. The molecular formula shared by all disaccharides is:

- A. $C_6H_{12}O_6$
- B. $C_{12}H_{22}O_{11}$
- C. $C_5H_{10}O_5$
- D. $C_{10}H_{20}O_{10}$

Answer: B. $C_{12}H_{22}O_{11}$

Module 4 – Glycosidic Bond

11. The bond that joins two monosaccharides to form a disaccharide is called:

- A. Peptide bond
- B. Hydrogen bond
- C. Glycosidic bond
- D. Ionic bond

Answer: C. Glycosidic bond

20. different classes of carboh...

12. Glycosidic bonds are formed during:

- A. Hydrolysis
- B. Condensation reactions
- C. Oxidation
- D. Deamination

Answer: B. Condensation reactions (*biological context directly linked to glycosidic bond formation*)

20. different classes of carboh...

Here are **important basic MCQs** created directly from your uploaded file:**

File: *Different Classes of Carbohydrates – Polysaccharides*

(Cited as:

21. different classes of carboh...

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on PPT content)

Module 1 – Definition of Polysaccharides

1. Polysaccharides are formed when:

- A. Two monosaccharides join
- B. Many monosaccharides join through glycosidic bonds
- C. Amino acids join through peptide bonds
- D. Fatty acids attach to glycerol

Answer: B. Many monosaccharides join through glycosidic bonds

21. different classes of carboh...

2. The general formula of polysaccharides is:

- A. $C_6H_{12}O_6$
- B. $C_6H_{10}O_{5n}$ or $C_x(H_2O)_y$
- C. $C_{12}H_{22}O_{11}$
- D. $C_5H_{10}O_5$

Answer: B. $C_6H_{10}O_{5n}$ or $C_x(H_2O)_y$

21. different classes of carboh...

Module 2 – Physicochemical Properties

3. Polysaccharides are generally:

- A. Sweet in taste
- B. Bitter in taste
- C. Not sweet in taste
- D. Salty in taste

Answer: C. Not sweet in taste

21. different classes of carboh...

4. Solubility of polysaccharides in water is:

- A. High
- B. Moderate
- C. Very little
- D. Complete

Answer: C. Very little

21. different classes of carboh...

5. Polysaccharides are:

- A. Reducing sugars
- B. Non-reducing sugars
- C. Always reducing
- D. Sometimes reducing

Answer: B. Non-reducing sugars

21. different classes of carboh...

Module 3 – Common Polysaccharides

6. Which of the following is a common polysaccharide?

- A. Sucrose
- B. Lactose
- C. Glycogen
- D. Maltose

Answer: C. Glycogen

21. different classes of carboh...

7. Starch is classified as a:

- A. Disaccharide
- B. Polysaccharide
- C. Monosaccharide
- D. Lipid polymer

Answer: B. Polysaccharide

21. different classes of carboh...

8. Cellulose is mainly found in:

- A. Animal muscles
- B. Bacterial membranes
- C. Plant cell walls
- D. Animal liver

Answer: C. Plant cell walls
(Implied by its common biological role)

21. different classes of carboh...

9. Chitin is a polysaccharide found in:

- A. Human blood
- B. Plant roots
- C. Fungal cell walls and arthropod exoskeletons
- D. Mammalian tissues

Answer: C. Fungal cell walls and arthropod exoskeletons
(Biological fact aligned with slide content listing chitin)

21. different classes of carboh...

Module 4 – Biological Roles of Polysaccharides

10. Glycogen serves mainly as:

- A. Structural support
- B. Water storage
- C. Fuel storage in animals
- D. A hormone

Answer: C. Fuel storage in animals

21. different classes of carboh...

11. Starch serves mainly as:

- A. Structural material in animals
- B. Energy storage in plants
- C. Genetic material
- D. Pigment source

Answer: B. Energy storage in plants
(Biological knowledge consistent with slide listing starch under fuel storage)

21. different classes of carboh...

12. The main structural polysaccharide in plants is:

- A. Glycogen
- B. Cellulose
- C. Starch
- D. Ribose

Answer: B. Cellulose

21. different classes of carboh...

13. The main structural polysaccharide in arthropods and fungi is:

- A. Starch
- B. Maltose
- C. Chitin
- D. Glucose

Answer: C. Chitin

21. different classes of carboh...

Here are **important basic MCQs** created directly from your uploaded file:**

File: *Carbohydrates as Energy Source and Structural Support*

(Cited as:

22. carbohydrates as energy sou...

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on PPT content)

MODULE 1 – Carbohydrates as an Energy Source

1. Carbohydrates provide approximately what percentage of daily energy needs?

- A. 10–15%
- B. 20–30%
- C. 60–65%
- D. 80–90%

Answer: C. 60–65%

22. carbohydrates as energy sou...

2. Recommended daily intake of carbohydrates is:

- A. 50–100 g
- B. 150–200 g
- C. 225–325 g
- D. 400–500 g

Answer: C. 225–325 g

22. carbohydrates as energy sou...

3. Glucose is primarily synthesized by:

- A. Animals
- B. Fungi
- C. Plants
- D. Humans

Answer: C. Plants (during photosynthesis)

22. carbohydrates as energy sou...

4. The most important source of energy for organisms is:

- A. Sucrose
- B. Cellulose
- C. Glucose
- D. Chitin

Answer: C. Glucose

22. carbohydrates as energy sou...

5. Other monosaccharides that provide energy include:

- A. Ribose & Deoxyribose
- B. Fructose & Galactose

C. Maltose & Sucrose
D. Glycogen & Cellulose
Answer: B. Fructose & Galactose
22. carbohydrates as energy sou...

MODULE 2 – Disaccharides as Energy

Source

6. Sucrose is obtained mainly from:

- A. Fruits only
- B. Milk and cheese
- C. Sugar cane and sugar beet
- D. Eggs

Answer: C. Sugar cane and sugar beet

22. carbohydrates as energy sou...

7. Lactose is commonly known as:

- A. Fruit sugar
- B. Milk sugar
- C. Table sugar
- D. Sweet potato sugar

Answer: B. Milk sugar

22. carbohydrates as energy sou...

8. Maltose is present in:

- A. Meat and eggs
- B. Rice
- C. Sweet potatoes and candies
- D. Milk

Answer: C. Sweet potatoes and candies

22. carbohydrates as energy sou...

9. Disaccharides must be broken down into _____ before absorption.

- A. Polysaccharides
- B. Amino acids
- C. Monosaccharides
- D. Lipids

Answer: C. Monosaccharides

22. carbohydrates as energy sou...

MODULE 3 – Polysaccharides as Energy

Source

10. Starch is digested into:

- A. Fructose
- B. Lactose
- C. Glucose
- D. Galactose

Answer: C. Glucose

22. carbohydrates as energy sou...

11. Glycogen is important because it:

- A. Provides structural support to plants
- B. Breaks quickly to provide glucose
- C. Is the main component of exoskeleton
- D. Provides sweetness to fruits

Answer: B. Breaks quickly to provide glucose

22. carbohydrates as energy sou...

12. Cellulose can be broken down into:

- A. Ribose
- B. Glucose
- C. Fructose
- D. Sucrose

Answer: B. Glucose

22. carbohydrates as energy sou...

MODULE 4 – Carbohydrates as Structural Support

13. Chitin provides support in:

- A. Plants
- B. Mammals
- C. Arthropods
- D. Viruses

Answer: C. Arthropods (e.g., insects, crustaceans)

22. carbohydrates as energy sou...

14. Chitin also provides protection from:

- A. Heat
- B. Predators
- C. Light
- D. Water loss

Answer: B. Predators

22. carbohydrates as energy sou...

15. The main structural polysaccharide in plant cell walls is:

- A. Glycogen
- B. Starch
- C. Cellulose
- D. Maltose

Answer: C. Cellulose

22. carbohydrates as energy sou...

16. The glycocalyx is found:

- A. Inside the nucleus
- B. Outside the cell membrane

- C. In chloroplasts
 - D. In mitochondria
- Answer:** B. Outside the cell membrane

22. carbohydrates as energy sou...

17. The glycocalyx helps in:

- A. DNA replication
- B. Cell recognition and protection
- C. ATP production
- D. Protein synthesis

Answer: B. Cell recognition and protection

22. carbohydrates as energy sou...

18. Extracellular material contributes to:

- A. Energy production
- B. Holding cells together
- C. Breaking down toxins
- D. Increasing sweetness

Answer: B. Holding cells together

22. carbohydrates as energy sou...

19. Extracellular material is present in all of the following EXCEPT:

- A. Fluid of joints
- B. Vitreous humor
- C. Tendons and ligaments
- D. Mitochondria

Answer: D. Mitochondria

22. carbohydrates as energy sou...

Here are **important basic MCQs** created directly from your uploaded file:**

File: Lipids – Chemical Composition and Occurrence

(Cited as:

23. Lipids, chemical compositio...

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on PPT content)

MODULE 1 – Definition of Lipids

1. Lipids are best defined as:

- A. Water-soluble biomolecules
- B. Polar molecules soluble in water
- C. Water-insoluble, non-polar biomolecules
- D. Carbohydrate derivatives

Answer: C. Water-insoluble, non-polar biomolecules

23. Lipids, chemical compositio...

2. Lipids dissolve readily in:

- A. Water
- B. Salt solutions
- C. Organic solvents
- D. Acids

Answer: C. Organic solvents

23. Lipids, chemical compositio...

MODULE 2 – Chemical Composition

3. Lipids are mainly composed of:

- A. C, H, O
- B. C, H, O with small amounts of N, S, P
- C. Only hydrogen and oxygen
- D. Carbon and nitrogen only

Answer: B. C, H, O with small amounts of N, S, P

23. Lipids, chemical compositio...

4. Lipids contain ____ compared to carbohydrates.

- A. More oxygen
- B. Less oxygen
- C. No hydrogen
- D. No carbon

Answer: B. Less oxygen

23. Lipids, chemical compositio...

5. Lipids have the highest number of:

- A. C–O bonds
- B. H–O bonds
- C. C–H bonds
- D. N–H bonds

Answer: C. C–H bonds

23. Lipids, chemical compositio...

MODULE 3 – Occurrence of Lipids

6. Lipids commonly occur in all of the following EXCEPT:

- A. Milk and butter
- B. Meat and eggs
- C. Dry fruits
- D. Table salt

Answer: D. Table salt

23. Lipids, chemical compositio...

7. Cooking oils such as sunflower and soybean oil are good sources of:

- A. Carbohydrates
- B. Lipids
- C. Proteins
- D. Minerals

Answer: B. Lipids

23. Lipids, chemical compositio...

8. Fried foods contain a high amount of:

- A. Vitamins
- B. Lipids
- C. Nucleic acids
- D. Starch only

Answer: B. Lipids

23. Lipids, chemical compositio...

MODULE 4 – Importance of Lipids

9. The richest source of energy among the biomolecules is:

- A. Proteins
- B. Carbohydrates
- C. Lipids
- D. Vitamins

Answer: C. Lipids

23. Lipids, chemical compositio...

10. Lipids form an integral part of:

- A. Ribosomes
- B. Mitochondrial matrix
- C. All cellular membranes
- D. Lysosomes only

Answer: C. All cellular membranes

23. Lipids, chemical compositio...

11. Lipids are important for all of the following EXCEPT:

- A. Hormone formation
- B. Insulation
- C. Protection
- D. DNA replication

Answer: D. DNA replication

23. Lipids, chemical compositio...

12. Lipids play a major role in:

- A. Cell signaling
- B. Photosynthesis
- C. Transcription only
- D. ATP synthesis only

Answer: A. Cell signaling

MODULE 5 – Types of Lipids

13. Which of the following is NOT one of the major types of lipids listed in the file?

- A. Triglycerides
- B. Phospholipids
- C. Steroids
- D. Amino acids

Answer: D. Amino acids

14. Triglycerides, phospholipids, and steroids are examples of:

- A. Carbohydrates
- B. Proteins
- C. Lipids
- D. Vitamins

Answer: C. Lipids

Here are **important basic MCQs** created directly from your uploaded file:**

File: Fatty Acids

(Cited as:

24. Fatty acids

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on PPT content)

MODULE 1 – Definition of Fatty Acids

1. Fatty acids are long chains of hydrocarbons bonded to a:

- A. Phosphate group
- B. Carboxylic group (–COOH)
- C. Sulfate group
- D. Hydroxyl group

Answer: B. Carboxylic group (–COOH)

24. Fatty acids

2. Oxygen in fatty acids is present only in the form of:

- A. Hydroxyl groups
 - B. Carboxyl group
 - C. Carbonyl rings
 - D. Ester bonds
- Answer:** B. Carboxyl group

24. Fatty acids

3. Fatty acids have a maximum number of:

- A. C–O bonds
 - B. C–H bonds
 - C. N–H bonds
 - D. O–H bonds
- Answer:** B. C–H bonds

24. Fatty acids

MODULE 2 – Carbon Chain of Fatty Acids

4. Most fatty acids contain an _____ number of carbons.

- A. Odd
- B. Even
- C. Random
- D. Variable but always less than 10

Answer: B. Even

24. Fatty acids

5. The size of fatty acid carbon chains generally ranges from:

- A. 1–10 carbons
 - B. 4–36 carbons
 - C. 40–60 carbons
 - D. 60–100 carbons
- Answer:** B. 4–36 carbons

24. Fatty acids

6. The most common fatty acids contain how many carbons?

- A. 6–10
 - B. 10–14
 - C. 14–20
 - D. 22–30
- Answer:** C. 14–20

24. Fatty acids

MODULE 3 – Saturated vs. Unsaturated Fatty Acids

7. Saturated fatty acids have:

- A. Multiple double bonds
- B. One double bond
- C. Single covalent bonds between all carbons
- D. No C–H bonds

Answer: C. Single covalent bonds between all carbons

24. Fatty acids

8. Unsaturated fatty acids contain:

- A. Only single bonds
- B. One or more double bonds
- C. Only triple bonds
- D. No carbon bonds

Answer: B. One or more double bonds

24. Fatty acids

9. At room temperature, saturated fatty acids are generally:

- A. Gases
- B. Liquids
- C. Solids
- D. Plasma-like

Answer: C. Solids

24. Fatty acids

10. Unsaturated fatty acids are usually _____ at room temperature.

- A. Solid
- B. Liquid
- C. Gas
- D. Crystalline

Answer: B. Liquid

24. Fatty acids

MODULE 4 – Melting Point & Solubility

11. Melting point of fatty acids increases when:

- A. Double bonds increase
- B. Carbon chain decreases
- C. Carbon chain increases
- D. Solubility increases

Answer: C. Carbon chain increases

24. Fatty acids

12. Solubility of fatty acids in water:

- A. Increases with chain length
- B. Decreases with chain length
- C. Remains constant
- D. Is independent of chain length

Answer: B. Decreases with chain length

24. Fatty acids

13. Melting point decreases when:

- A. Double bonds are added
- B. C–H bonds increase
- C. Carbon chain increases
- D. Fatty acid loses COOH group

Answer: A. Double bonds are added

24. Fatty acids

Here are **important basic MCQs** created directly from your uploaded file:**

File: *Fatty Acids – Examples*

(Cited as:

25. Fatty acids examples

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on the PPT content)

MODULE 1 – Palmitic Acid

1. Palmitic acid contains how many carbon atoms?

A. 14

B. 16

C. 18

D. 20

Answer: B. 16

25. Fatty acids examples

2. Palmitic acid is a:

A. Polyunsaturated fatty acid

B. Monounsaturated fatty acid

C. Saturated fatty acid

D. Trans fatty acid

Answer: C. Saturated fatty acid

25. Fatty acids examples

3. The melting point of palmitic acid is:

A. 13.4°C

B. 62.9°C

C. 69.6°C

D. 76.5°C

Answer: B. 62.9°C

25. Fatty acids examples

MODULE 2 – Stearic Acid

4. Stearic acid has how many carbon atoms?

A. 16

B. 18

C. 20

D. 14

Answer: B. 18

25. Fatty acids examples

5. Stearic acid is commonly found in:

- A. Olive oil
- B. Human adipose tissue
- C. Animal fat
- D. Corn oil

Answer: C. Animal fat

25. Fatty acids examples

6. The melting point of stearic acid is:

- A. 62.9°C
- B. 69.6°C
- C. 76.5°C
- D. -11°C

Answer: B. 69.6°C

25. Fatty acids examples

MODULE 3 – Arachidic Acid

7. Arachidic acid contains how many carbon atoms?

- A. 14
- B. 16
- C. 18
- D. 20

Answer: D. 20

25. Fatty acids examples

8. Arachidic acid is found in:

- A. Olive oil
- B. Peanut and corn oil
- C. Human adipose tissues
- D. Milk lipids

Answer: B. Peanut and corn oil

25. Fatty acids examples

9. The melting point of arachidic acid is:

- A. 13.4°C
- B. 62.9°C
- C. 76.5°C
- D. -11°C

Answer: C. 76.5°C

25. Fatty acids examples

MODULE 4 – Palmitoleic Acid

10. Palmitoleic acid has how many carbons?

- A. 14
- B. 16
- C. 18
- D. 20

Answer: B. 16

25. Fatty acids examples

11. Palmitoleic acid is:

- A. Saturated
- B. Unsaturated with a double bond between C9 and C10
- C. Polyunsaturated
- D. Omega-3 fatty acid

Answer: B. Unsaturated with a double bond between C9 and C10

25. Fatty acids examples

12. Palmitoleic acid is commonly found in:

- A. Human adipose tissue
- B. Olive oil
- C. Corn oil
- D. Animal fat

Answer: B. Olive oil

25. Fatty acids examples

13. The melting point of palmitoleic acid is:

- A. 13.4°C
- B. -11°C
- C. 76.5°C
- D. 1-5°C

Answer: A. 13.4°C

25. Fatty acids examples

MODULE 5 – Oleic Acid

14. Oleic acid contains how many carbon atoms?

- A. 14
- B. 16
- C. 18
- D. 20

Answer: C. 18

25. Fatty acids examples

15. The double bond in oleic acid is between:

- A. C1 and C2
- B. C5 and C6
- C. C9 and C10
- D. C15 and C16

Answer: C. C9 and C10

25. Fatty acids examples

16. Oleic acid is abundant in:

- A. Olive oil
- B. Human adipose tissues
- C. Corn oil
- D. Wheat germ

Answer: B. Human adipose tissues

25. Fatty acids examples

MODULE 6 – Linoleic Acid

17. Linoleic acid is a _____ fatty acid.

- A. Saturated
- B. Monounsaturated
- C. Polyunsaturated, omega-6
- D. Polyunsaturated, omega-3

Answer: C. Polyunsaturated, omega-6

25. Fatty acids examples

18. The melting point of linoleic acid is:

- A. 13.4°C
- B. 76.5°C
- C. -11°C
- D. 1-5°C

Answer: D. 1-5°C

25. Fatty acids examples

MODULE 7 – Linolenic Acid

19. Linolenic acid is also called:

- A. Omega-6 fatty acid
- B. Omega-9 fatty acid
- C. Omega-3 fatty acid
- D. Saturated fatty acid

Answer: C. Omega-3 fatty acid

25. Fatty acids examples

20. The melting point of linolenic acid is:

- A. 62.9°C
- B. 69.6°C
- C. 76.5°C
- D. -11°C

Answer: D. -11°C

25. Fatty acids examples

Here are **important basic MCQs** created directly from your uploaded file:**

File: *Fats and Oils*

(Cited as:

26. Fats and oils

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on PPT content)

MODULE 1 – Triglycerides Basics

1. Triglycerides are also known as:

- A. Steroids
- B. Neutral fats or triacylglycerols
- C. Phospholipids
- D. Lipoproteins

Answer: B. Neutral fats or triacylglycerols

26. Fats and oils

2. Triglycerides are classified as the:

- A. Most complex lipids
- B. Simplest group of lipids
- C. Largest lipids
- D. Only animal lipids

Answer: B. Simplest group of lipids

26. Fats and oils

MODULE 2 – Chemical Composition of Triglycerides

3. The backbone of a triglyceride molecule is:

- A. Glucose
- B. Glycerol
- C. Cholesterol
- D. Ribose

Answer: B. Glycerol

26. Fats and oils

4. In triglycerides, fatty acids are attached to glycerol by:

- A. Peptide bonds
- B. Hydrogen bonds
- C. Ester linkages
- D. Ionic bonds

Answer: C. Ester linkages

26. Fats and oils

5. Fatty acids in triglycerides are mostly:

- A. 10–14 carbons long
- B. 16–20 carbons long
- C. 4–8 carbons long
- D. 1–3 carbons long

Answer: B. 16–20 carbons long

26. Fats and oils

MODULE 3 – Functions of Triglycerides

6. Triglycerides serve as:

- A. Enzymes
- B. Structural proteins
- C. Energy storage molecules
- D. Hormones

Answer: C. Energy storage molecules

26. Fats and oils

7. Compared to carbohydrates, triglycerides:

- A. Provide less energy
- B. Provide the same amount of energy
- C. Provide more energy
- D. Cannot be stored

Answer: C. Provide more energy

26. Fats and oils

8. Triglycerides require _____ for storage compared to carbohydrates.

- A. More water
- B. Less water
- C. The same amount of water
- D. No water

Answer: B. Less water

26. Fats and oils

9. One major function of triglycerides is:

- A. Photosynthesis
- B. DNA replication
- C. Insulation against heat loss
- D. Production of enzymes

Answer: C. Insulation against heat loss

26. Fats and oils

MODULE 4 – Fats

10. Fats are triacylglycerols that are _____ at room temperature.

- A. Liquid
- B. Solid
- C. Gaseous
- D. Semi-liquid

Answer: B. Solid

26. Fats and oils

11. Fats contain:

- A. Unsaturated fatty acids
- B. Saturated fatty acids
- C. Only polyunsaturated fatty acids
- D. No fatty acids

Answer: B. Saturated fatty acids

26. Fats and oils

12. Animal triacylglycerols are generally:

- A. Oils
- B. Fats
- C. Steroids
- D. Phospholipids

Answer: B. Fats

26. Fats and oils

MODULE 5 – Oils

13. Oils are triacylglycerols that are _____ at room temperature.

- A. Solid
- B. Liquid
- C. Crystalline
- D. Hard

Answer: B. Liquid

26. Fats and oils

14. Oils contain mostly:

- A. Saturated fatty acids
- B. Unsaturated fatty acids
- C. Branched fatty acids
- D. Trans fatty acids

Answer: B. Unsaturated fatty acids

26. Fats and oils

15. Plant triacylglycerols are generally:

- A. Fats
- B. Oils
- C. Sterols
- D. Vitamins

Answer: B. Oils

26. Fats and oils

Here are **important basic MCQs** created directly from your uploaded file:**

File: *Lipid Types – Phospholipids*

(Cited as:

27. lipid types, phospholipids

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on PPT content)

MODULE 1 – Definition of Phospholipids

1. Phospholipids are molecules in which glycerol combines with:

- A. Three fatty acids
- B. Two fatty acids, a phosphate group, and a nitrogen base

C. One fatty acid and two phosphate groups

D. Cholesterol

Answer: B. Two fatty acids, a phosphate group, and a nitrogen base

27. lipid types, phospholipids

2. Phospholipids are classified as:

A. Steroids

B. Neutral fats

C. Amphipathic lipids

D. Carbohydrates

Answer: C. Amphipathic lipids (*because they have both hydrophilic and hydrophobic parts*)

27. lipid types, phospholipids

MODULE 2 – Molecular Structure

3. Fatty acids are attached to glycerol via:

A. Peptide bonds

B. Hydrogen bonds

C. Ester (acyl) linkages

D. Ionic bonds

Answer: C. Ester (acyl) linkages

27. lipid types, phospholipids

4. On which carbon of glycerol does the phosphate group attach?

A. 1st carbon

B. 2nd carbon

C. 3rd carbon

D. It can attach anywhere

Answer: C. 3rd carbon

27. lipid types, phospholipids

5. The phosphate group in phospholipids is attached to:

A. A fatty acid

B. A nitrogen base

C. Cholesterol

D. Glucose

Answer: B. A nitrogen base

27. lipid types, phospholipids

MODULE 3 – Polarity & Structure

6. The head of a phospholipid is:

A. Hydrophobic and nonpolar

B. Hydrophilic and polar

C. Hydrophobic and polar

D. Hydrophilic and nonpolar

Answer: B. Hydrophilic and polar

27. lipid types, phospholipids

7. The fatty acid tails of a phospholipid are:

- A. Polar and hydrophilic
- B. Nonpolar and hydrophobic
- C. Polar but hydrophobic
- D. Nonpolar but hydrophilic

Answer: B. Nonpolar and hydrophobic

27. lipid types, phospholipids

MODULE 4 – Functions of Phospholipids

8. The main biological role of phospholipids is to:

- A. Store energy in animals
- B. Form cell and organelle membranes
- C. Act as enzymes
- D. Produce hormones

Answer: B. Form cell and organelle membranes

27. lipid types, phospholipids

9. Phospholipids form a lipid bilayer because:

- A. Fatty acids attract water strongly
- B. Both ends of the molecule are hydrophobic
- C. The head is hydrophilic and the tail is hydrophobic
- D. Nitrogen base repels water

Answer: C. The head is hydrophilic and the tail is hydrophobic

27. lipid types, phospholipids

MODULE 5 – Types of Phospholipids

10. One type of phospholipid mentioned in the slides is:

- A. Cholesterol
- B. Phosphatidic acid
- C. Triglyceride
- D. Sphingomyelin

Answer: B. Phosphatidic acid

27. lipid types, phospholipids

Here are **important basic MCQs** created directly from your uploaded file:**

File: *Lipid Types – Steroids*

(Cited as required:

28. lipid types, steroids

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on PPT content)

MODULE 1 – Definition of Steroids

1. Steroids are:

- A. Carbohydrates with long chains
 - B. Naturally occurring lipids with four fused carbon rings
 - C. Proteins with double bonds
 - D. Phospholipids involved in membranes
- Answer:** B. Naturally occurring lipids with four fused carbon rings

28. lipid types, steroids

2. Which of the following is an example of a steroid?

- A. Glucose
 - B. Cholesterol
 - C. Triglyceride
 - D. Phosphatidic acid
- Answer:** B. Cholesterol

28. lipid types, steroids

3. Steroids typically include:

- A. Enzymes
 - B. DNA
 - C. Estrogen and testosterone
 - D. Pepsin
- Answer:** C. Estrogen and testosterone

28. lipid types, steroids

MODULE 2 – Molecular Structure of Steroids

4. Steroids consist of how many fused rings?

- A. 2
 - B. 3
 - C. 4
 - D. 5
- Answer:** C. 4

28. lipid types, steroids

5. In the steroid nucleus, how many rings are six-sided?

- A. 1
- B. 2
- C. 3
- D. 4

Answer: C. 3

28. lipid types, steroids

6. How many total carbons form the steroid nucleus?

- A. 12
- B. 14
- C. 16
- D. 17

Answer: D. 17

28. lipid types, steroids

MODULE 3 – Functions of Steroids

7. Cholesterol is an important component of:

- A. Ribosomes
- B. Cell membranes
- C. Cytosol
- D. Lysosomes

Answer: B. Cell membranes

28. lipid types, steroids

8. Testosterone functions in:

- A. Regulating calcium levels
- B. Secondary sex characteristics in males
- C. Breaking down lipids
- D. Energy storage

Answer: B. Secondary sex characteristics in males

28. lipid types, steroids

9. Estrogen is responsible for:

- A. Digestion
- B. Secondary sex characteristics in females
- C. Muscle contraction
- D. Blood clotting

Answer: B. Secondary sex characteristics in females

28. lipid types, steroids

10. Vitamin D plays a role in:

- A. Lipid breakdown
- B. Uptake and regulation of calcium levels
- C. Carbohydrate digestion
- D. DNA replication

Answer: B. Uptake and regulation of calcium levels

28. lipid types, steroids

Here are **important basic MCQs** created directly from your uploaded file:**

File: Nucleic Acids – DNA & RNA

(Cited properly:

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on PPT content + fundamental concepts directly implied by the slides)

MODULE 1 – Introduction to Nucleic Acids

1. The two major types of nucleic acids are:

- A. ATP and ADP
- B. DNA and RNA
- C. Glucose and glycogen
- D. Lipids and proteins

Answer: B. DNA and RNA

29-Nucleic Acids DNA And RNA

2. Nucleic acids function mainly to:

- A. Store and transmit genetic information
- B. Provide cell structure
- C. Act as enzymes
- D. Produce energy directly

Answer: A. Store and transmit genetic information

29-Nucleic Acids DNA And RNA

MODULE 2 – Bases of Nucleic Acids

3. Nitrogenous bases are classified into:

- A. Amino acids and peptides
- B. Purines and pyrimidines
- C. Monosaccharides and disaccharides
- D. Steroids and phospholipids

Answer: B. Purines and pyrimidines

29-Nucleic Acids DNA And RNA

4. Adenine and guanine are examples of:

- A. Pyrimidines
- B. Purines
- C. Amino acids
- D. Lipids

Answer: B. Purines

29-Nucleic Acids DNA And RNA

5. Cytosine is a:

- A. Purine
- B. Phospholipid

C. Pyrimidine
D. Fatty acid
Answer: C. Pyrimidine

29-Nucleic Acids DNA And RNA

MODULE 3 – Prime Ends (5' & 3')

6. The 5' (five-prime) end of a nucleic acid strand contains:

- A. A free hydroxyl group
 - B. A free phosphate group
 - C. A nitrogen atom
 - D. A sugar ring only
- Answer:** B. A free phosphate group

29-Nucleic Acids DNA And RNA

7. The 3' (three-prime) end of a nucleic acid strand contains:

- A. A free nitrogen base
 - B. A free hydroxyl (–OH) group
 - C. A double-bonded oxygen
 - D. A methyl group
- Answer:** B. A free hydroxyl (–OH) group

29-Nucleic Acids DNA And RNA

8. Nucleic acid strands grow by adding new nucleotides to the:

- A. 5' end only
 - B. 3' end only
 - C. Middle of the chain
 - D. Phosphate backbone
- Answer:** B. 3' end only

29-Nucleic Acids DNA And RNA

MODULE 4 – DNA vs. RNA (implied foundational differences)

9. DNA contains the sugar:

- A. Ribose
 - B. Deoxyribose
 - C. Glucose
 - D. Sucrose
- Answer:** B. Deoxyribose

29-Nucleic Acids DNA And RNA

10. RNA differs from DNA because it contains the base:

- A. Thymine
- B. Adenine

C. Uracil
D. Guanine
Answer: C. Uracil

29-Nucleic Acids DNA And RNA

Here are **important basic MCQs** created directly from your uploaded file:**

File: *Prokaryotic and Eukaryotic Cells*

(Cited properly:

30. Prokaryotic and eukaryotic ...

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on PPT content)

MODULE 1 – Definitions

1. The term *prokaryotic* comes from Greek words meaning:

- A. True nucleus
- B. Before nucleus
- C. Without cytoplasm
- D. Double membrane

Answer: B. Before nucleus

30. Prokaryotic and eukaryotic ...

2. Prokaryotic cells lack:

- A. Cytoplasm
- B. Cell membrane
- C. Membrane-bounded nucleus
- D. Ribosomes

Answer: C. Membrane-bounded nucleus

30. Prokaryotic and eukaryotic ...

3. Eukaryotic cells contain:

- A. No DNA
- B. A membrane-bounded nucleus
- C. Only one organelle
- D. No membranes at all

Answer: B. A membrane-bounded nucleus

30. Prokaryotic and eukaryotic ...

MODULE 2 – Organelles

4. Prokaryotic cells DO NOT have:

- A. Mitochondria
- B. Ribosomes
- C. Cell membranes
- D. DNA

Answer: A. Mitochondria

30. Prokaryotic and eukaryotic ...

5. Eukaryotic cells contain:

- A. No membrane-bounded organelles
- B. Many membrane-bounded organelles
- C. Only ribosomes
- D. Only cell walls

Answer: B. Many membrane-bounded organelles

30. Prokaryotic and eukaryotic ...

MODULE 3 – Examples of Organisms

6. Bacteria belong to which group?

- A. Eukaryotes
- B. Prokaryotes
- C. Both A and B
- D. Neither

Answer: B. Prokaryotes

30. Prokaryotic and eukaryotic ...

7. Cyanobacteria are:

- A. Fungi
- B. Animals
- C. Prokaryotes
- D. Viruses

Answer: C. Prokaryotes

30. Prokaryotic and eukaryotic ...

8. Amoeba and Paramecium are examples of:

- A. Prokaryotes
- B. Viruses
- C. Eukaryotes
- D. Bacteria

Answer: C. Eukaryotes

30. Prokaryotic and eukaryotic ...

MODULE 4 – Eukaryotic Organisms

9. Animals, plants, and fungi belong to:

- A. Prokaryotes
- B. Eukaryotes
- C. Only plants
- D. Neither category

Answer: B. Eukaryotes

30. Prokaryotic and eukaryotic ...

MODULE 5 – Comparison Questions

10. Which cell type has a *true nucleus*?

- A. Prokaryotic
- B. Eukaryotic
- C. Viral
- D. None

Answer: B. Eukaryotic

30. Prokaryotic and eukaryotic ...

11. Which cell type evolved first?

- A. Eukaryotic
- B. Prokaryotic
- C. Both appeared simultaneously
- D. Animal cells only

Answer: B. Prokaryotic

(Implied from “pro = before nucleus” concept)

30. Prokaryotic and eukaryotic ...

12. Which group is **MORE** complex in structure?

- A. Prokaryotes
- B. Eukaryotes
- C. Both same
- D. Viruses

Answer: B. Eukaryotes

30. Prokaryotic and eukaryotic ...

Here are **important basic MCQs** created directly from your uploaded file:

File: *General Features of Eukaryotic Cells*

(Cited properly:

31. General features of eukaryo...

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on PPT content)

MODULE 1 – Definition of Eukaryotic Cells

1. Eukaryotic cells are defined as cells in which genetic material is:

- A. Floating freely in cytoplasm
- B. Enclosed in a membranous nucleus
- C. Absent
- D. Enclosed in ribosomes

Answer: B. Enclosed in a membranous nucleus

31. General features of eukaryo...

2. Eukaryotic cells are found in how many kingdoms?

- A. One
- B. Three
- C. Four out of five
- D. All five

Answer: C. Four out of five

31. General features of eukaryo...

MODULE 2 – Structure of Eukaryotic Cells

3. Eukaryotic cells have how many basic parts?

- A. One
- B. Two
- C. Three
- D. Four

Answer: C. Three

31. General features of eukaryo...

4. Which of the following is *NOT* one of the three basic parts of a eukaryotic cell?

- A. Plasma membrane
- B. Cytoplasm
- C. Nucleus
- D. Cell wall

Answer: D. Cell wall

31. General features of eukaryo...

5. Eukaryotic cells contain:

- A. No organelles
- B. Only non-membranous organelles
- C. Membrane-bounded organelles
- D. Only ribosomes

Answer: C. Membrane-bounded organelles

31. General features of eukaryo...

6. Chromosomes in eukaryotic cells are:

- A. Circular
- B. Linear
- C. Absent
- D. Single-stranded only

Answer: B. Linear

31. General features of eukaryo...

MODULE 3 – Division of Labor (Compartmentalization)

7. Division of labor in eukaryotic cells is achieved through:

- A. Lack of organelles
 - B. Compartmentalization
 - C. Diffusion only
 - D. Cell shrinkage
- Answer:** B. Compartmentalization

31. General features of eukaryo...

8. Lysosomes are responsible for:

- A. Energy production
 - B. Digestion
 - C. Cell secretion
 - D. DNA replication
- Answer:** B. Digestion

31. General features of eukaryo...

9. Mitochondria perform which function?

- A. Digestion
 - B. Cell secretion
 - C. Energy production
 - D. Protein folding
- Answer:** C. Energy production

31. General features of eukaryo...

10. Golgi apparatus is involved in:

- A. Respiration
 - B. Cell secretion
 - C. DNA repair
 - D. Protein breakdown
- Answer:** B. Cell secretion

31. General features of eukaryo...

Here are **important basic MCQs** created directly from your uploaded file:

File: *Comparison of Plant and Animal Cells*

(Cited properly:

32. Comparison of plant and ani...

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on PPT content)

MODULE 1 – General Differences

1. Plant and animal cells differ mainly in the presence of:

- A. Nucleus
- B. Cytoplasm
- C. Cell wall
- D. Cell membrane

Answer: C. Cell wall

32. Comparison of plant and ani...

2. Which type of cell contains chloroplasts?

- A. Animal cell
- B. Plant cell
- C. Both
- D. Neither

Answer: B. Plant cell

32. Comparison of plant and ani...

MODULE 2 – Organelles Unique to Plants

3. Large central vacuole is a characteristic of:

- A. Only animal cells
- B. Only plant cells
- C. Both
- D. Neither

Answer: B. Only plant cells

32. Comparison of plant and ani...

4. Chlorophyll is present in:

- A. Chloroplasts
- B. Mitochondria
- C. Nucleus
- D. Ribosomes

Answer: A. Chloroplasts

32. Comparison of plant and ani...

MODULE 3 – Organelles Unique to Animals

5. Which organelle is present only in animal cells?

- A. Cell wall
- B. Chloroplast
- C. Centrioles
- D. Vacuole

Answer: C. Centrioles

32. Comparison of plant and ani...

6. Lysosomes are more commonly found in:

- A. Plant cells
- B. Animal cells
- C. Both equally
- D. Neither

Answer: B. Animal cells

32. Comparison of plant and ani...

MODULE 4 – Shape and Size

7. Plant cells are generally:

- A. Round and flexible
- B. Rectangular and rigid
- C. Irregular in shape
- D. Long and thread-like

Answer: B. Rectangular and rigid

32. Comparison of plant and ani...

8. Animal cells are usually:

- A. Box-shaped
- B. Round or irregular
- C. Always rectangular
- D. Without a membrane

Answer: B. Round or irregular

32. Comparison of plant and ani...

MODULE 5 – Other Key Differences

9. The structure responsible for maintaining turgor pressure in plant cells is the:

- A. Mitochondrion
- B. Nucleus
- C. Central vacuole
- D. Cell membrane

Answer: C. Central vacuole

32. Comparison of plant and ani...

10. The organelle responsible for energy production in both plant and animal cells is:

- A. Chloroplast
- B. Mitochondrion
- C. Centriole
- D. Golgi apparatus

Answer: B. Mitochondrion

32. Comparison of plant and ani...

Here are **important basic MCQs** created directly from your uploaded file:

File: *Introduction to Plasma Membrane*

(Cited properly:

33.Introduction to plasma membr...

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on PPT content)

MODULE 1 – Basic Concept of Membrane

1. Biological membranes are described as:

- A. Solid rigid structures
- B. Only barriers
- C. More than barriers
- D. Only channels

Answer: C. More than barriers

33.Introduction to plasma membr...

2. Membranes are mainly composed of:

- A. Triglycerides only
- B. Phospholipids
- C. Steroids only
- D. Carbohydrates

Answer: B. Phospholipids

33.Introduction to plasma membr...

MODULE 2 – Amphipathic Nature

3. Phospholipids are termed *amphipathic* because they contain:

- A. Only hydrophobic regions
- B. Only hydrophilic regions
- C. Both hydrophilic and hydrophobic regions
- D. No charged regions

Answer: C. Both hydrophilic and hydrophobic regions

33.Introduction to plasma membr...

4. The hydrophobic region of a phospholipid is formed by:

- A. Phosphate group
- B. Nitrogen base
- C. Fatty acid tails
- D. Water molecules

Answer: C. Fatty acid tails

33.Introduction to plasma membr...

MODULE 3 – Membrane Assembly

5. Phospholipids spontaneously form a bilayer due to:

- A. Enzymes
- B. Free-energy considerations
- C. Active transport
- D. Protein synthesis

Answer: B. Free-energy considerations

33.Introduction to plasma membr...

6. A typical fatty acid tail length ranges from:

- A. 1–10 carbons
- B. 14–24 carbons
- C. 30–40 carbons
- D. 2–5 carbons

Answer: B. 14–24 carbons

33.Introduction to plasma membr...

7. Membranes contain at least one fatty acid tail that is:

- A. Saturated
- B. Trans-unsaturated
- C. Unsaturated and cis
- D. Branched

Answer: C. Unsaturated and cis

33.Introduction to plasma membr...

MODULE 4 – Membrane Properties

8. Biological membranes are:

- A. Static and rigid
- B. Dynamic structures
- C. Hard crystalline structures
- D. Completely impermeable

Answer: B. Dynamic structures

33.Introduction to plasma membr...

9. Membranes maintain physical integrity because:

- A. Lipids create a barrier to hydrophobic materials
- B. Lipids create a barrier to hydrophilic materials such as water and ions
- C. Proteins form a thick wall
- D. Water pushes them into shape

Answer: B. Lipids create a barrier to hydrophilic materials such as water and ions

33.Introduction to plasma membr...

10. Membranes play vital roles in:

- A. Protein synthesis only
- B. Cell–cell interactions
- C. DNA replication
- D. Cell wall formation

Answer: B. Cell–cell interactions

33.Introduction to plasma membr...

Here are **important basic MCQs** created directly from your uploaded file:**

File: *Structure of Cell Membrane – Fluid Mosaic Model*

(Cited properly:

34. Structure of cell membrane,...

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on PPT content)

MODULE 1 – Basics of Cell Membrane

1. The cell membrane separates the cell from:

- A. Nucleus
 - B. Cytoplasm
 - C. External environment
 - D. Mitochondria
- Answer:** C. External environment

34. Structure of cell membrane,...

2. Movement of materials across the cell occurs through the:

- A. Ribosomes
 - B. Cell membrane
 - C. Nucleus
 - D. Cytoskeleton
- Answer:** B. Cell membrane

34. Structure of cell membrane,...

MODULE 2 – Unit Membrane Model

3. The unit membrane model describes the membrane as having:

- A. One lipid layer
 - B. Four layers
 - C. Three layers
 - D. Two protein layers only
- Answer:** B. Four layers

34. Structure of cell membrane,...

4. According to the unit membrane model, lipids are:

- A. Inside the nucleus
 - B. Sandwiched between two protein layers
 - C. Scattered randomly
 - D. Not part of the membrane
- Answer:** B. Sandwiched between two protein layers

34. Structure of cell membrane,...

MODULE 3 – Fluid Mosaic Model

5. The fluid mosaic model was proposed in:

- A. 1955
- B. 1960

C. 1972
D. 2000
Answer: C. 1972

34. Structure of cell membrane,...

6. Scientists who proposed the fluid mosaic model were:

- A. Watson and Crick
 - B. Singer and Nicolson
 - C. Franklin and Wilkins
 - D. Schleiden and Schwann
- Answer:** B. Singer and Nicolson

34. Structure of cell membrane,...

7. According to the fluid mosaic model, the membrane is:

- A. Rigid
 - B. Static
 - C. Fluid in nature
 - D. Crystalline
- Answer:** C. Fluid in nature

34. Structure of cell membrane,...

8. In the fluid mosaic model, lipids form a:

- A. Single layer
 - B. Bilayer
 - C. Protein coat
 - D. Carbohydrate sheet
- Answer:** B. Bilayer

34. Structure of cell membrane,...

9. In the fluid mosaic model, proteins are:

- A. Only on the surface
 - B. Embedded in a mosaic pattern
 - C. Absent
 - D. Found only inside the nucleus
- Answer:** B. Embedded in a mosaic pattern

34. Structure of cell membrane,...

Here are **important basic MCQs** created directly from your uploaded file:

File: *Structure of Cell Membrane – Fluid Mosaic Model (Part II)*

(Cited properly:

35. Structure of cell membrane,...

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on PPT content)

MODULE 1 – Fluid Mosaic Model Basics

1. According to the fluid mosaic model, phospholipids form a:

- A. Single layer
 - B. Bilayer due to hydrophobic tails
 - C. Protein-only structure
 - D. Carbohydrate mesh
- Answer:** B. Bilayer due to hydrophobic tails

35. Structure of cell membrane,...

2. In the fluid mosaic model, proteins are described as:

- A. Fixed in place
 - B. Floating like icebergs in a sea of lipids
 - C. Forming a continuous outer layer
 - D. Arranged only on the inner membrane
- Answer:** B. Floating like icebergs in a sea of lipids

35. Structure of cell membrane,...

3. The membrane is described as fluid because it:

- A. Is made of water
 - B. Constantly shifts and changes
 - C. Contains no proteins
 - D. Has rigid lipid layers
- Answer:** B. Constantly shifts and changes

35. Structure of cell membrane,...

4. Despite its fluid nature, the membrane maintains:

- A. No structure
 - B. A uniform structure
 - C. A crystalline form
 - D. A rigid outer layer
- Answer:** B. A uniform structure

35. Structure of cell membrane,...

MODULE 2 – Importance of Fluid Mosaic Model

5. One major importance of the fluid nature of membranes is that they:

- A. Are locked and cannot transport materials
 - B. Allow flexible transport of materials
 - C. Prevent all molecular movement
 - D. Support only active transport
- Answer:** B. Allow flexible transport of materials

35. Structure of cell membrane,...

6. Because membrane proteins are not arranged in a continuous layer:

- A. Only proteins can cross the membrane
- B. Lipid-soluble molecules can cross where proteins are absent
- C. Nothing can pass through the membrane
- D. Only ions can cross

Answer: B. Lipid-soluble molecules can cross where proteins are absent

35. Structure of cell membrane,...

7. The fluid mosaic model states that membrane proteins are:

- A. Rigid structures
 - B. Evenly spaced and immovable
 - C. Free to move within the lipid bilayer
 - D. Not part of the membrane
- Answer:** C. Free to move within the lipid bilayer

35. Structure of cell membrane,...

MODULE 3 – Lipids and Proteins in Membrane Function

8. Lipids in the membrane create a barrier mainly to:

- A. Hydrophobic materials
 - B. Hydrophilic materials like water and ions
 - C. All molecules equally
 - D. Proteins only
- Answer:** B. Hydrophilic materials like water and ions
(Supported by fluid mosaic model concept in Part I & II)

35. Structure of cell membrane,...

9. The “mosaic” part of the model refers to:

- A. Arrangement of carbohydrates
 - B. Different types of proteins embedded in the membrane
 - C. Shape of phospholipids
 - D. Random arrangement of DNA
- Answer:** B. Different types of proteins embedded in the membrane

35. Structure of cell membrane,...

Here are **important basic MCQs** created directly from your uploaded file:

File: Chemical Composition of Cell Membrane

(Cited properly:

36. Chemical composition of cel...

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on PPT content)

MODULE 1 – Components of Cell

Membrane

1. The cell membrane is the:

- A. Innermost part of the cell
- B. Outermost boundary of the cell
- C. Only boundary of the nucleus
- D. Part of the cytoskeleton

Answer: B. Outermost boundary of the cell

36. Chemical composition of cel...

2. The cell membrane is mainly composed of:

- A. Lipids and DNA
- B. Proteins and RNA
- C. Phospholipids, cholesterol, proteins, and carbohydrates
- D. Only carbohydrates

Answer: C. Phospholipids, cholesterol, proteins, and carbohydrates

36. Chemical composition of cel...

MODULE 2 – Phospholipids

3. Phospholipids in the membrane form a:

- A. Single lipid sheet
- B. Bilayer
- C. Protein coat
- D. Triple membrane

Answer: B. Bilayer

36. Chemical composition of cel...

4. The phospholipid molecule has:

- A. A nonpolar head and polar tails
 - B. A polar head and nonpolar tails
 - C. Two polar heads
 - D. No charged regions
- Answer:** B. A polar head and nonpolar tails

36. Chemical composition of cel...

5. Phospholipid heads face:

- A. Toward the fatty acid tails
- B. Toward the outside environment and cytoplasm
- C. Toward the nucleus only
- D. Toward cholesterol molecules

Answer: B. Toward the outside environment and cytoplasm

36. Chemical composition of cel...

MODULE 3 – Cholesterol

6. Cholesterol is present in the plasma membrane of:

- A. Prokaryotes only
- B. Plants only
- C. Eukaryotes only
- D. All living cells

Answer: C. Eukaryotes only

36. Chemical composition of cel...

7. Cholesterol makes the membrane:

- A. More permeable to water-soluble substances
- B. Less permeable to water-soluble substances
- C. Completely impermeable
- D. More fluid and unstable

Answer: B. Less permeable to water-soluble substances

36. Chemical composition of cel...

8. Cholesterol contributes to membrane:

- A. Instability
- B. Rigidity and stability
- C. DNA repair
- D. Carbohydrate synthesis

Answer: B. Rigidity and stability

36. Chemical composition of cel...

MODULE 4 – Proteins in the Membrane

9. Proteins present on the membrane surface are called:

- A. Intrinsic proteins
- B. Peripheral proteins
- C. Cytoskeletal proteins
- D. Glycolipids

Answer: B. Peripheral proteins

36. Chemical composition of cel...

10. Proteins embedded inside the membrane are known as:

- A. Surface carbohydrates
- B. Intrinsic proteins
- C. Peripheral proteins
- D. Glycocalyx

Answer: B. Intrinsic proteins

36. Chemical composition of cel...

11. One major role of membrane proteins is to act as:

- A. Hormones
- B. Transporters

- C. Genetic material
 - D. Lipid producers
- Answer:** B. Transporters

36. Chemical composition of cel...

MODULE 5 – Carbohydrates in the Membrane

12. Surface carbohydrates occur as:

- A. DNA strands and RNA
- B. Glycoproteins and glycolipids
- C. Cytoskeletal fibers
- D. Fatty acid derivatives

Answer: B. Glycoproteins and glycolipids

36. Chemical composition of cel...

13. The carbohydrate-rich region on the outside of the cell is called:

- A. Cytoskeleton
- B. Nucleoid
- C. Glycocalyx
- D. Periplasm

Answer: C. Glycocalyx

36. Chemical composition of cel...

14. The glycocalyx is involved in:

- A. Energy production
- B. DNA replication
- C. Cell recognition and cell behavior
- D. Protein synthesis

Answer: C. Cell recognition and cell behavior

36. Chemical composition of cel...

Here are **important basic MCQs** created directly from your uploaded file:

File: Recognition and Adhesion

(Cited properly:

37-Recognition and adhesion

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on PPT content)

MODULE 1 – Cell Recognition & Adhesion

1. Cell recognition refers to a process in which:

- A. A cell divides into two identical cells
- B. One cell specifically binds to another cell

- C. Cells exchange genetic material
 - D. A cell destroys another cell
- Answer:** B. One cell specifically binds to another cell

37-Recognition and adhesion

2. Cell adhesion refers to:

- A. Cells floating freely
- B. Cells repelling each other
- C. Cells becoming “cemented” together
- D. Cells dissolving other cells

Answer: C. Cells becoming “cemented” together

37-Recognition and adhesion

3. Cell recognition and adhesion are essential for:

- A. DNA replication
- B. Tissue formation
- C. Photosynthesis
- D. Protein synthesis

Answer: B. Tissue formation

37-Recognition and adhesion

MODULE 2 – Junctions in Epithelial Cells

4. Tight junctions are found mainly in:

- A. Muscle cells
- B. Epithelial cells
- C. Neurons
- D. Red blood cells

Answer: B. Epithelial cells

37-Recognition and adhesion

5. Tight junctions prevent:

- A. Protein synthesis
- B. Passage of molecules between cells
- C. DNA movement across membranes
- D. Mitochondria from functioning

Answer: B. Passage of molecules between cells

37-Recognition and adhesion

6. Tight junctions also restrict the migration of:

- A. RNA and DNA
- B. Lipid molecules and membrane proteins
- C. Mitochondria
- D. Ribosomes

Answer: B. Lipid molecules and membrane proteins

37-Recognition and adhesion

MODULE 3 – Transport Across Membrane

7. Passive transport requires:

- A. ATP energy input
- B. Chemical modification of molecules
- C. No energy input
- D. Enzymatic digestion

Answer: C. No energy input

37-Recognition and adhesion

8. Active transport requires:

- A. No proteins
- B. Diffusion
- C. Input of energy
- D. Only water movement

Answer: C. Input of energy

37-Recognition and adhesion

Here are **important basic MCQs** created directly from your uploaded file:

File: *Functions of Cell Membrane*

(Cited properly:

38. Functions of cell membrane

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on PPT content)

MODULE 1 – Cellular Boundary

1. The cell membrane separates:

- A. Nucleus from cytoplasm
- B. Mitochondria from ribosomes
- C. Inside of the cell from the outside environment
- D. DNA from RNA

Answer: C. Inside of the cell from the outside environment

38. Functions of cell membrane

2. By forming a boundary, the cell membrane allows the cell to:

- A. Produce its own DNA
- B. Maintain a stable internal environment
- C. Divide uncontrollably
- D. Stop all transport

Answer: B. Maintain a stable internal environment

38. Functions of cell membrane

MODULE 2 – Compartmentalization

3. Compartmentalization refers to:

- A. The movement of ribosomes
- B. Separation of organelles from the general cytoplasm
- C. Protein synthesis
- D. Formation of ATP

Answer: B. Separation of organelles from the general cytoplasm

38. Functions of cell membrane

4. The purpose of compartmentalization is to allow organelles to:

- A. Change their shape
- B. Perform specific functions
- C. Destroy other organelles
- D. Remain inactive

Answer: B. Perform specific functions

38. Functions of cell membrane

MODULE 3 – Material Transport

5. The cell membrane regulates:

- A. Only the exit of materials
- B. Only the entry of materials
- C. Both entry and exit of materials
- D. No movement at all

Answer: C. Both entry and exit of materials

38. Functions of cell membrane

6. The membrane also controls movement between:

- A. Organelles
- B. Organ systems
- C. Tissues only
- D. Nucleus and chromosomes

Answer: A. Organelles

38. Functions of cell membrane

MODULE 4 – Site for Chemical Reactions

7. Some chemical reactions in the body take place on the surface of:

- A. Chromosomes
- B. Cell membranes
- C. Centrioles
- D. Golgi bodies

Answer: B. Cell membranes

38. Functions of cell membrane

8. A reaction example occurring on membranes is:

- A. Photosynthesis
- B. Nerve impulse activity
- C. DNA replication
- D. Cell division

Answer: B. Nerve impulse activity

38. Functions of cell membrane

MODULE 5 – Identification Markers

9. Cell membranes contain molecules that act as:

- A. Enzymes for respiration
- B. Identification markers
- C. Ribosome producers
- D. ATP generators

Answer: B. Identification markers

38. Functions of cell membrane

10. Identification markers help in recognizing:

- A. DNA strands
- B. Different types of cells
- C. Minerals in blood
- D. Amount of water in the cell

Answer: B. Different types of cells

38. Functions of cell membrane

MODULE 6 – Homeostasis

11. Homeostasis means:

- A. Rapid cell division
- B. Maintenance of a relatively constant internal environment
- C. Production of proteins
- D. Destruction of waste materials

Answer: B. Maintenance of a relatively constant internal environment

38. Functions of cell membrane

12. Homeostasis is maintained by a membrane that is:

- A. Fully permeable
- B. Impermeable
- C. Selectively permeable
- D. Not involved in transport

Answer: C. Selectively permeable

38. Functions of cell membrane

Here are **important basic MCQs** created directly from your uploaded file:

File: Transport Across Membranes

(Cited properly:

39-Transport across membranes

)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on PPT content)

MODULE 1 – Osmosis

1. Osmosis is a type of:

- A. Active transport
- B. Passive transport
- C. Endocytosis
- D. Exocytosis

Answer: B. Passive transport

39-Transport across membranes

2. Osmosis involves movement of:

- A. Ions
- B. Water molecules
- C. Proteins
- D. Carbohydrates

Answer: B. Water molecules

39-Transport across membranes

3. In osmosis, water moves across membranes based on:

- A. Type of solute particles
- B. Number of solute particles
- C. Temperature of the cell
- D. Presence of ATP

Answer: B. Number of solute particles

39-Transport across membranes

MODULE 2 – Aquaporins

4. Aquaporins are:

- A. Lipids
- B. Channel proteins
- C. Enzymes
- D. Hormones

Answer: B. Channel proteins

39-Transport across membranes

5. The inside of an aquaporin channel contains:

- A. Nonpolar amino acids
- B. Polar amino acids and water
- C. DNA strands
- D. ATP molecules

Answer: B. Polar amino acids and water

39-Transport across membranes

6. The outside surface of aquaporins contains:

- A. Polar amino acids
- B. Carbohydrates
- C. Nonpolar amino acids
- D. RNA molecules

Answer: C. Nonpolar amino acids

39-Transport across membranes

MODULE 3 – Active Transport

7. Active transport requires:

- A. No energy
- B. Direct input of ATP always
- C. Energy input
- D. Osmosis

Answer: C. Energy input

39-Transport across membranes

8. Primary active transport uses:

- A. ATP directly
- B. ATP indirectly
- C. Only concentration gradients
- D. Osmosis

Answer: A. ATP directly

39-Transport across membranes

9. Secondary active transport uses:

- A. ATP directly
 - B. ATP indirectly (ion concentration gradient)
 - C. Only passive movement of ions
 - D. No energy at all
- Answer:** B. ATP indirectly (ion concentration gradient)

39-Transport across membranes

Here are **important basic MCQs** created directly from your uploaded file:**

File: *Cell Wall*

Cited properly:

40-Cell Wall

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on PPT content)

MODULE 1 – Introduction to Cell Wall

1. The cell wall is the:

- A. Innermost covering of all cells
 - B. Outermost covering in many organisms
 - C. Outer covering only in animal cells
 - D. Protective layer inside the nucleus
- Answer:** B. Outermost covering in many organisms

40-Cell Wall

2. The cell wall is present in:

- A. Only plant cells
 - B. Only prokaryotic cells
 - C. Prokaryotic cells, fungi, and plant cells
 - D. All types of cells including animal cells
- Answer:** C. Prokaryotic cells, fungi, and plant cells

40-Cell Wall

3. Compared to the cell membrane, the cell wall is:

- A. Softer and more flexible
 - B. More rigid and tough
 - C. Identical in structure
 - D. Found only around organelles
- Answer:** B. More rigid and tough

40-Cell Wall

MODULE 2 – Functions of Cell Wall

4. One major function of the cell wall is to:

- A. Carry out respiration
 - B. Provide rigidity and strength
 - C. Synthesize proteins
 - D. Transport ATP
- Answer:** B. Provide rigidity and strength

40-Cell Wall

5. The cell wall protects the cell from:

- A. Light
 - B. Mechanical damage
 - C. DNA mutations
 - D. Enzyme deficiency
- Answer:** B. Mechanical damage

40-Cell Wall

MODULE 3 – Types of Plant Cell Walls

6. The primary cell wall is found in:

- A. Mature cells
- B. Growing cells
- C. Dead cells only
- D. Animal cells

Answer: B. Growing cells

40-Cell Wall

7. The primary cell wall is mainly composed of:

- A. Protein and lipids
- B. DNA and RNA
- C. Cellulose, hemicellulose, and pectin
- D. Lignin and keratin

Answer: C. Cellulose, hemicellulose, and pectin

40-Cell Wall

8. The secondary cell wall is found:

- A. Outside the primary cell wall
- B. Beneath the primary cell wall
- C. Only in dividing cells
- D. In all animal cells

Answer: B. Beneath the primary cell wall

40-Cell Wall

9. The secondary cell wall contains cellulose and:

- A. Pectin
- B. Lipids
- C. Lignin
- D. Proteins

Answer: C. Lignin

40-Cell Wall

10. The secondary cell wall provides:

- A. Flexibility
- B. Extra strength
- C. Water absorption
- D. Enzyme activity

Answer: B. Extra strength

40-Cell Wall

Here are **important basic MCQs** created directly from your uploaded file:

File: *Cytoplasm and Its Functions*

Cited properly:

41. Cytoplasm and its functions

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on PPT content)

MODULE 1 – Definition of Cytoplasm

1. Cytoplasm is the portion of the cell located:

- A. Inside the nucleus only
- B. Outside the plasma membrane
- C. Outside the nucleus and inside the plasma membrane
- D. Only in plant cells

Answer: C. Outside the nucleus and inside the plasma membrane

41. Cytoplasm and its functions

MODULE 2 – Ectoplasm & Endoplasm

2. The clear, outer layer of the cytoplasm just below the plasma membrane is called:

- A. Cytosol
- B. Endoplasm
- C. Ectoplasm
- D. Enzyme layer

Answer: C. Ectoplasm

41. Cytoplasm and its functions

3. The dense, inner portion of the cytoplasm is known as:

- A. Cytosol
- B. Endoplasm
- C. Ectoplasm
- D. Nucleoplasm

Answer: B. Endoplasm

41. Cytoplasm and its functions

MODULE 3 – Cytosol & Organelles

4. The semifluid portion of the cytoplasm is called:

- A. Cytosol
- B. Ectoplasm
- C. Matrix
- D. Plasma

Answer: A. Cytosol

41. Cytoplasm and its functions

5. Organelles are described as:

- A. Protein particles floating freely
- B. Membranous and non-membranous structures performing specific functions
- C. Only membrane-bound structures
- D. Non-functional cell parts

Answer: B. Membranous and non-membranous structures performing specific functions

41. Cytoplasm and its functions

MODULE 4 – Functions of Cytoplasm

6. Organelles present in the cytoplasm are involved in:

- A. Photosynthesis only
 - B. Protein synthesis, digestion, ATP production
 - C. DNA replication only
 - D. Cell division only
- Answer:** B. Protein synthesis, digestion, ATP production

41. Cytoplasm and its functions

7. The cytoplasm also stores:

- A. Only water
- B. Molecules required for various cellular activities
- C. DNA and RNA
- D. Only waste materials

Answer: B. Molecules required for various cellular activities

41. Cytoplasm and its functions

Here are **important basic MCQs** created directly from your uploaded file:

File: *Cytoplasmic Organelles – Endoplasmic Reticulum (ER)*

Cited properly:

42. Cytoplasmic Organelles, Endo...

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on PPT content)

MODULE 1 – Definition & Nature of ER

1. The endoplasmic reticulum (ER) is a:

- A. Non-membranous organelle
- B. Network of membrane-bounded sheets, sacs, and tubules
- C. Group of ribosomes only
- D. Part of the nucleus

Answer: B. Network of membrane-bounded sheets, sacs, and tubules

42. Cytoplasmic Organelles, Endo...

MODULE 2 – Location of ER

2. The ER is located:

- A. Only near the plasma membrane
- B. Only near the nucleus
- C. Throughout the cytoplasm
- D. Only in plant cells

Answer: C. Throughout the cytoplasm

42. Cytoplasmic Organelles, Endo...

3. On one side, the ER is attached to the:

- A. Golgi apparatus
- B. Plasma membrane
- C. Outer membrane of the nucleus
- D. Ribosomes

Answer: C. Outer membrane of the nucleus

42. Cytoplasmic Organelles, Endo...

4. On the other side, the ER may be connected to the:

- A. Chloroplast membrane
- B. Plasma membrane
- C. Mitochondria
- D. Lysosomes

Answer: B. Plasma membrane

42. Cytoplasmic Organelles, Endo...

MODULE 3 – Structure of ER

5. The flattened sac-like or tubular structures of the ER are called:

- A. Grana
- B. Cristae
- C. Cisternae
- D. Lamellae

Answer: C. Cisternae

42. Cytoplasmic Organelles, Endo...

6. The inner space of the ER tubules is known as the:

- A. Matrix
- B. Cytosol
- C. Lumen
- D. Vacuole

Answer: C. Lumen

42. Cytoplasmic Organelles, Endo...

MODULE 4 – Types of ER

7. The ER is classified into how many types?

- A. One
- B. Two
- C. Three
- D. Four

Answer: B. Two

42. Cytoplasmic Organelles, Endo...

8. The two types of endoplasmic reticulum are:

- A. Inner and outer ER
- B. Large and small ER
- C. Rough ER (RER) and Smooth ER (SER)
- D. Light ER and dark ER

Answer: C. Rough ER (RER) and Smooth ER (SER)

42. Cytoplasmic Organelles, Endo...

Here are **important basic MCQs** created directly from your uploaded file:

File: *Endoplasmic Reticulum (II)*

Cited properly:

43. Endoplasmic Reticulum(II)

IMPORTANT BASIC MCQs (WITH ANSWERS)

(Strictly based on PPT content)

MODULE 1 – Types of Endoplasmic Reticulum

1. Rough Endoplasmic Reticulum (RER) appears rough because:

- A. It has folds
- B. Ribosomes are attached to its surface
- C. It stores lipids
- D. It contains vacuoles

Answer: B. Ribosomes are attached to its surface

43. Endoplasmic Reticulum(II)

2. Smooth Endoplasmic Reticulum (SER) appears smooth because:

- A. It has no membrane
- B. It breaks down proteins
- C. Ribosomes are absent on its surface
- D. It is connected to mitochondria

Answer: C. Ribosomes are absent on its surface

43. Endoplasmic Reticulum(II)

MODULE 2 – Functions of ER

3. The ER provides _____ to the cell.

- A. Genetic instructions
- B. Mechanical support
- C. Photosynthesis
- D. Blood circulation

Answer: B. Mechanical support

43. Endoplasmic Reticulum(II)

4. The ER helps in circulating:

- A. Blood
- B. Air
- C. Materials throughout the cytoplasm
- D. DNA inside the nucleus

Answer: C. Materials throughout the cytoplasm

43. Endoplasmic Reticulum(II)

5. The ER stores:

- A. DNA and RNA
- B. Enzymes and proteins
- C. Ribosomes only
- D. Vitamins only

Answer: B. Enzymes and proteins

43. Endoplasmic Reticulum(II)

6. One major function of the SER is:

- A. Lipid production
- B. Protein synthesis
- C. Chromosome replication
- D. DNA packaging

Answer: A. Lipid production

43. Endoplasmic Reticulum(II)

7. Detoxification of organic molecules occurs in the:

- A. Rough ER
- B. Smooth ER
- C. Ribosomes
- D. Nucleus

Answer: B. Smooth ER

43. Endoplasmic Reticulum(II)

8. Calcium storage is a function of:

- A. RER
- B. SER
- C. Lysosomes
- D. Mitochondria

Answer: B. SER

43. Endoplasmic Reticulum(II)

Here are **important and basic MCQs** created directly from the content of your uploaded file
“Cytoplasmic Organelles: Golgi Apparatus”

44. Cytoplasmic Organelles, Golg...

.

These MCQs are suitable for exams, quizzes, and quick revision.

MCQs on Golgi Apparatus (with Answers)

1. The Golgi apparatus was first reported by which scientist?

- A. Robert Hooke
- B. Camillo Golgi
- C. Schwann
- D. Schleiden

Answer: B. Camillo Golgi

2. The Golgi apparatus is mainly involved in:

- A. Photosynthesis
- B. Protein synthesis
- C. Sorting, packaging, and secretion of proteins and lipids
- D. Respiration

Answer: C. Sorting, packaging, and secretion of proteins and lipids

3. Golgi apparatus is also known as:

- A. Ribosome
- B. Cytoskeleton
- C. Golgi complex
- D. Endoplasmic reticulum

Answer: C. Golgi complex

4. How many cisternae typically make up a Golgi apparatus?

- A. 1–2
- B. 3–7
- C. 10–20
- D. 50

Answer: B. 3–7

5. Cisternae are:

- A. Fluid-filled channels of mitochondria
- B. The tubular vesicles making up the Golgi apparatus
- C. Enzymes for protein digestion
- D. Ribosome-rich sacs

Answer: B. The tubular vesicles making up the Golgi apparatus

6. The side of the Golgi apparatus where materials enter from the ER is called:

- A. Maturing face
- B. Apical face
- C. Forming face
- D. Distal face

Answer: C. Forming face

7. The side of the Golgi apparatus where materials leave is known as:

- A. Forming face
- B. Basal face
- C. Inner face
- D. Maturing face

Answer: D. Maturing face

8. Proteins are synthesized in the lumen of which organelle before entering the Golgi apparatus?

- A. Mitochondria
- B. Lysosome
- C. Smooth ER
- D. Endoplasmic reticulum

Answer: D. Endoplasmic reticulum

9. The small packets that carry proteins from ER to Golgi apparatus are called:

- A. Ribosomes
- B. Transfer vesicles
- C. Secretory vesicles

D. Lysosomes

Answer: B. Transfer vesicles

10. Secretory vesicles are responsible for:

- A. ATP production
- B. Storing genetic material
- C. Moving modified proteins to cell membrane for secretion
- D. Breaking down toxins

Answer: C. Moving modified proteins to cell membrane for secretion

11. Golgi apparatus is most abundant in which type of cells?

- A. Muscle cells
- B. Secreting cells (e.g., nerve cells)
- C. Blood cells
- D. Bone cells

Answer: B. Secreting cells (e.g., nerve cells)

12. After proteins are modified in the Golgi apparatus, they are packed into:

- A. Ribosomes
- B. Transfer vesicles
- C. Secretory vesicles
- D. Microtubules

Answer: C. Secretory vesicles

13. Which of the following is NOT a function of the Golgi apparatus?

- A. Sorting proteins
- B. Chemical modification of proteins
- C. Respiration
- D. Packaging proteins

Answer: C. Respiration

14. Proteins are concentrated and chemically modified in the:

- A. Ribosomes
- B. Golgi apparatus
- C. Nucleus
- D. Lysosome

Answer: B. Golgi apparatus

15. Transfer vesicles originate from the:

- A. Nucleus
- B. Mitochondria
- C. Endoplasmic reticulum
- D. Cell membrane

Answer: C. Endoplasmic reticulum

Here are **important basic MCQs** created from your uploaded file “**Cytoplasmic Organelles: Lysosomes**”

45. Lysosomes

.

These MCQs are suitable for quizzes, exams, and revision.

MCQs on Lysosomes (with Answers)

1. Lysosomes are surrounded by how many membranes?

- A. None
- B. Single membrane
- C. Double membrane
- D. Triple membrane

Answer: B. Single membrane

2. Lysosomes contain which type of enzymes?

- A. Respiratory enzymes
- B. Hydrolytic enzymes
- C. Photosynthetic enzymes
- D. Ribosomal enzymes

Answer: B. Hydrolytic enzymes

3. What is the main role of lysosomes?

- A. Protein synthesis
- B. Intracellular digestion
- C. Cell division
- D. ATP production

Answer: B. Intracellular digestion

4. Lysosomal enzymes work best under which conditions?

- A. Basic pH
- B. Neutral pH
- C. Acidic pH
- D. High temperature

Answer: C. Acidic pH

5. Which of the following biomolecules can lysosomes digest?

- A. Carbohydrates
- B. Lipids
- C. Proteins
- D. All of the above

Answer: D. All of the above

6. The hydrolytic enzymes present in lysosomes are also called:

- A. Oxidases
- B. Acid hydrolases
- C. Amylases
- D. Kinases

Answer: B. Acid hydrolases

7. Lysosomes can selectively digest and recycle their own cytoplasm or organelles. This process is called:

- A. Phagocytosis
- B. Exocytosis
- C. Autophagy
- D. Endocytosis

Answer: C. Autophagy

8. A phagosome is formed by the fusion of a food vacuole with:

- A. Golgi body
- B. Lysosome
- C. Mitochondria

D. Ribosome

Answer: B. Lysosome

9. In phagosome, what happens to the ingested material?

- A. It becomes toxic
- B. It is digested and absorbed
- C. It is stored permanently
- D. It is converted into ATP

Answer: B. It is digested and absorbed

10. The process by which undigested material is removed from the cell is called:

- A. Endocytosis
- B. Pinocytosis
- C. Exocytosis
- D. Osmosis

Answer: C. Exocytosis

11. Lysosomes are commonly known as:

- A. Powerhouse of cell
- B. Suicidal bags of the cell
- C. Protein factories
- D. Cell boundary

Answer: B. Suicidal bags of the cell (*common biological term*)

12. Lysosomes are involved in digestion of which of the following?

- A. Nucleic acids
- B. Carbohydrates
- C. Proteins and lipids
- D. All of the above

Answer: D. All of the above

13. Which organelle directly participates in forming a phagosome?

- A. Nucleus
- B. Ribosome
- C. Lysosome
- D. Golgi apparatus

Answer: C. Lysosome

14. Lysosomes help in recycling worn-out organelles. This function is related to:

- A. Homeostasis
- B. Autophagy
- C. Glycolysis
- D. Photosynthesis

Answer: B. Autophagy

15. The contents inside lysosomes remain active due to:

- A. High temperature
- B. Presence of double membrane
- C. Acidic internal environment
- D. Oxygen-rich conditions

Answer: C. Acidic internal environment

Here are **important basic MCQs** created from your uploaded file “**Cytoplasmic Organelles: Ribosomes**”

46. Ribosomes

MCQs on Ribosomes (with Answers)

1. Ribosomes are involved in which major cellular process?

- A. Lipid synthesis
- B. Protein synthesis
- C. DNA replication
- D. Respiration

Answer: B. Protein synthesis

2. Ribosomes are:

- A. Single-membrane structures
- B. Double-membrane structures
- C. Non-membrane-bounded structures
- D. Plastids

Answer: C. Non-membrane-bounded structures

3. Ribosomes consist of how many subunits?

- A. One
- B. Two
- C. Three
- D. Four

Answer: B. Two

4. The size of ribosomes in prokaryotes is:

- A. 80S
- B. 50S
- C. 70S
- D. 30S

Answer: C. 70S

5. The subunits of 70S ribosomes are:

- A. 20S + 30S
- B. 30S + 50S
- C. 40S + 60S
- D. 80S + 20S

Answer: B. 30S + 50S

6. Eukaryotic ribosomes are of which type?

- A. 20S
- B. 50S
- C. 70S
- D. 80S

Answer: D. 80S

7. The subunits of 80S ribosomes are:

- A. 20S + 30S
- B. 30S + 50S
- C. 40S + 60S
- D. 60S + 70S

Answer: C. 40S + 60S

8. Chemically, ribosomes are made up of:

- A. Only proteins
- B. Only RNA
- C. RNA and proteins in almost equal amounts
- D. DNA and proteins

Answer: C. RNA and proteins in almost equal amounts

9. Ribosomes attach to which molecule during protein synthesis?

- A. DNA
- B. tRNA
- C. mRNA
- D. Lysosome

Answer: C. mRNA

10. Ribosomes may be found:

- A. Only in the nucleus
- B. Free in cytoplasm or attached to endoplasmic reticulum
- C. Only in mitochondria
- D. Only on plasma membrane

Answer: B. Free in cytoplasm or attached to endoplasmic reticulum

11. A cluster of ribosomes attached to a single mRNA molecule is called:

- A. Chromatin
- B. Polysome
- C. Centrosome
- D. Lysosome

Answer: B. Polysome

12. The primary function of ribosomal RNA (rRNA) is to:

- A. Store genetic information
- B. Provide energy for cell
- C. Help in protein synthesis
- D. Transport lipids

Answer: C. Help in protein synthesis

13. Ribosomes are located in which cellular region?

- A. Cytoplasm
- B. Nucleus
- C. Vacuoles
- D. Plastids

Answer: A. Cytoplasm

14. Ribosomes attached to endoplasmic reticulum form:

- A. Smooth ER
- B. Rough ER
- C. Golgi apparatus
- D. Lysosomes

Answer: B. Rough ER

15. The main role of large ribosomal subunit is:

- A. Binding mRNA
- B. Binding tRNA and forming peptide bonds
- C. Transporting proteins

D. DNA replication

Answer: B. Binding tRNA and forming peptide bonds

Here are **important basic MCQs** created from your uploaded file “**Cytoplasmic Organelles: Vacuole**”

47. Vacuole

.

These MCQs are simple, clear, and helpful for quick revision.

MCQs on Vacuoles (with Answers)

1. Vacuoles are mainly involved in the storage of:

- A. DNA
- B. Water and salts
- C. Proteins only
- D. Enzymes

Answer: B. Water and salts

2. Vacuoles are:

- A. Non-membrane structures
- B. Double-membrane organelles
- C. Single-membrane organelles
- D. Protein bodies

Answer: C. Single-membrane organelles

3. Animal cells usually contain:

- A. One large central vacuole
- B. No vacuole
- C. Many small vacuoles
- D. Only contractile vacuoles

Answer: C. Many small vacuoles

4. The membrane of vacuole is specialized for:

- A. Energy production
- B. Transport of water and salts
- C. Protein synthesis
- D. Photosynthesis

Answer: B. Transport of water and salts

5. Vacuoles vary in:

- A. Only shape
- B. Only number
- C. Both shape and number
- D. Neither shape nor number

Answer: C. Both shape and number

6. Which type of vacuole removes waste water from protozoa and sponges?

- A. Food vacuole
- B. Contractile vacuole
- C. Central vacuole
- D. Storage vacuole

Answer: B. Contractile vacuole

7. Food vacuole is involved in:

- A. Respiration
- B. Digestion and storage
- C. Protein synthesis
- D. DNA replication

Answer: B. Digestion and storage

8. Vacuoles are found in which type of cells?

- A. Only plant cells
- B. Only bacteria
- C. Only fungi
- D. Both plant and animal cells

Answer: D. Both plant and animal cells

9. Which of the following is *not* a role of vacuole?

- A. Storage of water
- B. Digestion of food
- C. Removal of waste water
- D. Protein synthesis

Answer: D. Protein synthesis

10. The contractile vacuole plays a major role in:

- A. Removing excess water
- B. Storing pigments
- C. Producing energy
- D. Forming ribosomes

Answer: A. Removing excess water

Here are **important basic MCQs** created from your uploaded file “**Cilia and Flagella: Occurrence and Function**”

48. cilia and flagell, occurenc...

.

These MCQs cover **occurrence, structure, and functions** of cilia and flagella.

MCQs on Cilia & Flagella (with Answers)

1. Cilia in the trachea function to:

- A. Move sperm
- B. Push mucus and dust toward the mouth
- C. Absorb nutrients
- D. Produce hormones

Answer: B. Push mucus and dust toward the mouth

2. In fallopian tubes, cilia help in:

- A. Fertilization
- B. Moving the egg toward the uterus
- C. Formation of placenta
- D. Formation of hormones

Answer: B. Moving the egg toward the uterus

3. In protozoa, cilia mainly help in:

- A. Respiration
- B. Reproduction
- C. Locomotion
- D. Digestion

Answer: C. Locomotion

4. Which organisms contain cilia for locomotion?

- A. Bacteria only
- B. Plants only
- C. Protozoa and invertebrate larvae
- D. Mammals only

Answer: C. Protozoa and invertebrate larvae

5. Flagella are commonly found in:

- A. Trachea
- B. Fungi, algae, plants, animals (gametes)
- C. Invertebrate larvae only
- D. Fallopian tube

Answer: B. Fungi, algae, plants, animals (gametes)

6. Which of the following contains flagellated protozoa?

- A. Trachea
- B. Amoeba
- C. Euglena
- D. Hydra

Answer: C. Euglena

7. In bacteria, flagella function mainly for:

- A. Food storage
- B. Locomotion
- C. Protein synthesis
- D. Digestion

Answer: B. Locomotion

8. The flagellum in gametes (e.g., sperm) is used for:

- A. Sensory perception
- B. ATP production
- C. Locomotion
- D. Digestion

Answer: C. Locomotion

9. Cilia can be found in all of the following EXCEPT:

- A. Trachea
- B. Fallopian tubes
- C. Flagellated protozoa
- D. Invertebrate larvae

Answer: C. Flagellated protozoa (they contain *flagella*, not cilia)

10. Flagella may function as:

- A. Excretory organelles
- B. Sensory organelles
- C. Digestive sacs
- D. Lysosomes

Answer: B. Sensory organelles

11. Cilia and flagella are similar in that both:

- A. Are used for locomotion
- B. Are present only in plants
- C. Help in photosynthesis

D. Are found only in protozoa

Answer: A. Are used for locomotion

12. Which statement is TRUE about cilia in trachea?

A. They help in gas exchange

B. They trap dust and move it upward

C. They form sperm

D. They produce mucus

Answer: B. They trap dust and move it upward

13. Flagella of bacteria are mainly used for:

A. Attachment only

B. Movement of food vacuoles

C. Swimming

D. Regulating temperature

Answer: C. Swimming

14. Cilia vs Flagella: Which are generally shorter and more numerous?

A. Cilia

B. Flagella

Answer: A. Cilia

15. Flagella in algae and plants are found mostly in:

A. Stem cells

B. Gametes

C. Roots

D. Leaves

Answer: B. Gametes

Here are **important basic MCQs** created from your uploaded file “Cilia and Flagella: Structure”

49. cilia and flagell, structure

.

These questions cover **size, structure, axoneme, and basal bodies.**

MCQs on Cilia & Flagella Structure (with Answers)

1. Cilia are generally:

A. Longer than flagella

B. Smaller than flagella

C. Equal in length to flagella

D. Absent in animals

Answer: B. Smaller than flagella

2. Flagella are typically how many times longer than cilia?

A. 2–3 times

B. 5–20 times

C. 50 times

D. 100 times

Answer: B. 5–20 times

3. Structurally, cilia and flagella are:

- A. Completely different
- B. Same in structure
- C. Membrane-less structures
- D. Found only in plants

Answer: B. Same in structure

4. The axoneme is also known as:

- A. Cytoplasm
- B. Axial filament
- C. Cell membrane
- D. Nucleus

Answer: B. Axial filament

5. Axoneme contains how many central microtubules?

- A. 1
- B. 2
- C. 9
- D. 12

Answer: B. 2

6. The typical microtubule arrangement in cilia and flagella axoneme is:

- A. $9 + 0$
- B. $9 + 1$
- C. $9 + 2$
- D. $9 + 3$

Answer: C. $9 + 2$

7. The outer microtubules in axoneme are arranged as:

- A. Nine pairs in a circle
- B. Nine triplets in a circle
- C. Two pairs in a circle
- D. A single ring of microfilaments

Answer: A. Nine pairs in a circle

8. Dynein arms project toward:

- A. The nucleus
- B. The plasma membrane
- C. Neighboring doublets
- D. Central microtubules

Answer: C. Neighboring doublets

9. Which of the following connects outer microtubules to central microtubules?

- A. Axial filaments
- B. Radial spokes
- C. Ribosomes
- D. Basal granules

Answer: B. Radial spokes

10. Basal bodies contain how many microtubule triplets?

- A. 4 triplets
- B. 6 triplets
- C. 9 triplets
- D. 12 triplets

Answer: C. 9 triplets

11. Basal bodies differ from axoneme because basal bodies have:

- A. A 9 + 2 arrangement
- B. A 9 + 0 arrangement
- C. Only two central tubules
- D. No microtubules

Answer: B. A 9 + 0 arrangement

(Nine triplets with no central pair)

12. The outer membrane of cilia and flagella is actually the continuation of:

- A. Mitochondrial membrane
- B. Nuclear membrane
- C. Golgi membrane
- D. Cell membrane

Answer: D. Cell membrane

13. The internal space inside cilia and flagella is called:

- A. Matrix
- B. Cytoplasm
- C. Nucleoplasm
- D. Vacuole

Answer: A. Matrix

14. Which of the following is NOT present in the basal body?

- A. Nine triplets
- B. Central pair of microtubules
- C. Microtubule structure
- D. Base of cilia/flagella

Answer: B. Central pair of microtubules

15. The central microtubules are connected to outer doublets by:

- A. Dynein arms
- B. Radial spokes
- C. Triplet fibers
- D. Chromosomes

Answer: B. Radial spokes

Here are **important basic MCQs** created from your uploaded file “Centrioles”

50. centrioles

.

These MCQs cover **definition, structure, and functions** of centrioles.

MCQs on Centrioles (with Answers)

1. Centrioles are located:

- A. In the nucleus
- B. Free in cytoplasm
- C. Near the nucleus
- D. In chloroplasts

Answer: C. Near the nucleus

2. Centrioles are mainly involved in:

- A. ATP production
- B. Cell division
- C. Photosynthesis

D. Protein synthesis

Answer: B. Cell division

3. How many centrioles are present in a typical animal cell?

A. One

B. Two (a pair)

C. Four

D. Five

Answer: B. Two (a pair)

4. The two centrioles in a cell are arranged:

A. Parallel to each other

B. At right angles to each other

C. End-to-end

D. Randomly

Answer: B. At right angles to each other

5. Centrioles duplicate:

A. After cell division

B. Before cell division

C. During cytokinesis only

D. Only in plants

Answer: B. Before cell division

6. The structure of centrioles consists of:

A. 9 doublets

B. 9 triplets of microtubules

C. 9 + 2 arrangement

D. No microtubules

Answer: B. 9 triplets of microtubules

7. Microtubules in centrioles radiate:

A. In random directions

B. From center like spokes of a wheel

C. Parallel to each other

D. From the nucleus outward

Answer: B. From center like spokes of a wheel

8. Centrioles help in forming:

A. Ribosomes

B. Lysosomes

C. Mitotic apparatus / spindle fibers

D. Mitochondria

Answer: C. Mitotic apparatus / spindle fibers

9. The mitotic spindle helps in:

A. Separating chromosomes

B. Forming nucleolus

C. ATP production

D. Protein storage

Answer: A. Separating chromosomes

10. Centrioles also play a role in the formation of:

- A. Cilia and flagella
- B. Golgi bodies
- C. Smooth ER
- D. Nucleus

Answer: A. Cilia and flagella

11. Basal bodies and centrioles share which structural feature?

- A. 9 + 2 arrangement
- B. 9 triplets of microtubules
- C. No microtubules
- D. Presence of a central pair of tubules

Answer: B. 9 triplets of microtubules

12. Basal bodies differ from axoneme because basal bodies have:

- A. Central microtubule pair
- B. No central pair
- C. 9 + 2 pattern
- D. No microtubules

Answer: B. No central pair

13. In cell division, centrioles:

- A. Form chromosomes
- B. Destroy spindle fibers
- C. Organize spindle fibers
- D. Produce ribosomes

Answer: C. Organize spindle fibers

14. Centrioles are absent in:

- A. Animal cells
- B. Protozoa
- C. Most plant cells
- D. Fungi

Answer: C. Most plant cells (*general biological fact*)

15. The appearance of centriole microtubules resembles:

- A. A ladder
- B. A wheel with spokes
- C. A triangular prism
- D. A spiral

Answer: B. A wheel with spokes

Here are **important basic MCQs** created from your uploaded file “**Cytoplasmic Organelles: Mitochondria**”

51. Cytoplasmic Organelles, Mito...

.

These questions cover **definition, structure, and functions**.

MCQs on Mitochondria (with Answers)

1. Mitochondria are involved mainly in:

- A. Protein synthesis
- B. Energy production
- C. Photosynthesis

D. Cell division

Answer: B. Energy production

2. Mitochondria are bounded by:

A. No membrane

B. Single membrane

C. Double membrane

D. Triple membrane

Answer: C. Double membrane

3. The number of mitochondria in a cell depends on:

A. Cell shape

B. Cell size only

C. Energy requirements of the cell

D. Age of organism

Answer: C. Energy requirements of the cell

4. The inner membrane of mitochondria is:

A. Smooth

B. Flat

C. Folded into cristae

D. Absent

Answer: C. Folded into cristae

5. Cristae are:

A. Outer membrane structures

B. Finger-like projections of inner membrane

C. Cytoplasmic filaments

D. Products of ribosomes

Answer: B. Finger-like projections of inner membrane

6. The space between cristae is called:

A. Cytosol

B. Matrix

C. Stroma

D. Vacuole

Answer: B. Matrix

7. The mitochondrial matrix contains:

A. Circular DNA and ribosomes

B. Nucleus

C. RNA only

D. Golgi vesicles

Answer: A. Circular DNA and ribosomes

8. Mitochondria are often called:

A. Kitchen of the cell

B. Protein factory of the cell

C. Power house of the cell

D. Control center of the cell

Answer: C. Power house of the cell

9. ATP is produced in mitochondria mainly through:

- A. Glycolysis
- B. Krebs cycle and electron transport chain
- C. Photosynthesis
- D. Fermentation

Answer: B. Krebs cycle and electron transport chain

10. Which of the following processes occurs in mitochondria?

- A. Fatty acid oxidation
- B. Mitosis
- C. Protein packaging
- D. Photosynthesis

Answer: A. Fatty acid oxidation

11. The outer membrane of mitochondria is:

- A. Highly folded
- B. Completely absent
- C. Flat and smooth
- D. Full of ribosomes

Answer: C. Flat and smooth

12. Mitochondria can be:

- A. Only spherical
- B. Spherical to elongated in shape
- C. Only rod-shaped
- D. Flat structures

Answer: B. Spherical to elongated in shape

13. Mitochondrial DNA is:

- A. Linear
- B. Circular
- C. Absent
- D. Only found in plants

Answer: B. Circular

14. The energy currency produced by mitochondria is:

- A. NADPH
- B. GTP
- C. ATP
- D. FADH₂

Answer: C. ATP

15. Which organelle is mainly responsible for oxidation of fatty acids?

- A. Golgi apparatus
- B. Mitochondria
- C. Ribosomes
- D. Lysosomes

Answer: B. Mitochondria

Here are **important basic MCQs** created from your uploaded file “**The Miracle Workers (Introduction to Molecular Cell Biology – Enzymes & Nitrogenase)**”

52-The Miracle Workers

.

These MCQs cover **enzymes, nitrogenase, and energy conversions** exactly as presented in the file.

MCQs from *The Miracle Workers*

(Enzymes, Nitrogenase, Energy Conversion)

1. What is one of the biggest "miracles of life" described in the file?

- A. Life produces light
- B. Life performs slow and extreme-condition reactions quickly at physiological conditions
- C. Life stops chemical reactions
- D. Life works only in extreme environments

Answer: B

2. The enzyme responsible for converting nitrogen (N₂) into ammonia (NH₃) is:

- A. ATP synthase
- B. Nitrogenase
- C. Amylase
- D. Catalase

Answer: B

3. Nitrogenase converts nitrogen gas (N₂) into ammonia using:

- A. No energy
- B. Only oxygen
- C. Protons, electrons, and ATP
- D. Heat alone

Answer: C

(H⁺, e⁻, and ATP are required)

4. How many ATP molecules are consumed in the nitrogenase reaction to convert N₂ to NH₃?

- A. 2
- B. 8
- C. 12
- D. 16

Answer: D (16 ATP)

5. The industrial process for ammonia synthesis requires:

- A. Low temperature and low pressure
- B. 150–250 atm and 450–500°C
- C. No heat
- D. Only sunlight

Answer: B

6. Industrial ammonia synthesis gives a yield of:

- A. 1–5%
- B. 10–20%
- C. 50–60%
- D. 90–95%

Answer: B

7. The biological nitrogen fixation reaction produces:

- A. Only NH₃
- B. NH₃ and O₂
- C. NH₃, H₂, ADP, and Pi

D. NH_3 and CO_2

Answer: C

8. Enzymes enable life to:

- A. Slow down all reactions
- B. Stop metabolic processes
- C. Carry out reactions rapidly without extreme conditions
- D. Replace ATP

Answer: C

9. Nitrogenase activity requires:

- A. Oxygen-rich environment
- B. Heat above 500°C
- C. Anaerobic (oxygen-sensitive) conditions
- D. Chlorophyll

Answer: C (*Biological fact: nitrogenase is oxygen-sensitive*)

10. The general role of enzymes in biological systems is to:

- A. Increase activation energy
- B. Allow reactions to occur only at high temperatures
- C. Act as catalysts to speed up reactions
- D. Convert ATP into DNA

Answer: C

.
These MCQs cover **energy types, thermodynamics, and free energy concepts**.

MCQs on Energy & Thermodynamics (with Answers)

1. Energy is defined as:

- A. The mass of an object
- B. The capacity to do work
- C. The ability to reproduce
- D. The ability to create matter

Answer: B

2. Potential energy refers to:

- A. Energy of motion
- B. Heat energy
- C. Energy of position or stored in chemical bonds
- D. Energy of sound

Answer: C

3. Kinetic energy includes:

- A. Chemical bond energy
- B. Stored fat energy
- C. Energy of motion, heat, light, electricity
- D. Water pressure

Answer: C

4. Energy conversion efficiency in biological systems is:

- A. 100%
- B. 90%
- C. 50%
- D. Less than 100%

Answer: D

5. The First Law of Thermodynamics states that:

- A. Energy increases with entropy
- B. Energy can be created
- C. Energy cannot be created or destroyed
- D. Energy is always lost

Answer: C

6. The Second Law of Thermodynamics states that:

- A. Energy becomes more organized
- B. Free energy increases over time
- C. Unusable energy (entropy) increases
- D. Entropy remains constant

Answer: C

7. According to the Second Law, the amount of energy available to do work (free energy):

- A. Always increases
- B. Always decreases
- C. Remains unchanged
- D. Becomes zero

Answer: B

8. In the equation $H = G + TS$, the term “H” represents:

- A. Entropy
- B. Free energy
- C. Enthalpy (total energy)
- D. Temperature

Answer: C

9. In the equation $G = H - TS$, “G” refers to:

- A. Unusable energy
- B. Entropy
- C. Free energy
- D. Heat

Answer: C

10. In the expression $H = G + TS$, “S” stands for:

- A. Speed
- B. Stored energy
- C. Entropy
- D. Stability

Answer: C

11. “T” in the formula $H = G + TS$ represents:

- A. Time
- B. Temperature (absolute)
- C. Tension

D. Total mass

Answer: B

12. As entropy increases, the amount of usable energy:

- A. Increases
- B. Decreases
- C. Becomes infinite
- D. Remains constant

Answer: B

13. Which of the following is an example of kinetic energy?

- A. Chemical bonds
- B. A stretched bow
- C. Moving car or flowing water
- D. Stored food energy

Answer: C

14. A molecule of glucose contains:

- A. Only kinetic energy
- B. Only thermal energy
- C. Potential energy stored in chemical bonds
- D. No usable energy

Answer: C

15. The unusable portion of energy in a system is associated with:

- A. Mass
- B. Entropy
- C. Potential energy
- D. Structure of molecules

Answer: B

These MCQs cover ΔG , ΔH , ΔS , thermodynamics equations, and reaction interpretation, exactly as presented.

MCQs on Free Energy (ΔG)

(with Answers)

1. The equation that relates total, usable, and unusable energy is:

- A. $E = mc^2$
- B. $H = G + TS$
- C. $G = H \times S$
- D. $\Delta G = T - HS$

Answer: B

2. In the formula $H = G + TS$, “G” represents:

- A. Entropy
- B. Temperature
- C. Free energy
- D. Heat of reaction

Answer: C

3. The term “S” in the equation $H = G + TS$ stands for:

- A. Solubility
- B. Surface tension
- C. Entropy
- D. Speed

Answer: C

4. Free energy is calculated using the equation:

- A. $G = H + TS$
- B. $G = H - TS$
- C. $G = TS - H$
- D. $G = H \times S$

Answer: B

5. Which values cannot be measured directly?

- A. ΔG
- B. Absolute H, G, or S
- C. Temperature
- D. Entropy change

Answer: B

6. The correct expression for change in free energy is:

- A. $\Delta G = \Delta H + T\Delta S$
- B. $\Delta G = T - \Delta S$
- C. $\Delta G = \Delta H - T\Delta S$
- D. $\Delta G = H - S$

Answer: C

7. If ΔG is negative (–ve), it means the reaction:

- A. Requires energy
- B. Releases energy
- C. Stops immediately
- D. Is impossible

Answer: B

8. If ΔG is positive (+ve), the reaction:

- A. Occurs spontaneously
- B. Releases energy
- C. Requires free energy input
- D. Results in no energy change

Answer: C

9. In living systems, ΔG can depend significantly on:

- A. pH only
- B. Light intensity
- C. Changes in entropy (ΔS)
- D. Molecular weight

Answer: C

10. During hydrolysis, entropy (ΔS) is typically:

- A. Zero
- B. Negative
- C. Positive
- D. Undefined

Answer: C

****11. For the combustion example given:**



The ΔG of the reaction is:

- A. +300
- B. -300
- C. 0
- D. -100

Answer: B (-300)

12. ΔG (reaction) is calculated as:

- A. $G(\text{reactants}) - G(\text{products})$
- B. $G(\text{products}) - G(\text{reactants})$
- C. $H(\text{products}) - H(\text{reactants})$
- D. $S(\text{products}) - S(\text{reactants})$

Answer: B

13. Water molecules next to a hydrophobic molecule:

- A. Become more disordered
- B. Are restricted in forming hydrogen bonds
- C. Form stronger ionic bonds
- D. Gain kinetic energy

Answer: B

14. Hydrogen bonding between water molecules:

- A. Increases energy state
- B. Reduces energy state
- C. Has no effect on energy
- D. Stops spontaneous reactions

Answer: B

15. Which equation represents the relationship between enthalpy and free energy change?

- A. $\Delta G = \Delta H - T\Delta S$
- B. $\Delta G = TS - H$
- C. $H = G - TS$
- D. $S = H - G$

Answer: A

MCQs on Exergonic & Endergonic Reactions

(With Answers – Based on the file)

1. Endergonic reactions have a ΔG that is:

- A. Zero
- B. Negative
- C. Positive
- D. Always constant

Answer: C

2. Endergonic reactions require:

- A. No energy input
- B. Release of energy
- C. Free energy to proceed
- D. Very low temperature

Answer: C

3. Exergonic reactions have a ΔG that is:

- A. Positive
- B. Negative
- C. Zero
- D. Always high

Answer: B

4. Exergonic reactions:

- A. Take in energy
- B. Require external energy to proceed
- C. Release free energy
- D. Never reach equilibrium

Answer: C

5. At equilibrium, the value of ΔG is:

- A. +1
- B. -1
- C. 0
- D. Undefined

Answer: C

6. At equilibrium, which statement is TRUE?

- A. No molecule changes state
- B. Amounts of reactants and products remain constant
- C. Only forward reaction occurs
- D. Only reverse reaction occurs

Answer: B

7. A reaction will proceed spontaneously only if ΔG is:

- A. Positive
- B. Negative
- C. Zero
- D. Undefined

Answer: B

8. In a reversible reaction at equilibrium:

- A. Forward reaction stops
- B. Reverse reaction stops
- C. Forward and reverse reactions occur at same rate
- D. Both reactions stop completely

Answer: C

9. The equilibrium point for exergonic reactions lies:

- A. Toward reactants
- B. In the middle
- C. Toward completion (more products formed)
- D. Toward equal reactants and products

Answer: C

10. For the example in the file (@0.02M, 25°C, pH 7), the ΔG value is:

- A. +1.7 kcal/mol
- B. -1.7 kcal/mol
- C. Zero
- D. -7 kcal/mol

Answer: B

11. If ΔG is positive, the reaction:

- A. Occurs spontaneously
- B. Requires free energy input
- C. Releases heat
- D. Never reaches equilibrium

Answer: B

12. Which of the following is TRUE about exergonic reactions?

- A. $\Delta G > 0$
- B. Products have higher free energy than reactants
- C. They release free energy
- D. They cannot reach equilibrium

Answer: C

13. In an endergonic reaction, the products have:

- A. Lower free energy than reactants
- B. Higher free energy than reactants
- C. Equal energy to reactants
- D. Undefined energy

Answer: B

14. At equilibrium, molecules:

- A. Stop reacting entirely
- B. Continue to change states but overall amounts stay constant
- C. Increase in number
- D. Disappear

Answer: B

15. ΔG of a reaction is calculated from:

- A. Reactant concentration only
- B. Product concentration only
- C. Difference in free energy of products and reactants
- D. Temperature alone

Answer: C

MCQs from “ATP Hydrolysis”

1. ATP Hydrolysis — Energy

Q1. Hydrolysis of ATP releases approximately how much free energy?

- A. -2 kcal/mol
- B. -7 kcal/mol
- C. -12 kcal/mol
- D. -20 kcal/mol

Correct Answer: C. -12 kcal/mol

2. ATP as Energy Currency

Q2. ATP is commonly referred to as the cell's:

- A. Genetic currency
- B. Energy currency
- C. Information molecule
- D. Signaling molecule

Correct Answer: B. Energy currency

3. Coupling Reactions

Q3. ATP hydrolysis is often used to:

- A. Increase entropy
- B. Lower pH
- C. Drive energetically unfavorable reactions
- D. Produce heat only

Correct Answer: C. Drive energetically unfavorable reactions

4. Speed of Reaction

Q4. According to the slide, degradation of a 600–poly-arginine chain takes how long without enzymes?

- A. 1 second
- B. 1 minute
- C. 1 day
- D. 7 years

Correct Answer: D. 7 years

Q5. Carboxypeptidase catalyzes the same degradation reaction in approximately:

- A. 7 years
- B. 1 day
- C. Half a second
- D. Half a minute

Correct Answer: C. Half a second

5. Activation Energy

Q6. Reactants need to overcome which barrier to convert into products?

- A. Structural barrier
- B. Concentration barrier
- C. Activation energy (E_a)
- D. Thermal equilibrium barrier

Correct Answer: C. Activation energy (E_a)

Q7. Adding a small amount of energy to a reaction allows reactants to:

- A. Decrease temperature
- B. Cross the activation energy barrier
- C. Increase ΔG
- D. Stop the reaction

Correct Answer: B. Cross the activation energy barrier

6. Thermodynamics

Q8. Which statement is TRUE regarding ΔG and reaction speed?

- A. ΔG determines how fast a reaction occurs

- B. ΔG is independent of reaction speed
- C. A more negative ΔG means a faster reaction
- D. Reaction speed determines ΔG

Correct Answer: B. ΔG is independent of reaction speed

7. Reaction Example

Q9. The example $\text{CH}_4 + \text{O}_2$ in the slide illustrates a reaction that:

- A. Does not need activation energy
- B. Cannot release energy
- C. Requires activation energy to proceed
- D. Is endergonic only

Correct Answer: C. Requires activation energy to proceed

Basic MCQs from ENZYMES (Based on Slides)

1. Enzymes as Biological Catalysts

Q1. Enzymes are classified as:

- A. Carbohydrates
- B. Lipids
- C. Proteins (mostly)
- D. Steroids

Correct Answer: C. Proteins (mostly)

(Ribozymes are exceptions — they are RNA.)

2. Free Energy (ΔG)

Q2. According to the slide, the more negative (–ve) the ΔG of a reaction:

- A. The slower the reaction proceeds
- B. The reaction proceeds further toward completion
- C. The reaction stops early
- D. ΔG determines the reaction rate

Correct Answer: B. The reaction proceeds further toward completion

Q3. ΔG tells us:

- A. How fast a reaction occurs
- B. The exact kinetic energy of molecules
- C. The extent to which a reaction goes to completion
- D. The activation energy of the reaction

Correct Answer: C. The extent to which a reaction goes to completion

3. Reaction Rate & ΔG

Q4. Which of the following is TRUE?

- A. ΔG determines the reaction rate
- B. ΔG does not tell us the rate of a reaction
- C. A high ΔG makes reactions faster
- D. All reactions with $\Delta G = 0$ are fast

Correct Answer: B. ΔG does not tell us the rate of a reaction

4. Activation Energy (E_a)

Q5. At normal temperatures, very few molecules have enough kinetic energy to react because:

- A. ΔG is too high
- B. Enzymes are absent
- C. The activation energy barrier is high
- D. They are unstable

Correct Answer: C. The activation energy barrier is high

Q6. Enzymes speed up reactions mainly by:

- A. Increasing ΔG
- B. Increasing temperature
- C. Lowering the activation energy
- D. Adding more reactants

Correct Answer: C. Lowering the activation energy

5. Types of Catalysts

Q7. Ribozymes are unique because they are:

- A. DNA molecules that act as enzymes
- B. Lipids that speed up reactions
- C. RNA molecules with catalytic activity
- D. Proteins that bind RNA

Correct Answer: C. RNA molecules with catalytic activity

Basic MCQs from Enzyme Mechanisms

1. Enzyme Specificity

Q1. Reactants that bind to an enzyme are known as:

- A. Products
- B. Substrates
- C. Cofactors
- D. Inhibitors

Correct Answer: B. Substrates

Q2. The active site of an enzyme consists of:

- A. Only catalytic site
- B. Only binding site
- C. Catalytic site + binding site
- D. Regulatory site only

Correct Answer: C. Catalytic site + binding site

Q3. The specificity of enzyme–substrate binding depends mainly on:

- A. pH only
- B. Concentration of substrate
- C. Three-dimensional shape and environment of the active site
- D. Temperature only

Correct Answer: C. Three-dimensional shape and environment of the active site

2. Enzymes and Energy Barriers

Q4. Enzymes lower the activation energy for:

- A. Only forward reactions
- B. Only reverse reactions
- C. Both forward and reverse reactions
- D. Only exergonic reactions

Correct Answer: C. Both forward and reverse reactions

Q5. When enzymes lower activation energy, what happens to equilibrium and ΔG ?

- A. Both ΔG and equilibrium change
- B. ΔG changes but equilibrium does not
- C. Equilibrium changes but ΔG does not
- D. Neither ΔG nor equilibrium changes

Correct Answer: D. Neither ΔG nor equilibrium changes

3. Enzyme Mechanisms – Induced Fit

Q6. According to the induced-fit model, when a substrate binds to the enzyme, the enzyme:

- A. Remains rigid
- B. Breaks down immediately
- C. Changes shape to facilitate catalysis
- D. Becomes inactive

Correct Answer: C. Changes shape to facilitate catalysis

Q7. Induced fit modifies the active site in order to:

- A. Reduce enzyme concentration
- B. Expose or align catalytic regions
- C. Destroy substrate
- D. Prevent product formation

Correct Answer: B. Expose or align catalytic regions

4. Water Exclusion & Catalysis Example

Q8. Water exclusion at the active site helps transfer phosphate from ATP to glucose by:

- A. Increasing ATP hydrolysis into ADP
- B. Preventing ATP hydrolysis into ADP
- C. Breaking down glucose
- D. Removing enzymes from reaction

Correct Answer: B. Preventing ATP hydrolysis into ADP

Q9. The exclusion of water during induced fit is important because it:

- A. Inactivates ATP
- B. Blocks substrate entry
- C. Enhances catalytic efficiency
- D. Stops enzyme activity

Correct Answer: C. Enhances catalytic efficiency

Basic MCQs from ENZYME HELPERS

1. Cofactors

Q1. Cofactors are best described as:

- A. Organic molecules that permanently bind enzymes
- B. Inorganic ions essential for enzyme function
- C. Proteins required for enzyme folding
- D. Lipids bound to enzymes

Correct Answer: B. Inorganic ions essential for enzyme function

Q2. Which of the following are examples of enzyme cofactors?

- A. Na, K, Ca
- B. Cu, Zn, Fe
- C. C, H, O
- D. ATP, ADP

Correct Answer: B. Cu, Zn, Fe

2. Coenzymes

Q3. Coenzymes are:

- A. Permanent inorganic groups
- B. Carbon-containing molecules that bind temporarily
- C. Metals required for catalysis
- D. Non-functional enzyme fragments

Correct Answer: B. Carbon-containing molecules that bind temporarily

Q4. Coenzymes during enzymatic reactions are:

- A. Never changed
- B. Chemically changed temporarily
- C. Permanently destroyed
- D. Unaffected and not involved

Correct Answer: B. Chemically changed temporarily

3. Prosthetic Groups

Q5. A prosthetic group is:

- A. A temporary helper molecule
- B. A non-protein component permanently attached to an enzyme
- C. A protein subunit of the enzyme
- D. A degraded form of coenzyme

Correct Answer: B. A non-protein component permanently attached to an enzyme

Q6. Which of the following is an example of a prosthetic group?

- A. ATP
- B. Heme group
- C. Glucose
- D. Zinc ion only

Correct Answer: B. Heme group

4. Apoenzyme vs. Holoenzyme

Q7. An enzyme without its prosthetic group is called:

- A. Holoenzyme
- B. Isoenzyme
- C. Apoenzyme
- D. Coenzyme

Correct Answer: C. Apoenzyme

Q8. An enzyme with its prosthetic group bound is called:

- A. Holoenzyme
- B. Apoenzyme
- C. Substrate
- D. Ribosome

Correct Answer: A. Holoenzyme

Basic MCQs from ENZYME REGULATION

1. General Enzyme Regulation

Q1. Enzyme activity can be influenced by factors that:

- A. Only activate enzymes
- B. Only inhibit enzymes
- C. Can either activate or inhibit enzymes
- D. Have no effect on enzymes

Correct Answer: C. Can either activate or inhibit enzymes

2. Types of Inhibitors

Q2. Irreversible inhibitors usually inactivate enzymes by:

- A. Competing with the substrate
- B. Modifying the active site permanently
- C. Binding to the substrate
- D. Binding at an allosteric site temporarily

Correct Answer: B. Modifying the active site permanently

Q3. Competitive inhibitors resemble the substrate and:

- A. Bind at a site different from the active site
- B. Allow catalysis to occur faster
- C. Bind the active site but do not undergo reaction
- D. Permanently destroy the enzyme

Correct Answer: C. Bind the active site but do not undergo reaction

Q4. Competitive inhibition is:

- A. Always permanent
- B. Irreversible
- C. Reversible
- D. Caused by changes in temperature only

Correct Answer: C. Reversible

Q5. Noncompetitive inhibitors affect enzymes by:

- A. Competing for the active site
- B. Permanently modifying the enzyme
- C. Binding at a non-active site and causing conformational change
- D. Increasing substrate concentration

Correct Answer: C. Binding at a non-active site and causing conformational change

3. Feedback Inhibition

Q6. Feedback (end-product) inhibition typically occurs in pathways such as:

- A. Glycolysis
- B. Citric Acid Cycle
- C. Photosynthesis
- D. DNA replication

Correct Answer: B. Citric Acid Cycle

4. Substrate Concentration

Q7. Increasing substrate concentration generally affects enzyme activity by:

- A. Decreasing product formation
- B. Not affecting reaction rate
- C. Increasing reaction rate until saturation
- D. Denaturing the enzyme

Correct Answer: C. Increasing reaction rate until saturation

5. Physical Factors

Q8. Physical factors such as fever influence enzyme activity by:

- A. Having no effect
- B. Increasing enzyme synthesis only
- C. Altering enzyme structure and function
- D. Increasing ΔG of reaction

Correct Answer: C. Altering enzyme structure and function

Basic MCQs from *The Nucleus*

1. Structure of the Nucleus

Q1. The nucleus is the:

- A. Energy-producing organelle
- B. Control center of the cell
- C. Site of protein synthesis
- D. Site of lipid production

Correct Answer: B. Control center of the cell

Q2. Nuclear EMs shown in the slide refer to:

- A. Electron micrographs of mitochondria
- B. Electron micrographs of nuclei
- C. Light microscopy of chromosomes
- D. Fluorescent images of ribosomes

Correct Answer: B. Electron micrographs of nuclei

2. Chromatin

Q3. Chromatin is composed of:

- A. DNA only
- B. RNA only
- C. DNA and proteins
- D. Lipids and carbohydrates

Correct Answer: C. DNA and proteins

Q4. Euchromatin is generally:

- A. Highly condensed and inactive
- B. Lightly packed and transcriptionally active
- C. Only found in bacteria
- D. Visible only during cell division

Correct Answer: B. Lightly packed and transcriptionally active

Q5. Heterochromatin is characterized by:

- A. Loosely packed DNA
- B. High transcriptional activity
- C. Densely packed and inactive regions of chromatin
- D. Being present only in gametes

Correct Answer: C. Densely packed and inactive regions of chromatin

3. Chromosomes

Q6. Chromosomes are formed when chromatin:

- A. Becomes highly condensed during cell division
- B. Is completely degraded
- C. Separates from the nucleus
- D. Turns into ribosomes

Correct Answer: A. Becomes highly condensed during cell division

Q7. During interphase, nuclear DNA mainly exists as:

- A. Chromosomes only
- B. Euchromatin and heterochromatin
- C. RNA bundles
- D. Centrosomes

Correct Answer: B. Euchromatin and heterochromatin

Basic MCQs from *The Nuclear Envelope*

1. Nuclear Envelope Basics

Q1. The nuclear envelope is best described as:

- A. A single membrane separating nucleus and cytoplasm
- B. A double membrane separating nucleus and cytoplasm
- C. A protein coat around the nucleus
- D. A membrane formed by ribosomes

Correct Answer: B. A double membrane separating nucleus and cytoplasm

Q2. The nuclear envelope functions to:

- A. Join mitochondria together
- B. Separate nucleus from cytoplasm
- C. Produce proteins
- D. Store lipids

Correct Answer: B. Separate nucleus from cytoplasm

2. Structure of Nuclear Membrane

Q3. Which statement is TRUE about the outer nuclear membrane?

- A. It is discontinuous and isolated
- B. It is fused with the Golgi apparatus

C. It is continuous with the endoplasmic reticulum

D. It is thicker than the inner membrane

Correct Answer: C. It is continuous with the endoplasmic reticulum

Q4. The outer and inner nuclear membranes are:

A. Completely fused

B. Separate from each other

C. Identical in function

D. Connected by ribosomes

Correct Answer: B. Separate from each other

3. Nuclear Pores

Q5. Nuclear pores are located:

A. On the nucleolus only

B. On the outer membrane only

C. Near or on the nuclear envelope

D. Only in dividing cells

Correct Answer: C. Near or on the nuclear envelope

Q6. The number of nuclear pores:

A. Is always fixed in all cells

B. Ranges from few to thousands

C. Does not exceed 10 in any cell

D. Remains constant throughout life

Correct Answer: B. Ranges from few to thousands

Q7. An example of a high number of nuclear pores given in the slide is:

A. Muscle cell (500 pores)

B. Egg cell (3000 pores)

C. Neuron (100 pores)

D. Liver cell (50 pores)

Correct Answer: B. Egg cell (3000 pores)

Q8. Nuclear pores are composed of:

A. Carbohydrates only

B. Globular and filamentous proteins

C. Lipids only

D. RNA molecules

Correct Answer: B. Globular and filamentous proteins

Q9. The main function of nuclear pores is to:

A. Control transport between nucleus and cytoplasm

B. Store chromosomes

C. Produce ribosomes

D. Strengthen the nuclear membrane

Correct Answer: A. Control transport between nucleus and cytoplasm

Basic MCQs from *The Nucleoplasm*

1. Definition of Nucleoplasm

Q1. Nucleoplasm is the:

- A. Fluid inside mitochondria
- B. Soluble portion of the nucleus enclosed by the nuclear envelope
- C. Outer part of the nucleus
- D. Solid matrix of cytoplasm

Correct Answer: B. Soluble portion of the nucleus enclosed by the nuclear envelope

2. Composition of Nucleoplasm

Q2. The nucleoplasm contains all of the following EXCEPT:

- A. Nucleotides
- B. RNA molecules
- C. Proteins
- D. Ribosomes for translation

Correct Answer: D. Ribosomes for translation

Q3. Enzymes present in the nucleoplasm are mainly involved in:

- A. Lipid synthesis
- B. Cellular respiration
- C. DNA replication and transcription
- D. Protein folding

Correct Answer: C. DNA replication and transcription

Q4. The ground substance of nucleoplasm includes:

- A. Only water
- B. Only phospholipids
- C. Nucleotides, RNA, proteins, and enzymes
- D. Only DNA

Correct Answer: C. Nucleotides, RNA, proteins, and enzymes

3. Activities in the Nucleoplasm

Q5. Which of the following activities takes place in the nucleoplasm?

- A. Translation
- B. DNA replication
- C. ATP production
- D. Protein secretion

Correct Answer: B. DNA replication

Q6. Transcription occurs in the:

- A. Cytoplasm
- B. Ribosomes
- C. Nucleoplasm
- D. Cell membrane

Correct Answer: C. Nucleoplasm

Q7. RNA processing is an activity that occurs in the:

- A. Mitochondria
- B. Nucleoplasm
- C. Endoplasmic reticulum
- D. Golgi apparatus

Correct Answer: B. Nucleoplasm

Basic MCQs from *The Nucleolus*

1. Definition & Occurrence

Q1. The nucleolus is a:

- A. Membrane-bound organelle
- B. Non-membrane bound structure inside the nucleus
- C. Part of the cytoplasm
- D. Membrane of the nucleus

Correct Answer: B. Non-membrane bound structure inside the nucleus

Q2. The nucleolus is present in:

- A. Dividing cells only
- B. Non-dividing cells
- C. Prokaryotic cells only
- D. Mitochondria

Correct Answer: B. Non-dividing cells

Q3. Most cells contain how many nucleoli?

- A. One
- B. Two or three
- C. Five
- D. Dozens

Correct Answer: B. Two or three

Q4. Some cells may contain thousands of nucleoli, such as:

- A. Red blood cells
- B. Plant root cells
- C. Amphibian egg cells
- D. Neurons

Correct Answer: C. Amphibian egg cells

2. Structure of Nucleolus

Q5. The nucleolus contains which two regions?

- A. Matrix and membrane region
- B. Granular and fibrillar region
- C. Vesicular and tubular region
- D. Dense and loose region

Correct Answer: B. Granular and fibrillar region

3. Functions of the Nucleolus

Q6. The main function of the nucleolus is:

- A. DNA replication
- B. Protein packaging
- C. rRNA synthesis
- D. Lipid synthesis

Correct Answer: C. rRNA synthesis

Q7. Ribosome assembly takes place in the:

- A. Cytoplasm only
- B. Nuclear envelope

C. Endoplasmic reticulum

D. Nucleolus

Correct Answer: D. Nucleolus

Q8. Which of the following does the nucleolus directly produce?

A. mRNA

B. rRNA

C. tRNA

D. DNA polymerase

Correct Answer: B. rRNA

(All questions are based ONLY on the content shown in the slides.)

Basic MCQs from *Chromosomes & Chromatin Material*

1. Genetic Material

Q1. All animals have which molecule as their genetic material?

A. RNA

B. DNA

C. Protein

D. Lipids

Correct Answer: B. DNA

Q2. A gene is best defined as:

A. A segment of RNA that makes proteins

B. A segment of DNA that encodes a specific protein

C. Any random sequence of nucleotides

D. A structure made only of histones

Correct Answer: B. A segment of DNA that encodes a specific protein

Q3. Besides proteins, genes also encode which molecules?

A. mRNA and lipids

B. Carbohydrates and enzymes

C. tRNA and rRNA

D. Centrosomes and spindle fibers

Correct Answer: C. tRNA and rRNA

2. Chromatin Material

Q4. Chromatin is composed of:

A. DNA only

B. RNA only

C. DNA and histone proteins

D. Lipids and carbohydrates

Correct Answer: C. DNA and histone proteins

Q5. The network of DNA packed with histone proteins in the nucleus is called:

A. Chromosomes

B. Cytoskeleton

C. Chromatin material

D. Ribosomes

Correct Answer: C. Chromatin material

3. Chromosomes

Q6. Chromosomes form when chromatin material:

A. Becomes loosely packed

B. Degrades into fragments

C. Condenses during cell division

D. Moves out of the nucleus

Correct Answer: C. Condenses during cell division

Q7. The number of chromosomes in a cell equals the:

A. Number of DNA molecules only

B. Number of chromatin fibers

C. Number of nucleoli

D. Number of ribosomes

Correct Answer: B. Number of chromatin fibers

Q8. Every animal species has:

A. A variable number of chromosomes

B. A fixed number of chromosomes

C. Changing chromosome numbers during life

D. No chromosomes in somatic cells

Correct Answer: B. A fixed number of chromosomes

Here are **important basic MCQs** created directly from your uploaded file “66_ Discovery of DNA.pptx”

66_ Discovery of DNA

(All MCQs are based ONLY on the content shown in the slides.)

Basic MCQs from *Discovery of DNA*

1. Early Discoveries

Q1. Who first discovered “nuclein” in 1869?

A. Watson

B. Avery

C. Friedrich Miescher

D. Hershey

Correct Answer: C. Friedrich Miescher

Q2. Albrecht Kossel is known for:

A. Discovering the double helix

B. Isolating nucleic acids and identifying the five nitrogenous bases

C. Demonstrating DNA transformation

D. Conducting bacteriophage experiments

Correct Answer: B. Isolating nucleic acids and identifying the five nitrogenous bases

Q3. Which nitrogenous bases did Kossel isolate?

A. A, T, C only

B. A, T, G, C & U

C. A, G, C only

D. U, G only

Correct Answer: B. A, T, G, C & U

2. Concepts of Hereditary Molecule

Q4. In 1927, Koltsov proposed that hereditary traits were carried by a molecule with:

A. A single strand

B. No replication ability

C. Two mirror strands replicating semi-conservatively

D. Only protein components

Correct Answer: C. Two mirror strands replicating semi-conservatively

3. Key Experiments in DNA Discovery

Q5. The transformation experiment performed in 1928 was conducted by:

A. Watson

B. Avery

C. Frederick Griffith

D. Hershey

Correct Answer: C. Frederick Griffith

Q6. The experiment by Avery, MacLeod, and McCarty (1944) demonstrated that the transforming principle is:

A. Protein

B. Lipid

C. RNA

D. DNA

Correct Answer: D. DNA

Q7. The Hershey–Chase experiment of 1952 used:

A. Mice and bacteria

B. Bacteriophages

C. Plants

D. Radioactive proteins only

Correct Answer: B. Bacteriophages

4. Structure of DNA

Q8. “Photo 51,” crucial for identifying DNA structure, was produced by:

A. Watson

B. Avery

C. Rosalind Franklin

D. Griffith

Correct Answer: C. Rosalind Franklin

Q9. Maurice Wilkins contributed to DNA discovery mainly by:

A. Conducting transformation experiments

B. Working on X-ray diffraction images

C. Creating bacteriophage models

D. Isolating nitrogenous bases

Correct Answer: B. Working on X-ray diffraction images

Q10. Watson & Crick proposed the double helix structure in:

A. 1944

- B. 1952
- C. 1953
- D. 1960

Correct Answer: C. 1953

Basic MCQs from *Nucleoside & Nucleotide*

1. Definitions

Q1. A molecule that contains a nitrogenous base, a sugar, and a phosphate group is called a:

- A. Nucleoside
- B. Nucleotide
- C. Amino acid
- D. Fatty acid

Correct Answer: B. Nucleotide

Q2. A molecule that lacks the phosphate group but contains a nitrogenous base and sugar is called a:

- A. Nucleotide
- B. Polymer
- C. Nucleoside
- D. Protein

Correct Answer: C. Nucleoside

2. Relationship Between Nucleotide & Nucleoside

Q3. Which formula is correct according to the slide?

- A. Nucleotide = Nucleoside – Phosphoric acid
- B. Nucleoside = Nucleotide + Phosphoric acid
- C. Nucleotide = Nucleoside + Phosphoric acid
- D. Nucleoside = Sugar + Phosphoric acid

Correct Answer: C. Nucleotide = Nucleoside + Phosphoric acid

Q4. Removing the phosphate group from a nucleotide results in a:

- A. Base
- B. Sugar
- C. Nucleoside
- D. Protein

Correct Answer: C. Nucleoside

3. Components

Q5. How many major components does a nucleotide contain?

- A. One
- B. Two
- C. Three
- D. Four

Correct Answer: C. Three

(Nitrogenous base, sugar, phosphate group)

Basic MCQs from *Types of Deoxyribonucleotides*

1. Number of Deoxyribonucleotides

Q1. How many types of deoxyribonucleotides are found in DNA?

- A. Two
- B. Three
- C. Four
- D. Five

Correct Answer: C. Four

2. Role in DNA

Q2. Deoxyribonucleotides serve as the:

- A. Structural units of RNA
- B. Structural units of DNA
- C. Building blocks of proteins
- D. Components of lipids

Correct Answer: B. Structural units of DNA

3. Basic Understanding

Q3. Deoxyribonucleotides differ from ribonucleotides mainly by:

- A. Presence of ribose sugar
- B. Absence of phosphate
- C. Having deoxyribose sugar
- D. Having uracil instead of thymine

Correct Answer: C. Having deoxyribose sugar

4. DNA Composition

Q4. The sequence of deoxyribonucleotides determines:

- A. The pH of the cell
- B. The structure and function of DNA
- C. Lipid metabolism
- D. ATP synthesis

Correct Answer: B. The structure and function of DNA

5. General Knowledge Check

Q5. Which of the following is TRUE about deoxyribonucleotides?

- A. They are not involved in genetic information storage
- B. They form proteins during translation
- C. They pair specifically to form the DNA double helix
- D. They do not contain nitrogenous bases

Correct Answer: C. They pair specifically to form the DNA double helix

(All questions are based ONLY on the content in the slides.)

Basic MCQs from *Types of Ribonucleotides*

1. Number of Ribonucleotides

Q1. How many major types of ribonucleotides are present in RNA?

- A. Two
- B. Three
- C. Four
- D. Five

Correct Answer: C. Four

2. Basis of Classification

Q2. The main difference among the four types of ribonucleotides is due to the:

- A. Type of sugar
- B. Type of nitrogenous base
- C. Number of phosphate groups
- D. Position of ribose

Correct Answer: B. Type of nitrogenous base

3. Poly-Ribonucleotide Formation

Q3. Ribonucleotides join together to form:

- A. Polypeptides
- B. Polysaccharides
- C. Poly-ribonucleotides
- D. Lipid bilayers

Correct Answer: C. Poly-ribonucleotides

4. Role in RNA

Q4. Ribonucleotides are the basic building blocks of:

- A. DNA
- B. RNA
- C. Proteins
- D. Carbohydrates

Correct Answer: B. RNA

5. Structure Check

Q5. Ribonucleotides contain which sugar?

- A. Deoxyribose
- B. Fructose
- C. Ribose
- D. Glucose

Correct Answer: C. Ribose

Basic MCQs from *Chemical Composition of DNA*

1. Components of DNA

Q1. DNA is a polymer of:

- A. Ribonucleotides
- B. Amino acids

C. Deoxyribonucleotides

D. Fatty acids

Correct Answer: C. Deoxyribonucleotides

Q2. A deoxyribonucleotide consists of how many components?

A. One

B. Two

C. Three

D. Four

Correct Answer: C. Three

(Deoxyribose, nitrogenous base, phosphoric acid)

2. Deoxyribose

Q3. Deoxyribose is what type of sugar?

A. Hexose

B. Disaccharide

C. Pentose derivative

D. Monosaccharide without carbon

Correct Answer: C. Pentose derivative

3. Nitrogenous Bases

Q4. Nitrogenous bases in DNA are divided into:

A. Proteins and lipids

B. Purines and pyrimidines

C. Sugars and acids

D. Phosphates and esters

Correct Answer: B. Purines and pyrimidines

Q5. Purines are:

A. Single-ring bases

B. Double-ring bases

C. Sugars

D. Phosphates

Correct Answer: B. Double-ring bases

Q6. Pyrimidines are:

A. Double-ring bases

B. Sugars

C. Single-ring bases

D. Proteins

Correct Answer: C. Single-ring bases

4. Phosphoric Acid

Q7. Phosphoric acid in DNA functions to:

A. Form peptide bonds

B. Connect nucleotides via phosphodiester bonds

C. Create hydrogen bonds

D. Bind ribosomes

Correct Answer: B. Connect nucleotides via phosphodiester bonds

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MCQs: Chemical Composition of RNA

1. RNA is a polymer of which basic units?

- A. Deoxyribonucleotides
- B. Ribonucleotides
- C. Amino acids
- D. Fatty acids

Correct Answer: B

2. Each ribonucleotide of RNA contains how many main components?

- A. One
- B. Two
- C. Three
- D. Four

Correct Answer: C

3. Which sugar is present in RNA?

- A. Glucose
- B. Ribose
- C. Deoxyribose
- D. Fructose

Correct Answer: B

4. Ribose sugar found in RNA contains how many carbon atoms?

- A. 3
- B. 4
- C. 5
- D. 6

Correct Answer: C

(Because Ribose is a pentose sugar.)

5. Which of the following is *not* a component of a ribonucleotide?

- A. Ribose sugar
- B. Nitrogenous base
- C. Phosphoric acid
- D. Deoxyribose sugar

Correct Answer: D

6. Nitrogenous bases in RNA include which of the following?

- A. Adenine, Guanine, Thymine, Cytosine
- B. Adenine, Guanine, Uracil, Cytosine
- C. Adenine, Thymine, Uracil, Cytosine
- D. Guanine, Thymine, Cytosine, Uracil

Correct Answer: B

7. Phosphoric acid in RNA is part of which structure?

- A. Nitrogenous base
- B. Sugar-phosphate backbone
- C. Ribose ring
- D. Base pairing region

Correct Answer: B

8. A ribonucleotide in RNA consists of ribose sugar, nitrogenous base, and

- _____.
- A. Water molecule
 - B. Phosphoric acid
 - C. Fatty acid
 - D. Protein

Correct Answer: B

9. Which of the following best describes RNA?

- A. Double-stranded polymer of deoxyribonucleotides
- B. Single-stranded polymer of ribonucleotides
- C. Protein polymer
- D. Carbohydrate chain

Correct Answer: B

10. The nitrogenous bases in RNA attach to which carbon of the ribose sugar?

- A. 1' carbon
- B. 2' carbon
- C. 3' carbon
- D. 5' carbon

Correct Answer: A

(Standard biochemical fact based on structure.)

MCQs: How Deoxyribonucleotides Join in DNA

1. Successive nucleotides in DNA are joined together through which type of linkage?

- A. Hydrogen bond
- B. Glycosidic bond
- C. Phospho-diester linkage
- D. Peptide bond

Correct Answer: C

2. Phosphodiester linkages form the backbone of which molecule?

- A. Proteins
- B. DNA
- C. Lipids
- D. Polysaccharides

Correct Answer: B

3. A phosphodiester bond connects which parts of adjacent nucleotides?

- A. Base to base
- B. Sugar to base
- C. Sugar of one nucleotide to the phosphate of the next
- D. Phosphate to amino acid

Correct Answer: C

4. Which two components are directly involved in forming a phosphodiester linkage?

- A. Nitrogenous base and hydrogen atom
- B. Phosphate group and pentose sugar
- C. Ribose and deoxyribose
- D. Two nitrogenous bases

Correct Answer: B

5. Phosphodiester linkages in DNA result in the formation of what structural feature?

- A. Double helix twist
- B. Sugar-phosphate backbone
- C. Base pair formation
- D. Protein synthesis

Correct Answer: B

6. What type of reaction typically forms phosphodiester bonds in DNA synthesis?

- A. Hydrolysis
- B. Dehydration (condensation)
- C. Oxidation
- D. Reduction

Correct Answer: B

7. The bond that joins nucleotides in a single strand of DNA is called:

- A. Peptide bond
- B. Ionic bond
- C. Phosphodiester bond
- D. Disulfide bond

Correct Answer: C

8. The phosphodiester bond forms between the 3' hydroxyl of one nucleotide and the:

- A. 5' phosphate of the next nucleotide
- B. 2' hydrogen atom
- C. Base of adjacent nucleotide
- D. 1' carbon of sugar

Correct Answer: A

MCQs: Structure of DNA & Chargaff's Rules

1. Who provided an important clue to the structure of DNA in the late 1940s?

- A. Watson and Crick
- B. Rosalind Franklin
- C. Erwin Chargaff
- D. Francis Collins

Correct Answer: C

2. According to Chargaff's first rule, the base composition of DNA:

- A. Is identical in all species
- B. Varies from one species to another
- C. Changes with age

D. Changes with nutrition

Correct Answer: B

3. Chargaff found that DNA isolated from different tissues of the same species:

A. Has different base compositions

B. Has the same base composition

C. Changes during development

D. Contains no purines

Correct Answer: B

4. According to Chargaff, the base composition of DNA in a species does NOT change due to:

A. Age

B. Nutritional state

C. Environmental changes

D. All of the above

Correct Answer: D

5. According to Chargaff's rule, in DNA the number of Adenine (A) residues is equal to:

A. Guanine (G)

B. Cytosine (C)

C. Thymine (T)

D. Uracil (U)

Correct Answer: C

6. In Chargaff's rules, the number of Guanine (G) residues equals the number of:

A. Adenine (A)

B. Cytosine (C)

C. Thymine (T)

D. Uracil (U)

Correct Answer: B

7. According to Chargaff, the total number of purines equals the total number of:

A. Sugars

B. Nucleotides

C. Pyrimidines

D. Phosphates

Correct Answer: C

8. Purines in DNA include which bases?

A. A and T

B. G and C

C. A and G

D. C and T

Correct Answer: C

9. Pyrimidines in DNA include which bases?

A. A and G

B. C and T

C. A and T

D. G and C

Correct Answer: B

10. Chargaff's work contributed directly to understanding which property of DNA?

A. Its ability to replicate

B. Its double-helix structure

C. Its base-pairing ratios

D. Its protein-coding ability

Correct Answer: C

MCQs: Watson & Crick Double Helix Model

1. Watson and Crick proposed the three-dimensional structure of DNA in which year?

A. 1945

B. 1950

C. 1953

D. 1960

Correct Answer: C

2. According to Watson and Crick, DNA consists of how many polynucleotide strands?

A. One

B. Two

C. Three

D. Four

Correct Answer: B

3. The two strands of DNA are coiled around the same axis to form a:

A. Left-handed helix

B. Zig-zag shape

C. Right-handed double helix

D. Flat ribbon

Correct Answer: C

4. In the double helix model, the hydrophilic sugar-phosphate backbone is located:

A. Inside the helix

B. Outside the helix

C. Between the base pairs

D. At only one end

Correct Answer: B

5. The nitrogenous bases in the DNA double helix are positioned:

A. On the outside

B. On the sugar-phosphate backbone

C. Inside the helix, stacked perpendicular to the axis

D. Randomly arranged

Correct Answer: C

6. The coiling of DNA strands creates which structural features?

- A. One large pocket
- B. Hydrogen tunnels
- C. Major and minor grooves
- D. Peptide folds

Correct Answer: C

7. According to Watson and Crick, base pairing occurs because the bases:

- A. Are identical in size
- B. Fit best within the helix structure
- C. Repel each other
- D. Pair randomly

Correct Answer: B

8. Which base pairs together in the Watson–Crick model?

- A. A–C and G–T
- B. A–U and G–C
- C. A–T and G–C
- D. A–G and T–C

Correct Answer: C

9. The two strands of DNA run in opposite 5' → 3' directions. This orientation is called:

- A. Parallel
- B. Linear
- C. Antiparallel
- D. Bidirectional

Correct Answer: C

10. The two antiparallel DNA strands are described as:

- A. Identical
- B. Oppositional
- C. Complementary
- D. Non-functional

Correct Answer: C

MCQs: Histone Proteins & DNA Packaging

1. The most abundant proteins present in chromatin are:

- A. Enzymes
- B. Lipoproteins
- C. Histones
- D. Collagen

Correct Answer: C

2. How many types of histones are mentioned in the file?

- A. Five
- B. Six
- C. Seven
- D. Nine

Correct Answer: D

(Two each of H2A, H2B, H3, H4 and one H1)

3. Histones fall into how many major classes?

- A. Three
- B. Four
- C. Five
- D. Six

Correct Answer: C

(H1, H2A, H2B, H3, H4)

4. Which histone type is present only as a single copy in the nucleosome structure?

- A. H2A
- B. H3
- C. H1
- D. H4

Correct Answer: C

5. A typical human cell contains approximately how many copies of each histone type?

- A. 600
- B. 60,000
- C. 600,000
- D. 60 million

Correct Answer: D

6. Histone proteins are rich in which amino acids?

- A. Glycine and serine
- B. Valine and leucine
- C. Arginine and lysine
- D. Alanine and proline

Correct Answer: C

7. The lysine-to-arginine ratio:

- A. Is identical for all histones
- B. Differs in each type of histone
- C. Has no effect on DNA binding
- D. Is always 1:1

Correct Answer: B

8. Why can histones bind easily to DNA?

- A. They have hydrophobic side chains
- B. They contain negatively charged amino acids
- C. Their positive charges attract DNA's negative phosphate groups
- D. They covalently bond with DNA

Correct Answer: C

9. The negative charge in DNA comes from:

- A. Nitrogenous bases
- B. Ribose sugar
- C. Phosphate groups

D. Hydrogen bonds

Correct Answer: C

10. What type of force helps stabilize chromatin structure?

A. Hydrogen bonding

B. Covalent bonding

C. Electrostatic attraction

D. Peptide linkages

Correct Answer: C

MCQs: Structures of RNA (mRNA & General RNA Features)

1. mRNA is formed from DNA as a:

A. Double-stranded molecule

B. Single-stranded molecule

C. Triple-stranded molecule

D. Circular molecule

Correct Answer: B

2. Soon after its formation, mRNA assumes a double-helical conformation mainly due to:

A. Hydrogen bonding

B. Base stacking interactions

C. Phosphodiester bonding

D. Ribose instability

Correct Answer: B

3. RNA molecules may contain self-complementary sequences that allow them to:

A. Bind only to proteins

B. Base-pair with RNA or DNA

C. Function without folding

D. Become double-stranded permanently

Correct Answer: B

4. Which statement about RNA secondary structure is correct?

A. RNA has a simple, fixed secondary structure

B. RNA has no regular secondary structure as a reference point

C. RNA always forms a perfect helix

D. RNA structure is identical to DNA structure

Correct Answer: B

5. The three-dimensional structures of many RNA molecules are:

A. Flat and identical

B. Simple and repeating

C. Complex and unique

D. Unstable and temporary

Correct Answer: C

6. Breaks in the RNA helix caused by mismatched or unmatched bases result in:

- A. Mutations only
- B. Bulges or internal loops
- C. Perfect base pairing
- D. More stable double helices

Correct Answer: B

7. Hairpin loops in RNA form between:

- A. Far-apart DNA strands
- B. Self-complementary sequences that are nearby
- C. RNA and protein complexes
- D. Ribose and phosphate regions

Correct Answer: B

8. Which type of loop appears when bases do not match properly on either strand?

- A. Hairpin loop
- B. Terminal loop
- C. Bulge or internal loop
- D. Peptide loop

Correct Answer: C

9. RNA can base-pair with:

- A. Only itself
- B. Only DNA
- C. Complementary RNA or DNA
- D. Only ribosomes

Correct Answer: C

MCQs: DNA Replication

1. According to the double-helical model of DNA, the two strands are:

- A. Identical
- B. Complementary
- C. Unrelated
- D. Non-functional

Correct Answer: B

2. Because the two strands are complementary, each strand can serve as a:

- A. Ribosome
- B. Template for its partner
- C. Protein-coding sequence
- D. Non-coding region

Correct Answer: B

3. Which model is accepted as the correct mechanism of DNA replication?

- A. Conservative
- B. Dispersive
- C. Semiconservative
- D. Progressive

Correct Answer: C

4. The Watson–Crick model proposed that DNA replication occurs by:

- A. Breaking both strands and forming new DNA randomly
- B. Keeping parental strands together
- C. Separating parental strands to serve as templates
- D. Fragmenting DNA fully before replication

Correct Answer: C

5. In semiconservative replication, each daughter DNA molecule contains:

- A. Two parental strands
- B. Two new strands
- C. One parental and one new strand
- D. No parental DNA

Correct Answer: C

6. Semiconservative replication is called so because:

- A. Both strands are destroyed
- B. One parental strand is conserved
- C. New strands are not formed
- D. Replication is incomplete

Correct Answer: B

7. In the conservative model of replication:

- A. Parental strands separate
- B. Parental strands stay together
- C. Both strands become fragmented
- D. New strands replace parental strands

Correct Answer: B

8. According to the conservative model, the newly formed daughter helix contains:

- A. One parental and one new strand
- B. Two parental strands
- C. Two completely new strands
- D. Only fragments of DNA

Correct Answer: C

9. The dispersive model of replication proposes that:

- A. DNA remains intact during replication
- B. DNA becomes fragmented and mixes old and new DNA in the same strand
- C. Only new DNA is made
- D. Replication happens without templates

Correct Answer: B

10. Which model suggests coexistence of old and new DNA within the same strand?

- A. Semiconservative
- B. Conservative
- C. Dispersive
- D. Random

Correct Answer: C

MCQs: Meselson–Stahl Experiment

1. The Meselson and Stahl experiment was performed in:

- A. 1945
- B. 1950
- C. 1958
- D. 1965

Correct Answer: C

2. The purpose of the Meselson–Stahl experiment was to distinguish between different:

- A. Types of RNA molecules
- B. Mechanisms of DNA replication
- C. Models of protein synthesis
- D. Gene regulation pathways

Correct Answer: B

3. Which organism did Meselson and Stahl use in their experiment?

- A. Yeast
- B. *E. coli*
- C. *Drosophila*
- D. Human cells

Correct Answer: B

4. DNA was labeled with heavy nitrogen by growing cells in a medium enriched with which isotope?

- A. ^{12}N
- B. ^{13}N
- C. ^{14}N
- D. ^{15}N

Correct Answer: D

5. Growing *E. coli* in ^{15}N medium made their DNA:

- A. Lighter
- B. Denser than normal
- C. Unstable
- D. Radioactive

Correct Answer: B

6. After labeling with ^{15}N , the bacteria were transferred to a medium containing primarily:

- A. ^{13}N
- B. ^{15}N
- C. ^{14}N
- D. No nitrogen

Correct Answer: C

7. Meselson and Stahl analyzed DNA densities using:

- A. Gel electrophoresis
- B. Southern blotting
- C. Density gradient centrifugation

D. Spectrophotometry

Correct Answer: C

8. The heavy nitrogen (^{15}N) label allowed scientists to:

- A. Observe base-pairing
- B. Distinguish old DNA from new DNA
- C. Measure mutation rates
- D. Break DNA strands

Correct Answer: B

9. Switching cells to ^{14}N medium for various lengths of time allowed tracking of:

- A. Protein folding
- B. Changes in ribosome structure
- C. DNA replication cycles
- D. RNA degradation

Correct Answer: C

10. The Meselson–Stahl experiment ultimately supported which DNA replication model?

- A. Conservative
- B. Dispersive
- C. Semiconservative
- D. Parallel

Correct Answer: C

(This conclusion is from well-known biological knowledge, though not directly written in these slides.)

MCQs: Chemistry & Mechanism of DNA Synthesis

1. How many key substrates are required for DNA synthesis?

- A. One
- B. Two
- C. Three
- D. Four

Correct Answer: B

(Deoxynucleoside triphosphates and primer:template junction)

2. Which of the following is NOT one of the four deoxynucleoside triphosphates?

- A. dGTP
- B. dATP
- C. dUTP
- D. dTTP

Correct Answer: C

3. In nucleoside triphosphates, the three phosphoryl groups are attached to the:

- A. 3' hydroxyl of deoxyribose
- B. 5' hydroxyl of deoxyribose
- C. Nitrogenous base

D. Sugar-phosphate backbone

Correct Answer: B

4. The innermost phosphoryl group in a nucleoside triphosphate is called the:

A. β -phosphate

B. γ -phosphate

C. α -phosphate

D. δ -phosphate

Correct Answer: C

5. The particular arrangement of ssDNA and dsDNA required for replication is called the:

A. Okazaki fragment

B. Replication fork

C. Primer:template junction

D. Double helix junction

Correct Answer: C

6. DNA chain extension occurs at the:

A. 5' end of the primer

B. 3' end of the primer

C. Middle of the template

D. Nitrogenous base

Correct Answer: B

7. The phosphodiester bond during DNA synthesis is formed by which type of reaction?

A. SN1

B. SN2

C. Hydrolysis

D. Oxidation

Correct Answer: B

8. In the SN2 reaction of DNA synthesis, the 3' OH of the primer attacks the:

A. β -phosphate

B. γ -phosphate

C. α -phosphate of the incoming nucleotide

D. Ribose ring

Correct Answer: C

9. The leaving group released during DNA polymerization is:

A. Monophosphate

B. Pyrophosphate

C. Deoxyribose

D. Nitrogen base

Correct Answer: B

10. What enzyme hydrolyzes pyrophosphate to drive the reaction forward?

A. DNA polymerase

B. Helicase

C. Pyrophosphatase

D. Ligase

Correct Answer: C

11. Hydrolysis of pyrophosphate results in:

- A. Formation of ATP
- B. Breaking of two high-energy phosphate bonds
- C. No energy release
- D. Formation of ribose

Correct Answer: B

12. DNA synthesis is energetically favorable because:

- A. It consumes no energy
- B. It is coupled to pyrophosphate hydrolysis
- C. Bases pair randomly
- D. It requires no enzymes

Correct Answer: B

13. The direction of DNA synthesis is always:

- A. $5' \rightarrow 3'$
- B. $3' \rightarrow 5'$
- C. Random
- D. Both $5' \rightarrow 3'$ and $3' \rightarrow 5'$

Correct Answer: A

14. The reaction of nucleotide addition during DNA synthesis is considered:

- A. Reversible
- B. Irreversible
- C. Equally reversible and irreversible
- D. Not energy-dependent

Correct Answer: B

MCQs: Mechanism of DNA Replication – Chemistry of DNA Synthesis (Part 2)

1. DNA synthesis requires how many key substrates?

- A. One
- B. Two
- C. Three
- D. Four

Correct Answer: B

(Deoxynucleoside triphosphates + Primer:template junction)

2. Which of the following is NOT one of the required deoxynucleoside triphosphates?

- A. dGTP
- B. dCTP
- C. dATP
- D. dUTP

Correct Answer: D

3. The three phosphoryl groups of a nucleoside triphosphate attach to the:

- A. 3' hydroxyl of deoxyribose
- B. 5' hydroxyl of deoxyribose
- C. Nitrogenous base
- D. Sugar-phosphate backbone

Correct Answer: B

4. The middle phosphoryl group of a nucleoside triphosphate is called the:

- A. α -phosphate
- B. β -phosphate
- C. γ -phosphate
- D. δ -phosphate

Correct Answer: B

5. The essential structure made of ssDNA and dsDNA required for replication is called:

- A. Primer:template junction
- B. Replication fork
- C. Okazaki fragment
- D. DNA bubble

Correct Answer: A

6. DNA polymerase extends the DNA chain at which end of the primer?

- A. 5' end
- B. 3' end
- C. Middle of the strand
- D. Both ends

Correct Answer: B

7. The phosphodiester bond forms through which type of reaction?

- A. SN1
- B. SN2
- C. Hydrolysis
- D. Oxidation

Correct Answer: B

8. During bond formation, the 3' OH group attacks which phosphate?

- A. γ -phosphate
- B. β -phosphate
- C. α -phosphate
- D. δ -phosphate

Correct Answer: C

9. The leaving group during DNA polymerization is:

- A. Phosphate
- B. Monophosphate
- C. Pyrophosphate
- D. Deoxyribose

Correct Answer: C

10. Which enzyme provides the driving force for DNA synthesis by hydrolyzing pyrophosphate?

- A. DNA polymerase
- B. Helicase

C. Pyrophosphatase

D. Ligase

Correct Answer: C

11. Hydrolysis of pyrophosphate results in the breaking of:

A. No energy bonds

B. One high-energy bond

C. Two high-energy phosphate bonds

D. Three high-energy bonds

Correct Answer: C

12. DNA synthesis is considered a coupled and highly favorable reaction because:

A. It releases no energy

B. K_{eq} is very high

C. It is reversible

D. It requires no enzymes

Correct Answer: B

13. The template strand determines:

A. The sugar in the nucleotide

B. Which nucleoside triphosphate is added

C. The number of phosphates in DNA

D. The length of the primer

Correct Answer: B

14. DNA synthesis is considered an:

A. Equilibrium reaction

B. Reversible reaction

C. Irreversible reaction

D. Variable reaction

Correct Answer: C

MCQs: Mechanism of DNA Polymerase

1. DNA synthesis is catalyzed by which enzyme?

A. RNA polymerase

B. DNA ligase

C. DNA polymerase

D. Helicase

Correct Answer: C

2. DNA polymerase uses how many active sites to add any of the four dNTPs?

A. Four

B. Two

C. One

D. Three

Correct Answer: C

3. DNA polymerase monitors the incoming nucleotide based on its ability to:

A. Bind sugar-phosphate backbone

B. Form correct A:T or G:C base pairs

- C. Fit perfectly into the pocket
- D. Release pyrophosphate

Correct Answer: B

4. Correct nucleotide addition occurs when which two groups align properly?

- A. 5'-OH of primer & β -phosphate
- B. 3'-OH of primer & α -phosphate of nucleotide
- C. 2'-OH of primer & α -phosphate
- D. 3'-OH of primer & γ -phosphate

Correct Answer: B

5. Incorrect base pairing leads to:

- A. Faster catalysis
- B. Equal rate of replication
- C. Much slower nucleotide addition
- D. No effect on synthesis

Correct Answer: C

6. DNA polymerase distinguishes between rNTPs and dNTPs by:

- A. Charge difference
- B. Van der Waals bonds
- C. Steric exclusion from the active site
- D. Hydrogen bonding

Correct Answer: C

7. The incorporation rate of rNTPs compared to dNTPs is:

- A. Equal
- B. Slightly lower
- C. 10-fold lower
- D. 1000-fold lower

Correct Answer: D

8. rNTPs are present in the cell at concentrations that are:

- A. Lower than dNTPs
- B. Equal to dNTPs
- C. Ten-fold higher than dNTPs
- D. Never used in DNA

Correct Answer: C

9. Why can't DNA polymerase accept rNTPs into its active site?

- A. They lack phosphate groups
- B. Their 2'-OH prevents fitting into the binding pocket
- C. They cannot form hydrogen bonds
- D. They are too small

Correct Answer: B

10. The amino acids that block the 2'-OH of rNTPs by occupying space in the nucleotide-binding pocket are called:

- A. Regulatory amino acids
- B. Catalytic amino acids
- C. Discriminator amino acids
- D. Repair amino acids

Correct Answer: C

11. The discriminator amino acids make which type of contact with the deoxyribose ring?

- A. Ionic bonds
- B. Covalent bonds
- C. Van der Waals contacts
- D. Hydrogen bonds

Correct Answer: C

MCQs: DNA Polymerase – Hand Model

1. Structural studies show that DNA polymerase holds DNA in a cleft resembling a:

- A. Closed fist
- B. Open palm
- C. Partially closed right hand
- D. Left hand

Correct Answer: C

2. According to the hand analogy, DNA polymerase has how many major domains?

- A. Two
- B. Three
- C. Four
- D. Five

Correct Answer: B

(Thumb, fingers, and palm)

3. The palm domain of DNA polymerase is primarily composed of:

- A. α -helices
- B. β -sheet
- C. Random coils
- D. Lipids

Correct Answer: B

4. The catalytic site of DNA polymerase is located in the:

- A. Thumb domain
- B. Finger domain
- C. Palm domain
- D. Wrist region

Correct Answer: C

5. The palm domain binds how many divalent metal ions?

- A. One
- B. Two
- C. Three
- D. Four

Correct Answer: B

6. The metal ions in the catalytic site are typically:

- A. Na^+ or K^+
- B. Mg^{2+} or Zn^{2+}
- C. Fe^{2+} or Cu^{2+}
- D. Ca^{2+} or Fe^{3+}

Correct Answer: B

7. One metal ion reduces the affinity of the primer's 3'-OH for hydrogen, creating:

- A. A hydroxide ion (OH^-)
- B. A 3'- O^{2-} nucleophile
- C. A water molecule
- D. A pyrophosphate

Correct Answer: B

8. The second metal ion stabilizes:

- A. Hydrogen bonds
- B. Ribose sugar
- C. β - and γ -phosphates and pyrophosphate
- D. DNA helicase

Correct Answer: C

9. Apart from catalysis, the palm domain also monitors:

- A. Nucleotide size
- B. Base pairing of newly added nucleotides
- C. DNA supercoiling
- D. Ribose orientation

Correct Answer: B

10. Which domain binds the incoming dNTP?

- A. Thumb
- B. Wrist
- C. Fingers
- D. Palm

Correct Answer: C

11. The fingers domain closes around the dNTP when:

- A. Any nucleotide binds
- B. A correct base pair is formed
- C. A mismatch occurs
- D. The DNA breaks

Correct Answer: B

12. The thumb domain mainly interacts with:

- A. Incoming dNTPs
- B. DNA that has been newly synthesized
- C. RNA primers
- D. Helicase

Correct Answer: B

13. One role of the thumb domain is to:

- A. Catalyze phosphodiester bond formation
- B. Maintain correct primer-active site position

- C. Bind metal ions
- D. Proofread nucleotides

Correct Answer: B

14. The strong interaction between the thumb and DNA helps polymerase to:

- A. Proofread errors
- B. Add many dNTPs efficiently
- C. Create RNA primers
- D. Stop replication

Correct Answer: B

15. The domain least involved directly in catalysis is the:

- A. Palm
- B. Fingers
- C. Thumb
- D. All domains equally

Correct Answer: C

MCQs: The Replication Fork

1. In a cell, both strands of DNA are replicated:

- A. One at a time
- B. Randomly
- C. At the same time
- D. Only during transcription

Correct Answer: C

2. Separation of the two DNA strands is required to create:

- A. RNA primers
- B. Template DNAs
- C. Ribosomes
- D. Chromosomes

Correct Answer: B

3. The junction between newly separated strands and unreplicated DNA is called:

- A. Replication bubble
- B. Replication fork
- C. Okazaki point
- D. Chromosomal junction

Correct Answer: B

4. The replication fork moves continuously toward the:

- A. RNA region
- B. Replicated DNA
- C. Duplex region of unreplicated DNA
- D. Nuclear membrane

Correct Answer: C

5. As the replication fork moves, it creates two:

- A. Ribosomes
- B. RNA strands
- C. ssDNA templates
- D. Helicase enzymes

Correct Answer: C

6. DNA synthesis always occurs by elongation of the:

- A. 5' end
- B. 3' end
- C. Middle of the strand
- D. Both ends

Correct Answer: B

7. The strand that is synthesized continuously is called the:

- A. Lagging strand
- B. Leading strand
- C. Template strand
- D. Coding strand

Correct Answer: B

8. The lagging strand is synthesized:

- A. Continuously
- B. In small fragments
- C. Without enzymes
- D. In the same direction as the fork

Correct Answer: B

9. The DNA fragments formed on the lagging strand are called:

- A. Polymerase fragments
- B. Watson fragments
- C. Okazaki fragments
- D. Replication units

Correct Answer: C

10. The length of Okazaki fragments in bacteria is typically:

- A. 50–100 nucleotides
- B. 100–400 nucleotides
- C. 500–700 nucleotides
- D. 1000–2000 nucleotides

Correct Answer: D

11. The length of Okazaki fragments in eukaryotes is typically:

- A. 10–50 nucleotides
- B. 100–400 nucleotides
- C. 1000–2000 nucleotides
- D. 3000–5000 nucleotides

Correct Answer: B

12. Okazaki fragments are described as:

- A. Permanent parts of DNA
- B. Transient intermediates that are later joined
- C. Nonfunctional DNA

D. RNA–DNA hybrids

Correct Answer: B

13. Lagging strand synthesis must wait until:

- A. The leading strand finishes
- B. The fork exposes enough template
- C. RNA polymerase binds
- D. Chromosomes separate

Correct Answer: B

14. Okazaki fragments are eventually joined to produce:

- A. A double-stranded RNA
- B. A continuous DNA strand
- C. A new replication fork
- D. A helicase molecule

Correct Answer: B

MCQs Based on the File: THE RNA PRIMER

1. DNA polymerases require which feature to begin DNA synthesis?

- A. Free 5'-phosphate
- B. Free 3'-OH group
- C. Double-stranded DNA end
- D. No requirement; they can start de novo

Correct Answer: B

Explanation: DNA polymerases cannot start new strands; they require a free 3'-OH group to extend.

2. Why does the cell use RNA polymerase during DNA replication?

- A. RNA polymerase replicates DNA faster
- B. RNA polymerase can start new chains de novo
- C. RNA polymerase proofreads better
- D. DNA polymerase is inactive during replication

Correct Answer: B

Explanation: RNA polymerases can initiate RNA chain synthesis without a primer.

3. Which enzyme synthesizes the RNA primer?

- A. Ligase
- B. Helicase
- C. Primase
- D. Topoisomerase

Correct Answer: C

Explanation: Primase is a specialized RNA polymerase that makes short RNA primers.

86. THE RNA PRIMER

4. The typical length of an RNA primer synthesized by primase is:

- A. 1–3 nucleotides
- B. 5–10 nucleotides

- C. 20–30 nucleotides
- D. 50 nucleotides

Correct Answer: B

5. What is the role of the RNA primer?

- A. To join Okazaki fragments
- B. To provide a free 3'-OH for DNA polymerase
- C. To unwind the DNA helix
- D. To degrade RNA from DNA-RNA hybrids

Correct Answer: B

6. Which strand requires only a single RNA primer?

- A. Lagging strand
- B. Leading strand
- C. Both strands equally
- D. Neither strand

Correct Answer: B

Explanation: The leading strand is synthesized continuously and requires only one primer.

7. Why does the lagging strand require multiple primers?

- A. It is synthesized continuously
- B. It has no access to DNA polymerase
- C. It is synthesized discontinuously as Okazaki fragments
- D. RNA primers degrade too quickly

Correct Answer: C

8. Each Okazaki fragment requires:

- A. One DNA helicase and one ligase
- B. One RNA primer
- C. No primer
- D. Two primers

Correct Answer: B

9. Primase activity is significantly increased when it associates with:

- A. DNA polymerase I
- B. DNA ligase
- C. DNA helicase
- D. Topoisomerase

Correct Answer: C

10. What is the function of DNA helicase in primer synthesis?

- A. It synthesizes primers
- B. It proofreads RNA primers
- C. It unwinds DNA to create ssDNA for primase
- D. It joins Okazaki fragments

Correct Answer: C

Explanation: Helicase unwinds DNA at the replication fork, exposing ssDNA for primase.

86. THE RNA PRIMER

11. Which statement best describes primase?

- A. It synthesizes long RNA chains
- B. It is a DNA-dependent DNA polymerase

- C. It is a specialized RNA polymerase that initiates primer synthesis
- D. It removes RNA primers after replication

Correct Answer: C

12. Approximately how many primers may be needed for lagging strand synthesis?

- A. Only 1–2
- B. Tens
- C. Hundreds
- D. Thousands

Correct Answer: C

Explanation: Because each Okazaki fragment needs a primer, hundreds may be required.

Basic MCQs from: THE DNA HELICASE

1. Why does DNA replication require DNA helicase?

- A. Because DNA polymerase cannot join nucleotides
- B. Because DNA polymerase cannot separate DNA strands
- C. Because DNA helicase synthesizes RNA primers
- D. Because DNA helicase proofreads DNA

Correct Answer: B

Explanation: DNA polymerase cannot separate duplex DNA; helicase performs this function.

87. THE DNA HELICASE

2. DNA helicases catalyze:

- A. Formation of phosphodiester bonds
- B. Separation of DNA strands at the replication fork
- C. DNA proofreading
- D. Ligating Okazaki fragments

Correct Answer: B

3. DNA helicases are typically:

- A. Monomeric proteins
- B. Dimeric proteins
- C. Hexameric ring-shaped proteins
- D. RNA–protein complexes

Correct Answer: C

Explanation: Helicases are hexameric and form a ring around ssDNA.

87. THE DNA HELICASE

4. The ring-shaped structure of helicase encircles:

- A. Double-stranded DNA
- B. One single DNA strand
- C. Both strands simultaneously
- D. RNA primer

Correct Answer: B

5. High processivity of DNA helicases is due to:

- A. Their ability to synthesize DNA
- B. Their ring structure encircling DNA

- C. Their proofreading ability
- D. Their strong binding to RNA primers

Correct Answer: B

Explanation: Encircling DNA prevents frequent dissociation.

87. THE DNA HELICASE

6. When does helicase usually dissociate from DNA?

- A. When the ring opens frequently
- B. When it reaches the end of a DNA strand
- C. When primase binds
- D. When DNA ligase interacts

Correct Answer: B

7. How is helicase initially loaded onto DNA?

- A. By DNA polymerase
- B. By spontaneous binding
- C. By special mechanisms that open the hexameric ring
- D. By RNA primers

Correct Answer: C

Explanation: Specialized systems open the helicase ring to place it around DNA.

87. THE DNA HELICASE

8. The direction in which a helicase moves along ssDNA is called:

- A. Activity
- B. Polarity
- C. Viscosity
- D. Specificity

Correct Answer: B

9. DNA helicase polarity is defined relative to:

- A. The displaced strand
- B. The complementary strand
- C. The DNA strand bound by the helicase
- D. The RNA primer attached

Correct Answer: C

Explanation: Polarity refers to the direction of movement on the bound strand.

87. THE DNA HELICASE

10. DNA helicases can move in which direction(s) along ssDNA?

- A. Only 5' → 3'
- B. Only 3' → 5'
- C. Either 5' → 3' or 3' → 5'
- D. Only 5' → 5'

Correct Answer: C

11. Which statement is TRUE about helicase dissociation?

- A. Ring opening is common
- B. Helicase rarely opens its ring to leave DNA
- C. Helicase dissociates after every few base pairs
- D. It cannot dissociate at all

Correct Answer: B

Explanation: Ring opening is rare; processivity is high.

87. THE DNA HELICASE

12. The helicase ring is positioned around DNA at the:

- A. Random location
- B. Promoter region
- C. Replication fork near ssDNA–dsDNA junction
- D. Telomere region

Correct Answer: C

Basic MCQs from: TOPOISOMERASES

1. What happens to DNA ahead of the replication fork due to helicase activity?

- A. It becomes negatively supercoiled
- B. It becomes positively supercoiled
- C. It becomes linearized
- D. It becomes denatured permanently

Correct Answer: B

88. TOPOISOMERASES

2. Why must supercoils be removed during DNA replication?

- A. They speed up replication
- B. They cause DNA melting
- C. They create strain that would stop replication
- D. They prevent RNA primer synthesis

Correct Answer: C

Explanation: The replication machinery would halt due to excess strain.

88. TOPOISOMERASES

3. Which enzymes remove supercoils ahead of the replication fork?

- A. Helicases
- B. Polymerases
- C. Topoisomerases
- D. Ligases

Correct Answer: C

88. TOPOISOMERASES

4. Topoisomerases act on which region of DNA?

- A. Replicated DNA behind the fork
- B. Unreplicated dsDNA ahead of the fork
- C. RNA primers
- D. DNA–RNA hybrids

Correct Answer: B

88. TOPOISOMERASES

5. How do topoisomerases relieve supercoiling?

- A. By unwinding DNA like helicase
- B. By breaking either one or both DNA strands
- C. By synthesizing new DNA
- D. By degrading RNA primers

Correct Answer: B

6. What unique feature of topoisomerase activity prevents DNA from being lost?

- A. They stay attached to DNA during the break
- B. They require ATP for all actions
- C. They only cut RNA
- D. They replicate DNA simultaneously

Correct Answer: A

Explanation: They break strands *without letting go* of the DNA.

88. TOPOISOMERASES

7. Topoisomerases pass how many strands through the break they create?

- A. None
- B. Only one RNA strand
- C. The same number of DNA strands
- D. A random number

Correct Answer: C

88. TOPOISOMERASES

8. Topoisomerases are sometimes referred to as:

- A. Helicase helpers
- B. Polymerase stabilizers
- C. Swivelase
- D. Ligase substitutes

Correct Answer: C

Explanation: Because they allow DNA to rotate and remove supercoils.

88. TOPOISOMERASES

9. What would happen if topoisomerases were absent?

- A. DNA replication would proceed normally
- B. Supercoil buildup would stop replication
- C. Primase could replace their function
- D. Polymerase would unwind the DNA itself

Correct Answer: B

10. Supercoils appear in DNA because:

- A. Primase adds too many primers
- B. Helicase unwinds DNA strands
- C. Ligase joins fragments too quickly
- D. Polymerase proofreads DNA

Correct Answer: B

Basic MCQs from: INITIATION OF REPLICATION

1. The very first requirement for forming a replication fork is:

- A. Binding of DNA polymerase
- B. Separation of the two DNA strands
- C. RNA primer removal
- D. Joining of Okazaki fragments

Correct Answer: B

89. INITIATION OF REPLICATION

2. Why is ssDNA essential during replication initiation?

- A. It is needed for helicase binding
- B. It prevents supercoiling
- C. It blocks primer formation
- D. It stabilizes Okazaki fragments

Correct Answer: A

Explanation: ssDNA is required for helicase binding AND acts as a template for RNA primer and DNA synthesis.

89. INITIATION OF REPLICATION

3. Strand separation for replication initiation occurs most easily at:

- A. Internal chromosome regions
- B. Telomeres (chromosome ends)
- C. Promoter regions
- D. Ribosomal RNA genes

Correct Answer: B

89. INITIATION OF REPLICATION

4. Even though unwinding is easiest at chromosome ends, DNA synthesis usually begins at:

- A. Chromosome tips only
- B. Random positions
- C. Internal chromosomal regions
- D. Only at GC-rich areas

Correct Answer: C

5. Why is internal unwinding essential in circular chromosomes?

- A. Because they lack ends
- B. Because they unwind automatically
- C. Because primase only works internally
- D. Because replication occurs only at telomeres

Correct Answer: A

89. INITIATION OF REPLICATION

6. The specific site where DNA unwinding and replication begin is called:

- A. Replication fork
- B. Helicase binding site
- C. Origin of Replication
- D. Primase box

Correct Answer: C

89. INITIATION OF REPLICATION

7. The number of origins of replication per chromosome may range from:

- A. One to two
- B. Dozens only
- C. As few as one to thousands
- D. Millions

Correct Answer: C

89. INITIATION OF REPLICATION

8. The template required for synthesis of RNA primer and new DNA is:

- A. dsDNA
- B. ssDNA
- C. RNA–DNA hybrid
- D. Protein scaffold

Correct Answer: B

9. Which enzyme depends on ssDNA for proper binding during replication initiation?

- A. DNA ligase
- B. DNA polymerase
- C. DNA helicase
- D. Telomerase

Correct Answer: C

89. INITIATION OF REPLICATION

10. What event marks the true beginning of DNA replication?

- A. Primer removal
- B. Fork progression
- C. DNA strand separation at the origin
- D. DNA ligase activation

Correct Answer: C

Basic MCQs from: THE REPLICON MODEL

1. Who proposed the Replicon Model of replication initiation?

- A. Watson and Crick
- B. Meselson and Stahl

C. Jacob, Brenner, and Cuzin
D. Griffith and Avery
Correct Answer: C

90. THE REPLICON MODEL

2. What did the Replicon Model define as a “replicon”?

- A. A replication enzyme
- B. A region of RNA
- C. All DNA replicated from a single origin
- D. A chromosome end

Correct Answer: C

90. THE REPLICON MODEL

3. In *E. coli*, the entire chromosome is considered a single replicon because:

- A. It replicates very slowly
- B. It has only one origin of replication
- C. It contains linear DNA
- D. It has multiple replication forks

Correct Answer: B

4. Eukaryotic chromosomes contain:

- A. Only one replicon
- B. No replicons
- C. Multiple replicons
- D. Replicons only at telomeres

Correct Answer: C

90. THE REPLICON MODEL

5. According to the Replicon Model, replication initiation is controlled by:

- A. Helicase and primase
- B. Replicator and initiator
- C. Ligase and polymerase
- D. Okazaki fragments

Correct Answer: B

6. The replicator is defined as:

- A. A protein that starts replication
- B. Cis-acting DNA sequences sufficient to initiate replication
- C. The location where helicase binds
- D. RNA sequences required for replication

Correct Answer: B

90. THE REPLICON MODEL

7. The origin of replication is:

- A. A protein complex that unwinds DNA
- B. Always identical to the replicator
- C. The physical site where DNA unwinding begins
- D. Only found in prokaryotes

Correct Answer: C

90. THE REPLICON MODEL

8. The difference between replicator and origin is that:

- A. The origin is always larger
- B. The replicator may include more sequences than the origin
- C. The replicator is only in eukaryotes
- D. The origin binds RNA polymerase

Correct Answer: B

90. THE REPLICON MODEL

9. The initiator protein functions by:

- A. Unwinding DNA
- B. Recognizing specific DNA elements in the replicator
- C. Synthesizing RNA primers
- D. Repairing DNA mismatches

Correct Answer: B

90. THE REPLICON MODEL

10. Which protein selects the sites that will become origins of replication?

- A. Helicase
- B. Polymerase
- C. Initiator protein
- D. Ligase

Correct Answer: C

90. THE REPLICON MODEL

11. Which is the ONLY sequence-specific DNA-binding protein involved in replication initiation?

- A. Primase
- B. DNA polymerase
- C. Helicase
- D. Initiator protein

Correct Answer: D

90. THE REPLICON MODEL

12. All other proteins involved in replication initiation (other than initiator):

- A. Bind DNA in a sequence-specific manner
- B. Do not bind DNA at all
- C. Bind DNA but not sequence-specifically
- D. Replace the initiator protein

Correct Answer: C

90. THE REPLICON MODEL

Basic MCQs from: FINISHING REPLICATION

1. Completion of DNA replication requires events that differ between:

- A. RNA and DNA viruses
- B. Leading and lagging strands
- C. Circular and linear chromosomes
- D. Prokaryotes only

Correct Answer: C

91. FINISHING REPLICATION

2. In circular chromosomes, replication fork machinery:

- A. Cannot begin replication
- B. Replicates the entire molecule
- C. Leaves gaps in the middle
- D. Only synthesizes RNA primers

Correct Answer: B

91. FINISHING REPLICATION

3. After replication of circular chromosomes, daughter DNA molecules are:

- A. Completely separated
- B. Linearized
- C. Topologically linked
- D. Degraded at the ends

Correct Answer: C

91. FINISHING REPLICATION

4. The topologically linked daughter molecules of circular DNA are called:

- A. Plasmids
- B. Catenanes
- C. Replicons
- D. Okazaki fragments

Correct Answer: B

91. FINISHING REPLICATION

5. Separation of circular daughter chromosomes requires:

- A. DNA ligase
- B. Helicase
- C. Type II Topoisomerases
- D. Primase

Correct Answer: C

91. FINISHING REPLICATION

6. The process of unlinking catenated circular chromosomes is called:

- A. Decatenation
- B. Termination

C. Denaturation
D. Polymerization
Correct Answer: A

7. Linear chromosomes cannot be fully replicated by the replication fork machinery because:

- A. DNA polymerase is too slow
- B. Primase cannot bind to ends
- C. The very ends cannot be duplicated
- D. Replication forks collapse in the middle

Correct Answer: C

91. FINISHING REPLICATION

8. Organisms with linear chromosomes have developed:

- A. Extra helicases for finishing replication
- B. Novel strategies to replicate chromosome ends
- C. Circular intermediates
- D. Larger origins of replication

Correct Answer: B

91. FINISHING REPLICATION

9. Why must circular DNA chromosomes be decatenated after replication?

- A. To remove primers
- B. To allow segregation into daughter cells
- C. To fix replication errors
- D. To initiate transcription

Correct Answer: B

10. Which enzyme resolves topological linkage between circular DNA molecules?

- A. Topoisomerase I
- B. Type II Topoisomerase
- C. RNA Polymerase
- D. Telomerase

Correct Answer: B

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Basic MCQs from: TYPE II TOPOISOMERASES

1. Type II topoisomerases have the ability to:

- A. Break only one DNA strand
- B. Break dsDNA and pass another dsDNA through the break
- C. Remove RNA primers
- D. Unwind DNA at the replication fork

Correct Answer: B

92. TYPE II TOPOISOMERASES

2. The main role of Type II topoisomerases in circular chromosomes is to:

- A. Make RNA primers
- B. Initiate replication
- C. Decatenate daughter chromosomes
- D. Repair mismatched bases

Correct Answer: C

92. TYPE II TOPOISOMERASES

3. Decatenation refers to:

- A. DNA unwinding
- B. Joining Okazaki fragments
- C. Separation of interlinked DNA molecules
- D. Polymerization of nucleotides

Correct Answer: C

4. Type II topoisomerase activity is also required for segregation of:

- A. Small plasmids only
- B. Only circular chromosomes
- C. Large linear chromosomes
- D. RNA molecules

Correct Answer: C

92. TYPE II TOPOISOMERASES

5. Even though linear chromosomes do not form topological links after replication, segregation problems arise because:

- A. They lack helicases
- B. Their DNA is folded into loops attached to a protein scaffold
- C. They cannot replicate completely
- D. They contain excessive RNA primers

Correct Answer: B

92. TYPE II TOPOISOMERASES

6. DNA loops attached to a protein scaffold in linear chromosomes cause issues similar to:

- A. RNA splicing
- B. Circular chromosome linkage
- C. DNA methylation
- D. Transcription termination

Correct Answer: B

7. Type II topoisomerases help resolve problems in large linear chromosomes by:

- A. Synthesizing telomeric repeats
- B. Binding origins of replication
- C. Separating linked DNA loops
- D. Activating DNA helicase

Correct Answer: C

92. TYPE II TOPOISOMERASES

8. Which statement best describes Type II topoisomerase action?

- A. Cuts only one DNA strand
- B. Needs no energy for action
- C. Passes a second DNA duplex through a ds break
- D. Works only during transcription

Correct Answer: C

Basic MCQs from: TELOMERASE

1. The difficulty in replicating the ends of linear chromosomes is called:

- A. Replicon mismatch problem
- B. Primer deletion problem
- C. End replication problem
- D. Lagging strand failure

Correct Answer: C

93. TELOMERASE

2. Why does the leading strand not face the end replication problem?

- A. It uses multiple primers
- B. It is synthesized from both ends
- C. It requires only one primer that reaches the extreme terminus
- D. It is replicated by telomerase directly

Correct Answer: C

93. TELOMERASE

3. The lagging strand cannot be fully replicated because:

- A. DNA polymerase is slow
- B. Helicase does not reach the end
- C. Multiple primers are required and the last primer cannot cover the final bases
- D. RNA primers cannot bind to ssDNA

Correct Answer: C

93. TELOMERASE

4. After removal of the last RNA primer on the lagging strand, what remains at the chromosome end?

- A. A gap repaired by DNA ligase
- B. A short region of unreplicated ssDNA
- C. A telomeric repeat
- D. A double-stranded blunt end

Correct Answer: B

93. TELOMERASE

5. Telomerase contains:

- A. Only proteins
- B. Only RNA

- C. Protein subunits + an RNA component
- D. DNA-binding helicase

Correct Answer: C

93. TELOMERASE

6. Like other DNA polymerases, telomerase extends:

- A. The 5' end of DNA
- B. The 3' end of DNA substrate
- C. Both ends simultaneously
- D. Only RNA primers

Correct Answer: B

7. What unique feature distinguishes telomerase from most DNA polymerases?

- A. It needs no dNTPs
- B. It does not require a template from external DNA
- C. It removes RNA primers
- D. It synthesizes Okazaki fragments

Correct Answer: B

93. TELOMERASE

8. The RNA component of telomerase acts as:

- A. A primer
- B. A helicase activator
- C. The template for adding telomeric sequences
- D. A ligase helper

Correct Answer: C

93. TELOMERASE

9. Telomerase elongates which strand at the chromosome end?

- A. Both strands equally
- B. Only the 5' end
- C. Only the 3'-OH of telomeric ssDNA
- D. Only the RNA primer strand

Correct Answer: C

10. Telomerase action produces newly synthesized DNA that is:

- A. Double-stranded
- B. RNA–DNA hybrid
- C. Single-stranded
- D. Methylated

Correct Answer: C

93. TELOMERASE

11. How is the 5' end of the chromosome extended if telomerase only elongates the 3' end?

- A. The cell ignores the missing 5' sequence
- B. Telomerase later switches to the 5' strand
- C. Lagging-strand machinery synthesizes new primers using the extended 3' end
- D. DNA polymerase I directly fills in the gap

Correct Answer: C

93. TELOMERASE

12. The purpose of telomerase extension is to:

- A. Prevent helicase from unwinding ends
- B. Provide extra template for lagging-strand synthesis
- C. Remove last RNA primer
- D. Strengthen chromosomal proteins

Correct Answer: B

93. TELOMERASE

.

Basic MCQs from: Dilemma 1, 2 & 3

1. Dilemma 1 in DNA replication is related to:

- A. The leading strand
- B. The lagging strand
- C. RNA transcription
- D. DNA repair mechanisms

Correct Answer: B

94. Dilemma 1

2. Human telomeres contain which repeated sequence?

- A. AATCCC
- B. TTAGGG
- C. GCGCGC
- D. ATTTAA

Correct Answer: B

94. Dilemma 1

3. Approximately how many times is the TTAGGG sequence repeated in human telomeres?

- A. 100 times
- B. 250 times
- C. 2500 times
- D. 10,000 times

Correct Answer: C

94. Dilemma 1

4. How many base pairs are typically lost per cell division due to telomere shortening?

- A. 1–10 bp
- B. 20–30 bp
- C. 50–200 bp
- D. 500–1000 bp

Correct Answer: C

94. Dilemma 1

5. DNA polymerase has an intrinsic error rate of:

- A. 1 in 100 bp
- B. 1 in 1,000 bp
- C. 1 in 10^6 bp
- D. 1 in 10^{12} bp

Correct Answer: C

94. Dilemma 1

6. How many mutations per cell division result from the natural DNA polymerase error rate?

- A. 1 mutation/division
- B. 10 mutations/division
- C. 100 mutations/division
- D. 1000 mutations/division

Correct Answer: D

94. Dilemma 1

7. DNA polymerase proofreading efficiency recognizes mispaired bases at a frequency of approximately:

- A. 1 in 100 bp
- B. 1 in 1,000 bp
- C. 1 in 10,000 bp
- D. 1 in 100,000 bp

Correct Answer: C

94. Dilemma 1

8. After proofreading and repair mechanisms, the overall error rate of DNA replication becomes:

- A. 10^{-3}
- B. 10^{-5}
- C. 10^{-8}
- D. 10^{-10}

Correct Answer: D

94. Dilemma 1

9. Excision DNA repair functions to:

- A. Add telomeric repeats
- B. Remove RNA primers
- C. Remove abnormal chemically damaged bases
- D. Replicate the lagging strand

Correct Answer: C

94. Dilemma 1

10. Which dilemma is directly associated with telomere shortening?

- A. Dilemma 1
- B. Dilemma 2
- C. Dilemma 3
- D. Both 1 and 3

Correct Answer: B

Basic MCQs from: Dilemma 2 & 3

1. The repeated telomeric sequence shown in Dilemma 2 is:

- A. AATCCC
- B. TTAGGG
- C. GGGATT
- D. TTAACC

Correct Answer: B

95. Dilemma 2,3

2. Approximately how many times is the telomeric sequence repeated?

- A. 200 times
- B. 500 times
- C. 2500 times
- D. 10,000 times

Correct Answer: C

95. Dilemma 2,3

3. How many base pairs are typically lost at each cell division due to telomere shortening?

- A. 1–5 bp
- B. 20–30 bp
- C. 50–200 bp
- D. 500 bp

Correct Answer: C

95. Dilemma 2,3

4. The typical replicative lifespan of a cell (number of divisions possible) is:

- A. 5–10 divisions
- B. 10–15 divisions
- C. 20–30 divisions
- D. 100–200 divisions

Correct Answer: C

95. Dilemma 2,3

5. The intrinsic error rate of DNA polymerase is approximately:

- A. 1 in 10^3 bp
- B. 1 in 10^6 bp

C. 1 in 10^9 bp

D. 1 in 10^{12} bp

Correct Answer: B

95. Dilemma 2,3

6. An error rate of 1 in 10^6 base pairs results in approximately how many mutations per cell division?

A. 1 mutation

B. 10 mutations

C. 100 mutations

D. 1000 mutations

Correct Answer: D

95. Dilemma 2,3

7. DNA polymerase proofreading recognizes mismatched bases with a frequency of:

A. 1 in 100 bp

B. 1 in 1000 bp

C. 1 in 10,000 bp

D. 1 in 100,000 bp

Correct Answer: C

95. Dilemma 2,3

8. After proofreading and repair, the overall error rate of DNA replication becomes:

A. 10^{-3}

B. 10^{-5}

C. 10^{-7}

D. 10^{-10}

Correct Answer: D

95. Dilemma 2,3

9. Excision DNA repair removes:

A. Normal bases

B. RNA primers

C. Abnormal chemically damaged bases

D. Telomeric repeats

Correct Answer: C

95. Dilemma 2,3

10. Which dilemma is directly related to mutation errors during DNA replication?

A. Dilemma 1

B. Dilemma 2

C. Dilemma 3

D. Both 2 and 3

Correct Answer: C

Basic MCQs from: The Central Dogma

1. The central dogma describes the flow of genetic information from:

- A. Protein → RNA → DNA
- B. RNA → DNA → Protein
- C. DNA → RNA → Protein
- D. Protein → DNA → RNA

Correct Answer: C

96. The Central Dogma

2. What must occur before a cell divides?

- A. Transcription must stop
- B. DNA must be replicated
- C. Protein synthesis must double
- D. RNA must be degraded

Correct Answer: B

96. The Central Dogma

3. Replication is the process in which:

- A. RNA is produced from DNA
- B. DNA makes identical copies of itself
- C. Proteins regulate gene expression
- D. DNA is translated into protein

Correct Answer: B

96. The Central Dogma

4. Which statement is TRUE regarding information flow in cells?

- A. Protein → RNA flow is common
- B. Protein → DNA flow is possible
- C. RNA → DNA flow can occur in special cases
- D. RNA → Protein flow does not occur

Correct Answer: C

96. The Central Dogma

5. The enzyme responsible for “reverse” flow of information from RNA to DNA is:

- A. DNA polymerase
- B. RNA polymerase
- C. Reverse transcriptase
- D. Ligase

Correct Answer: C

96. The Central Dogma

6. According to the central dogma, proteins:

- A. Can act as templates for RNA synthesis
- B. Control DNA replication
- C. Never serve as templates for RNA
- D. Are needed to start transcription

Correct Answer: C

96. The Central Dogma

7. The hypothesis that DNA is the template for RNA was widely accepted by:

- A. 1940
- B. 1953
- C. 1965
- D. 1975

Correct Answer: B

96. The Central Dogma

8. After RNA is synthesized, it moves to the cytoplasm to:

- A. Copy DNA
- B. Form chromosomes
- C. Determine amino acid arrangement in proteins
- D. Repair damaged DNA

Correct Answer: C

96. The Central Dogma

9. Who coined the term “Central Dogma” in 1956?

- A. Watson
- B. Crick
- C. Franklin
- D. Hershey

Correct Answer: B

96. The Central Dogma

10. Which processes correspond to the three major steps of genetic information flow?

- A. Translation → Transcription → Duplication
- B. Duplication → Translation → Transcription
- C. Duplication → Transcription → Translation
- D. Transcription → Duplication → Translation

Correct Answer: C

96. The Central Dogma

11. How is the directionality of information flow in the central dogma represented?

- A. Two arrows pointing both ways
- B. One arrow from protein to RNA
- C. Two unidirectional arrows
- D. Circular arrows

Correct Answer: C

12. Reverse flow of information (RNA → DNA):

- A. Happens frequently
- B. Happens only during transcription
- C. Is extremely rare compared to normal flow
- D. Occurs only in bacteria

Correct Answer: C

Basic MCQs from: Synthesis of RNA upon DNA Templates

1. The enzymes that synthesize RNA using DNA templates are called:

- A. DNA polymerases
- B. RNA ligases
- C. RNA polymerases
- D. Reverse transcriptases

Correct Answer: C

2. RNA polymerases require which molecule to act as the template?

- A. RNA
- B. DNA
- C. Protein
- D. Ribosome

Correct Answer: B

3. Which nucleotides are used by RNA polymerase as precursors?

- A. ATP, GTP, CTP, UTP
- B. ATP, TTP, GTP, CTP
- C. dATP, dGTP, dCTP, dTTP
- D. NADH, FADH, GTP, UTP

Correct Answer: A

4. Evidence that DNA determines the correct ribonucleotide sequence came from experiments showing:

- A. RNA length varies with protein size
- B. RNA AU/GC ratio matches DNA AT/GC ratio
- C. RNA is always double-stranded
- D. RNA never binds DNA

Correct Answer: B

98. Synthesis of RNA upon DNA T...

5. During transcription, how many DNA strands serve as the template?

- A. Both strands
- B. Only one strand
- C. Three alternating strands
- D. Neither strand

Correct Answer: B

98. Synthesis of RNA upon DNA T...

6. Why does RNA polymerase use only one DNA strand?

- A. The two strands give identical messages
- B. Using both strands would produce conflicting messages
- C. The second strand is always degraded
- D. The enzyme cannot bind double-stranded DNA

Correct Answer: B

98. Synthesis of RNA upon DNA T...

7. RNA synthesis proceeds in which direction?

- A. 3' → 5'
- B. 5' → 3'
- C. Both directions simultaneously
- D. From the center outward

Correct Answer: B

98. Synthesis of RNA upon DNA T...

8. Pulse-chase experiments using radioactive ribonucleotides showed that:

- A. mRNA is synthesized in the cytoplasm
- B. mRNA remains permanently in the nucleus
- C. mRNA is synthesized in the nucleus and moves to the cytoplasm
- D. RNA cannot leave the nucleus

Correct Answer: C

98. Synthesis of RNA upon DNA T...

9. In pulse-chase experiments, newly synthesized mRNA becomes labeled because:

- A. DNA is radioactive
- B. Radioactive ribonucleotides are incorporated during transcription
- C. Ribosomes absorb radioactive atoms
- D. The nucleus traps all labeled molecules

Correct Answer: B

10. How long did it take for most labeled mRNA to move from the nucleus to the cytoplasm?

- A. A few seconds
- B. 10 minutes

C. About an hour
D. Several days
Correct Answer: C

98. Synthesis of RNA upon DNA T...

Basic MCQs from: *Establishing the Genetic Code*

1. How many amino acids are commonly used in protein synthesis?

- A. 4
- B. 16
- C. 20
- D. 64

Correct Answer: C

99. Establishing the Genetic Co...

2. Two-nucleotide codons could specify only how many amino acids?

- A. 4
- B. 8
- C. 16
- D. 20

Correct Answer: C

99. Establishing the Genetic Co...

3. Triplet codons (groups of three nucleotides) provide how many permutations?

- A. 4
- B. 16
- C. 20
- D. 64

Correct Answer: D

99. Establishing the Genetic Co...

4. Co-linearity between DNA and protein structure was demonstrated by:

- A. Nirenberg and Matthaei
- B. Yanofsky and Brenner
- C. Khorana and Crick
- D. Watson and Crick

Correct Answer: B

99. Establishing the Genetic Co...

5. The first genetic analyses establishing that codons consist of three nucleotides were performed in 1961 by:

- A. Nirenberg and Matthaei
- B. Yanofsky and Watson
- C. Brenner and Crick
- D. Khorana and Brenner

Correct Answer: C

99. Establishing the Genetic Co...

6. The experiment that identified UUU as the codon for phenylalanine used:

- A. DNA templates
- B. Poly-A RNA
- C. Poly-U synthetic RNA
- D. Ribosomes only

Correct Answer: C

99. Establishing the Genetic Co...

7. The work of Nirenberg and Matthaei demonstrated that UUU codes for:

- A. Alanine
- B. Valine
- C. Phenylalanine
- D. Glycine

Correct Answer: C

99. Establishing the Genetic Co...

8. Har Gobind Khorana contributed to cracking the code by synthesizing:

- A. Poly-U chains
- B. Complex repeating polynucleotides like AGUAGU
- C. Ribosomes
- D. DNA repair enzymes

Correct Answer: B

99. Establishing the Genetic Co...

9. Completion of the genetic code in 1966 revealed that how many codons specify amino acids?

- A. 20
- B. 3
- C. 61
- D. 64

Correct Answer: C

99. Establishing the Genetic Co...

10. Most amino acids are encoded by:

- A. A single codon
- B. Two codons only
- C. More than one codon
- D. No codons

Correct Answer: C

99. Establishing the Genetic Co...

Basic MCQs from: TRANSCRIPTION

1. Gene expression includes which of the following processes?

- A. Replication only
- B. Transcription, RNA processing, and translation
- C. Translation only
- D. DNA repair and recombination

Correct Answer: B

100. TRANSCRIPTION

2. Transcription is most similar to which cellular process?

- A. Translation
- B. DNA replication
- C. Cell division
- D. DNA repair

Correct Answer: B

100. TRANSCRIPTION

3. Both transcription and DNA replication involve enzymes that:

- A. Use RNA as a template
- B. Synthesize nucleic acids complementary to a DNA template
- C. Require primers to initiate synthesis
- D. Produce double-stranded products

Correct Answer: B

100. TRANSCRIPTION

4. A major difference between transcription and replication is that transcription uses:

- A. Deoxyribonucleotides
- B. Ribonucleotides
- C. Amino acids
- D. dNTPs only

Correct Answer: B

100. TRANSCRIPTION

5. RNA polymerase differs from DNA polymerase because RNA polymerase:

- A. Requires a primer
- B. Cannot initiate synthesis
- C. Initiates transcription de novo (without primer)
- D. Synthesizes DNA

Correct Answer: C

100. TRANSCRIPTION

6. During transcription, the RNA product:

- A. Remains base-paired to the DNA template
- B. Immediately degrades
- C. Does not remain base-paired to the DNA template
- D. Forms double-stranded RNA

Correct Answer: C

100. TRANSCRIPTION

7. Transcription accuracy is lower than replication because:

- A. RNA polymerase lacks extensive proofreading
- B. RNA contains uracil
- C. DNA polymerase works more slowly
- D. RNA is unstable

Correct Answer: A

100. TRANSCRIPTION

8. The error rate of transcription is approximately:

- A. 1 per 10 nucleotides
- B. 1 per 1,000 nucleotides
- C. 1 per 10,000 nucleotides
- D. 1 per 10 million nucleotides

Correct Answer: C

100. TRANSCRIPTION

9. Transcription differs from replication in that transcription:

- A. Copies the entire genome once per cell cycle
- B. Copies selected regions and can make many copies
- C. Requires helicase activity
- D. Occurs only during cell division

Correct Answer: B

100. TRANSCRIPTION

10. Replication must:

- A. Copy only selected regions
- B. Copy the entire genome once per cell division
- C. Occur multiple times in the same cell cycle
- D. Produce RNA molecules

Correct Answer: B

100. TRANSCRIPTION

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Basic MCQs from: RNA POLYMERASES

1. The primary function of RNA polymerase in all cells is:

- A. DNA replication
- B. RNA synthesis
- C. Protein synthesis
- D. DNA repair

Correct Answer: B

101. RNA POLYMERASES

2. Bacteria contain how many RNA polymerases?

- A. One
- B. Two
- C. Three
- D. Five

Correct Answer: A

101. RNA POLYMERASES

3. Eukaryotic cells contain how many major RNA polymerases?

- A. One
- B. Two
- C. Three (Pol I, Pol II, Pol III)
- D. Five

Correct Answer: C

101. RNA POLYMERASES

4. RNA polymerases IV and V are found only in:

- A. Bacteria
- B. Animals
- C. Plants
- D. Fungi

Correct Answer: C

101. RNA POLYMERASES

5. The function of RNA polymerases IV and V in plants is to transcribe:

- A. mRNA
- B. rRNA
- C. tRNA
- D. Small interfering RNAs

Correct Answer: D

101. RNA POLYMERASES

6. The bacterial RNA polymerase core enzyme contains how many different subunits?

- A. Two
- B. Three
- C. Four (α , β , β' , ω)
- D. Six

Correct Answer: C

101. RNA POLYMERASES

7. The bacterial RNA polymerase core contains how many copies of the α subunit?

- A. One
- B. Two
- C. Three
- D. Four

Correct Answer: B

101. RNA POLYMERASES

8. The overall structure of bacterial RNA polymerase and yeast Pol II resembles a:

- A. Double helix
- B. Hammer
- C. Crab claw
- D. Ribbon

Correct Answer: C

101. RNA POLYMERASES

9. The two “pincers” of the RNA polymerase crab claw structure are mainly formed by which subunits?

- A. α and β
- B. β and β'
- C. α and ω
- D. α and β'

Correct Answer: B

101. RNA POLYMERASES

10. The active site of RNA polymerase is located in the region called the:

- A. Polymerization tunnel
- B. Nucleotide pocket
- C. Active center cleft
- D. Replication fork

Correct Answer: C

101. RNA POLYMERASES

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Basic MCQs from: *Transcription Proceeds in a Series of Steps*

1. Transcription occurs in three major phases:

- A. Binding, processing, release
- B. Initiation, elongation, termination
- C. Replication, transcription, translation
- D. Loading, proofreading, sealing

Correct Answer: B

102. Transcription process

2. The DNA sequence that initially binds RNA polymerase is called:

- A. Operator
- B. Enhancer
- C. Promoter
- D. Codon

Correct Answer: C

102. Transcription process

3. When RNA polymerase binds the promoter and DNA is still double-stranded, the complex is called:

- A. Open complex
- B. Ternary complex
- C. Closed complex
- D. Initial transcribing complex

Correct Answer: C

102. Transcription process

4. The transition from closed to open complex involves:

- A. RNA polymerase leaving the DNA
- B. DNA strands separating over ~13 bp
- C. RNA strands annealing
- D. DNA replication beginning

Correct Answer: B

102. Transcription process

5. The unwound region of DNA used during transcription is called:

- A. Replication fork
- B. Transcription bubble
- C. DNA gap
- D. RNA window

Correct Answer: B

102. Transcription process

6. Transcription always occurs in which direction?

- A. 3' → 5'
- B. 5' → 3'
- C. Both directions
- D. Random direction

Correct Answer: B

102. Transcription process

7. During transcription, how many DNA strands act as a template?

- A. Both strands
- B. Only one strand

- C. Three alternating strands
- D. None

Correct Answer: B

102. Transcription process

8. Short RNA transcripts (<10 nucleotides) are often released during:

- A. Termination
- B. Initial transcription
- C. Elongation
- D. RNA processing

Correct Answer: B

102. Transcription process

9. The enzyme–DNA–RNA complex formed after promoter escape is called:

- A. Closed complex
- B. Open complex
- C. Ternary complex
- D. Initial transcribing complex

Correct Answer: C

102. Transcription process

10. RNA polymerase is considered to have “escaped the promoter” when:

- A. It releases the RNA product
- B. It synthesizes an RNA chain >10 nucleotides
- C. It binds the terminator sequence
- D. It unwinds the entire gene

Correct Answer: B

102. Transcription process

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Basic MCQs from: *Elongation & Termination in Transcription*

1. RNA polymerase enters the elongation phase after synthesizing approximately how many nucleotides?

- A. 2 nucleotides
- B. 5 nucleotides
- C. 10 nucleotides
- D. 50 nucleotides

Correct Answer: C

103. Elongation during Transcri...

2. Which of the following is *not* a function performed by RNA polymerase during elongation?

- A. DNA unwinding in front
- B. Re-annealing DNA behind
- C. Joining Okazaki fragments
- D. Proofreading

Correct Answer: C

103. Elongation during Transcri...

3. As RNA polymerase moves forward during elongation, it:

- A. Leaves behind single-stranded DNA permanently
- B. Dissociates the growing RNA from the DNA template
- C. Forms a permanent RNA–DNA hybrid
- D. Stops periodically every 3 nucleotides

Correct Answer: B

103. Elongation during Transcri...

4. Which activity is true for RNA polymerase but requires multiple enzymes during DNA replication?

- A. RNA synthesis
- B. DNA proofreading
- C. Unwinding and re-annealing DNA
- D. Helicase activation

Correct Answer: C

103. Elongation during Transcri...

5. Termination refers to:

- A. The formation of the transcription bubble
- B. The release of RNA polymerase and RNA transcript from DNA
- C. Stalling at promoter regions
- D. Conversion of RNA into protein

Correct Answer: B

103. Elongation during Transcri...

6. Termination signals in transcription:

- A. Are identical in all cells
- B. Are always recognized by proteins
- C. Can be well-characterized sequences or unknown signals
- D. Are only present in eukaryotes

Correct Answer: C

103. Elongation during Transcri...

7. Which of the following occurs *during* elongation?

- A. RNA polymerase binds promoter
- B. RNA polymerase escapes promoter
- C. RNA chain grows and separates from template
- D. RNA polymerase dissociates from DNA

Correct Answer: C

103. Elongation during Transcri...

8. Proofreading during transcription is carried out by:

- A. A dedicated proofreading enzyme
- B. DNA polymerase
- C. Helicase
- D. RNA polymerase itself

Correct Answer: D

103. Elongation during Transcri...

Basic MCQs from: *Transcription Cycle in Bacteria*

1. The bacterial core RNA polymerase can, in principle, initiate transcription anywhere on DNA, but actually initiates only at:

- A. Origins of replication
- B. Terminators
- C. Promoters
- D. Operators

Correct Answer: C

104. Transcription Cycle in Bac...

2. The factor required to convert core RNA polymerase into a promoter-specific holoenzyme is:

- A. ρ factor
- B. σ factor
- C. λ repressor
- D. Helicase

Correct Answer: B

104. Transcription Cycle in Bac...

3. The holoenzyme form of RNA polymerase initiates transcription:

- A. Randomly along DNA
- B. Only at promoters
- C. Only at terminators
- D. At replication origins

Correct Answer: B

104. Transcription Cycle in Bac...

4. The predominant σ factor in *E. coli* is:

- A. σ_{32}
- B. σ_{54}
- C. σ_{70}
- D. σ_{28}

Correct Answer: C

104. Transcription Cycle in Bac...

5. Promoters recognized by σ_{70} contain two conserved sequences located at:

- A. -10 and -50
- B. -20 and -40
- C. -10 and -35
- D. -5 and -25

Correct Answer: C

104. Transcription Cycle in Bac...

6. The conserved -10 and -35 promoter elements are separated by:

- A. 5-10 nucleotides
- B. 17-19 nucleotides
- C. 35-40 nucleotides
- D. 50 nucleotides

Correct Answer: B

104. Transcription Cycle in Bac...

7. σ_{70} factor contains how many major regions?

- A. 2
- B. 3
- C. 4
- D. 5

Correct Answer: C

104. Transcription Cycle in Bac...

8. Which σ_{70} region recognizes the -35 promoter element?

- A. Region 1
- B. Region 2
- C. Region 3
- D. Region 4

Correct Answer: D

104. Transcription Cycle in Bac...

9. Region 4 contains a DNA-binding motif known as:

- A. Zinc finger
- B. Helix-turn-helix
- C. Leucine zipper
- D. Beta-sheet clamp

Correct Answer: B

104. Transcription Cycle in Bac...

10. DNA melting during initiation is triggered at which promoter element?

- A. -35 region
- B. -10 region

- C. Shine–Dalgarno sequence
- D. Operator region

Correct Answer: B

104. Transcription Cycle in Bac...

11. The –10 region is stabilized in its melted state by:

- A. Zinc ions
- B. Aromatic amino acids in σ
- C. ρ factor
- D. tRNA binding

Correct Answer: B

104. Transcription Cycle in Bac...

12. Two bases of the non-template strand are stabilized in the open complex by:

- A. Binding to β subunit
- B. Insertion into pockets of the σ protein
- C. Interaction with RNA polymerase α subunits
- D. Phosphorylation

Correct Answer: B

104. Transcription Cycle in Bac...

13. The σ subunit is oriented in the holoenzyme so that:

- A. It cannot contact DNA
- B. Its DNA-binding regions face away from the enzyme body
- C. It binds only RNA
- D. It remains embedded inside the core enzyme

Correct Answer: B